

All available registered signals

Directed factor-mimicking portfolio catalogue

Abstract

This in-sample catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by directed raw Sharpe, highest first. Direction is the sign that makes comparison-window Sharpe non-negative.

Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes direction and Sharpe order and feeds all reported analyses. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only.

Construction Configuration

Setting	Effective value
Factors	88
Universe	shared security screen; refreshed monthly
Formation filters	price ≥ 1.0 ; 21d dollar volume $\geq$ none
Market-cap screen	top 37.5% monthly
Signal preparation	within-month winsorize, normalize, then demean
Composite	mean available normalized components; factor-specific coverage floor
Construction	one blanket profile
Rebalance / return	month-end / next calendar month simple return
Costs	effective configured costs; default 0 bps
Minimum sample	20 stocks/month; 24 common months; 90% factor coverage
Comparison window	2012-03-01 to 2026-05-01; factor- and control-complete
Winsor settings	1%-99%
Controls	MKT, SMB, VAL, MOM, BAB, QMJ
Catalogue note	Pinned gold contains 99/101 registered signals; unavailable and explicitly excluded: is_negative_working_capital, is_negative_net_operating_assets

Returns PCA

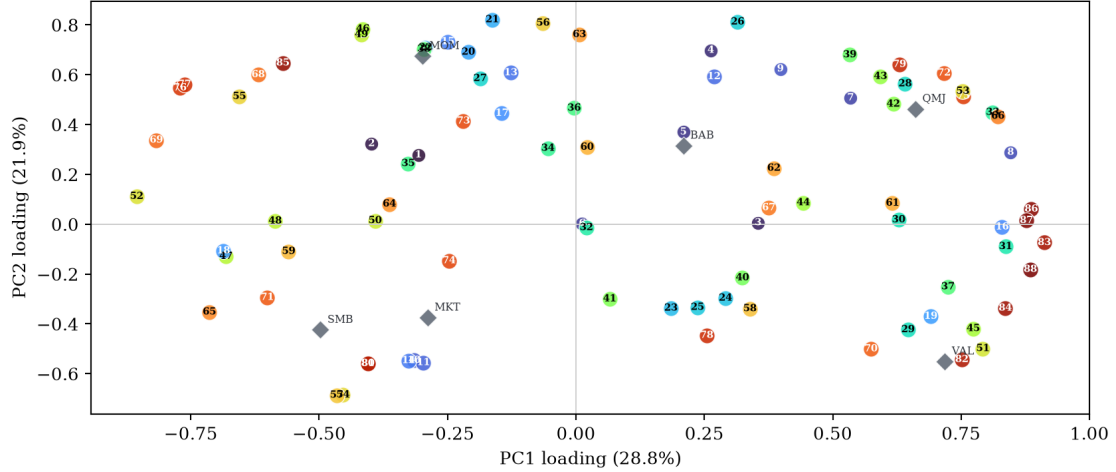


Figure: catalogue factors are numbered by directed-Sharpe index; attribution controls are named; $N=167$ common months.

Index

Idx	Factor	Dir.	N
1	surprisePercentage	+1	167
2	dollar_volume_21d	+1	167
3	is_positive_revenue	+1	167
4	amihud_illiq_21d	-1	167
5	surprise	+1	167
6	resid_ret_252v_1260c_SPY	-1	167
7	fcf_margin	+1	167
8	fcf	+1	167
9	cfoa	+1	167
10	asset_growth_yoy_sign_flip	+1	167
11	debt_issuance_yoy_sign_flip	+1	167
12	net_income_growth_yoy	+1	167
13	eps_growth_yoy	+1	167
14	equity_issuance_yoy_sign_flip	+1	167
15	momentum_21_252	+1	167
16	fcf_yield	+1	167
17	momentum_21_126	+1	167
18	accruals_total	-1	167
19	debt_to_equity	+1	167
20	momentum_126_252	+1	167
21	resid_momentum_21_252	+1	167
22	gpoa	+1	167
23	corr_1260d_FRENCH_CRSP	+1	167
24	corr_1260d_SPY	+1	167
25	corr_1260d_VTI	+1	167
26	high_52w_252d	+1	167
27	info_discreteness_21_252	-1	167
28	roe	+1	167
29	accruals_noa_proxy	-1	167
30	sga_to_revenue	-1	167
31	equity_issuance_yoy	-1	167
32	revenue_growth_yoy_sign_flip	+1	167
33	is_positive_net_income	+1	167
34	eps_growth_yoy_sign_flip	+1	167
35	dpo	-1	167
36	net_income_growth_yoy_sign_flip	+1	167
37	cash_earnings_ttm	+1	167
38	growth_cfoa_5y	+1	167
39	roa	+1	167
40	accruals_wc_proxy	-1	167
41	accruals_percent	-1	167
42	is_positive_income_tax	+1	167
43	roc	+1	167
44	ret_volume_corr_63d	-1	167
45	evs_ttm	-1	167
46	value_proxy_1260d	-1	167
47	accruals	-1	167
48	working_capital_proxy	+1	167
49	momentum_21_756	+1	167
50	dio	+1	167
51	ps_ttm	-1	167
52	rd_to_revenue	+1	167
53	net_margin	+1	167
54	BAB composite	-1	167
55	revenue_growth_yoy	+1	167
56	growth_roa_5y	+1	167
57	Beta composite	-1	167
58	is_negative_ev	+1	167
59	growth_acc_5y	+1	167
60	momentum_0_21	+1	167
61	debt_issuance_yoy	-1	167
62	skew_63d	-1	167
63	growth_roe_5y	+1	167
64	is_negative_equity	+1	167
65	roe_volatility	+1	167
66	operating_margin	+1	167
67	debt_to_assets	+1	167
68	growth_gpoa_5y	+1	167
69	asset_growth_yoy	+1	167
70	growth_gmar_5y	-1	167
71	ev	-1	167
72	vol_63d	-1	167
73	is_negative_invested_capital	-1	167
74	dso	+1	167
75	lottery_max_5_21d	-1	167
76	book_to_market	-1	167
77	pb	+1	167
78	gross_margin	-1	167
79	ivol_21d	-1	167
80	log_market_cap	-1	167
81	market_cap	-1	167
82	altman_z	-1	167
83	earnings_yield	+1	167
84	earnings_to_price	+1	167
85	pe_ttm	+1	167
86	cash_ratio	-1	167
87	current_ratio	-1	167
88	net_operating_assets_proxy	+1	167

Returns

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
surprisePercentage	4.99%	3.85%	1.30	4.96	-3.48%	0.258	4.05%	3.85%
dollar_volume_21d	6.03%	5.00%	1.21	4.47	-5.54%	0.451	3.94%	5.00%
is_positive_revenue	29.55%	25.32%	1.17	4.59	-27.00%	0.136	26.21%	25.32%
amihud_illiq_21d	6.02%	5.66%	1.06	3.73	-10.27%	0.581	2.92%	5.66%
surprise	3.59%	3.94%	0.91	3.71	-5.67%	0.217	2.04%	3.94%
resid_ret_252v_1260c_SPY	5.98%	6.72%	0.89	3.07	-13.56%	0.056	6.30%	6.72%
fcf_margin	4.39%	5.44%	0.81	2.34	-14.08%	0.414	2.25%	5.44%
fcf	4.89%	6.50%	0.75	2.25	-18.07%	0.684	3.05%	6.50%
cfoa	5.27%	7.03%	0.75	2.42	-18.48%	0.494	2.47%	7.03%
asset_growth_yoy_sign_flip	11.45%	16.05%	0.71	3.22	-27.35%	0.948	-0.48%	16.05%

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_issuance_yoy_sign_flip	11.42%	16.06%	0.71	3.22	-28.26%	0.950	-0.59%	16.06%
net_income_growth_yoy	3.72%	5.25%	0.71	2.27	-14.30%	0.327	0.99%	5.25%
eps_growth_yoy	4.35%	6.23%	0.70	2.55	-15.12%	0.334	1.64%	6.23%
equity_issuance_yoy_sign_flip	11.23%	16.09%	0.70	3.15	-28.01%	0.947	-0.60%	16.09%
momentum_21_252	8.08%	11.79%	0.69	2.82	-17.59%	0.920	2.03%	11.79%
fcf_yield	5.16%	8.02%	0.64	2.02	-23.23%	0.625	3.56%	8.02%
momentum_21_126	6.55%	10.19%	0.64	3.07	-10.59%	0.490	1.71%	10.19%
accruals_total	3.41%	5.86%	0.58	1.90	-22.16%	0.466	4.50%	5.86%
debt_to_equity	3.79%	6.94%	0.55	1.73	-18.24%	0.408	3.13%	6.94%
momentum_126_252	5.26%	10.49%	0.50	1.80	-22.41%	0.690	0.71%	10.49%
resid_momentum_21_252	5.93%	11.94%	0.50	2.03	-21.39%	0.864	0.89%	11.94%
gpoa	3.52%	7.45%	0.47	1.50	-18.84%	0.631	0.63%	7.45%
corr_1260d_FRENCH_CRSP	4.02%	8.67%	0.46	1.82	-15.28%	0.425	2.93%	8.67%
corr_1260d_SPY	3.81%	8.32%	0.46	1.76	-16.72%	0.403	2.49%	8.32%
corr_1260d_VTI	3.92%	8.58%	0.46	1.78	-15.86%	0.413	2.77%	8.58%
high_52w_252d	5.92%	13.30%	0.45	1.87	-23.17%	0.719	0.51%	13.30%
info_discreteness_21_252	2.76%	6.25%	0.44	1.76	-9.45%	0.388	0.42%	6.25%
roe	3.47%	7.97%	0.44	1.39	-22.59%	0.634	-0.65%	7.97%
accruals_noa_proxy	2.25%	5.26%	0.43	1.32	-22.11%	0.531	3.28%	5.26%
sga_to_revenue	2.47%	5.94%	0.42	1.16	-28.22%	0.483	1.79%	5.94%
equity_issuance_yoy	3.05%	7.49%	0.41	1.27	-22.25%	0.695	1.61%	7.49%
revenue_growth_yoy_sign_flip	12.04%	30.41%	0.40	1.63	-60.31%	0.071	10.88%	30.41%
is_positive_net_income	4.54%	11.51%	0.39	1.23	-32.56%	0.700	-0.12%	11.51%
eps_growth_yoy_sign_flip	5.77%	14.76%	0.39	1.49	-32.41%	0.106	2.48%	14.76%
dpo	1.57%	4.02%	0.39	1.29	-12.15%	0.226	1.00%	4.02%
net_income_growth_yoy_sign_flip	3.42%	9.06%	0.38	1.38	-21.06%	0.226	0.06%	9.06%
cash_earnings_ttm	2.24%	5.98%	0.37	1.31	-21.73%	0.695	3.63%	5.98%
growth_cfoa_5y	2.29%	6.18%	0.37	1.21	-13.64%	0.486	0.78%	6.18%
roa	3.06%	8.77%	0.35	1.18	-25.75%	0.621	-1.13%	8.77%
accruals_wc_proxy	1.26%	3.62%	0.35	1.22	-9.08%	0.118	1.81%	3.62%
accruals_percent	1.43%	4.20%	0.34	1.21	-12.77%	0.188	2.71%	4.20%
is_positive_income_tax	2.19%	6.52%	0.34	1.13	-21.50%	0.537	-0.35%	6.52%
roc	2.72%	8.68%	0.31	1.02	-28.26%	0.643	-0.96%	8.68%
ret_volume_corr_63d	1.62%	5.20%	0.31	1.23	-14.85%	0.164	0.55%	5.20%
evs_ttm	2.93%	9.44%	0.31	0.99	-32.18%	0.754	2.35%	9.44%
value_proxy_1260d	3.46%	11.58%	0.30	1.04	-38.66%	0.692	-1.11%	11.58%
accruals	1.58%	5.33%	0.30	1.07	-17.32%	0.362	3.24%	5.33%
working_capital_proxy	1.31%	4.51%	0.29	0.98	-13.61%	0.360	0.39%	4.51%
momentum_21_756	3.17%	11.41%	0.28	0.95	-36.06%	0.664	-2.08%	11.41%
dio	1.36%	5.00%	0.27	0.93	-17.69%	0.143	1.53%	5.00%
ps_ttm	2.83%	10.78%	0.26	0.83	-40.45%	0.796	2.88%	10.78%
rd_to_revenue	3.33%	12.97%	0.26	0.83	-48.91%	0.690	6.46%	12.97%
net_margin	1.90%	7.59%	0.25	0.77	-27.02%	0.643	-0.90%	7.59%
BAB_composite	3.76%	15.23%	0.25	0.92	-33.10%	0.816	5.71%	15.23%
revenue_growth_yoy	1.89%	8.15%	0.23	0.81	-31.05%	0.657	0.01%	8.15%
growth_roa_5y	1.53%	6.72%	0.23	0.76	-17.14%	0.471	-1.13%	6.72%
Beta_composite	3.32%	15.25%	0.22	0.81	-33.68%	0.815	5.26%	15.25%
is_negative_ev	2.52%	12.71%	0.20	0.75	-50.87%	0.168	5.81%	12.71%
growth_acc_5y	0.85%	4.78%	0.18	0.73	-14.82%	0.279	2.12%	4.78%
momentum_0_21	1.66%	9.48%	0.17	0.75	-29.20%	0.112	0.79%	9.48%
debt_issuance_yoy	0.73%	4.30%	0.17	0.53	-24.98%	0.322	0.38%	4.30%
skew_63d	0.80%	5.07%	0.16	0.59	-16.29%	0.144	-0.42%	5.07%
growth_roe_5y	0.92%	6.09%	0.15	0.50	-16.70%	0.430	-1.71%	6.09%
is_negative_equity	1.15%	7.59%	0.15	0.52	-32.27%	0.135	0.73%	7.59%
roe_volatility	0.94%	6.98%	0.13	0.47	-23.82%	0.586	1.85%	6.98%
operating_margin	1.00%	8.27%	0.12	0.35	-33.85%	0.657	-1.31%	8.27%
debt_to_assets	0.55%	5.14%	0.11	0.36	-18.31%	0.216	-1.08%	5.14%
growth_gpoa_5y	0.78%	7.27%	0.11	0.33	-24.76%	0.712	-0.79%	7.27%
asset_growth_yoy	0.70%	7.50%	0.09	0.28	-35.20%	0.729	-0.38%	7.50%
growth_gmar_5y	0.56%	6.17%	0.09	0.29	-26.44%	0.607	0.97%	6.17%
ev	0.55%	6.15%	0.09	0.36	-24.36%	0.710	1.31%	6.15%
vol_63d	1.33%	15.47%	0.09	0.30	-42.68%	0.770	-2.23%	15.47%
is_negative_invested_capital	0.76%	8.85%	0.09	0.34	-25.18%	0.107	-0.63%	8.85%
dso	0.36%	4.57%	0.08	0.34	-8.87%	0.136	1.08%	4.57%
lottery_max_5_21d	1.04%	13.43%	0.08	0.26	-43.32%	0.725	-2.00%	13.43%
book_to_market	0.68%	10.81%	0.06	0.19	-52.64%	0.825	-1.66%	10.81%
pb	0.68%	11.25%	0.06	0.19	-53.19%	0.822	-1.73%	11.25%
gross_margin	0.31%	5.60%	0.06	0.18	-26.26%	0.467	0.03%	5.60%
ivol_21d	0.60%	10.83%	0.06	0.20	-32.76%	0.650	-3.28%	10.83%
log_market_cap	0.32%	6.23%	0.05	0.21	-22.38%	0.775	1.22%	6.23%
market_cap	0.32%	6.23%	0.05	0.21	-22.38%	0.775	1.22%	6.23%
altman_z	0.34%	10.30%	0.03	0.10	-47.64%	0.713	2.31%	10.30%
earnings_yield	0.35%	10.53%	0.03	0.11	-44.36%	0.758	-1.08%	10.53%
earnings_to_price	0.27%	11.18%	0.02	0.08	-44.65%	0.749	-0.18%	11.18%
pe_ttm	0.22%	10.54%	0.02	0.08	-40.81%	0.709	-1.45%	10.54%
cash_ratio	0.09%	7.64%	0.01	0.04	-33.92%	0.702	-1.16%	7.64%
current_ratio	0.07%	7.03%	0.01	0.03	-28.57%	0.675	-0.32%	7.03%
net_operating_assets_proxy	0.09%	10.00%	0.01	0.03	-49.93%	0.805	0.33%	10.00%

Risk Loadings

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
surprisePercentage	0.06 (3.28)	0.07 (2.43)	-0.09 (-2.82)	0.08 (2.55)	-0.07 (-1.85)	0.08 (1.92)	3.73% (4.49)	0.266
dollar_volume_21d	0.12 (4.10)	-0.20 (-4.23)	-0.15 (-6.86)	0.07 (2.87)	-0.05 (-0.87)	-0.06 (-1.00)	3.48% (3.33)	0.463
is_positive_revenue	-0.02 (-0.09)	0.07 (0.30)	0.37 (2.26)	-0.02 (-0.11)	0.18 (0.70)	0.75 (2.59)	28.69% (4.03)	0.144
amihud_illiq_21d	0.07 (2.96)	-0.31 (-5.75)	-0.08 (-2.00)	0.13 (3.41)	-0.00 (-0.15)	0.20 (4.76)	3.09% (2.81)	0.581
surprise	0.08 (3.32)	0.02 (0.70)	0.01 (0.19)	0.10 (3.24)	-0.09 (-2.15)	0.21 (3.70)	1.90% (2.30)	0.218
resid_ret_252v_1260c_SPY	0.01 (0.12)	0.13 (1.58)	-0.05 (-0.93)	-0.09 (-1.20)	-0.02 (-0.20)	0.15 (1.60)	6.37% (3.26)	0.056
fcf_margin	0.02 (0.83)	-0.16 (-3.44)	-0.03 (-0.46)	0.04 (0.78)	0.03 (0.54)	0.28 (5.22)	2.77% (1.91)	0.421
fcf	0.04 (1.82)	-0.31 (-7.22)	0.19 (3.58)	0.06 (2.26)	-0.01 (-0.27)	0.31 (6.38)	3.37% (2.56)	0.686
cfoa	0.08 (2.44)	-0.19 (-3.27)	-0.03 (-0.87)	0.10 (2.71)	-0.13 (-2.18)	0.46 (7.08)	2.89% (1.92)	0.496
asset_growth_yoy_sign_flip	0.96 (45.85)	0.36 (8.92)	0.13 (5.39)	-0.07 (-2.36)	0.16 (2.94)	-0.18 (-4.12)	-1.42% (-1.44)	0.953
debt_issuance_yoy_sign_flip	0.97 (46.16)	0.35 (8.81)	0.15 (6.46)	-0.07 (-2.47)	0.17 (3.09)	-0.18 (-4.25)	-1.48% (-1.49)	0.954
net_income_growth_yoy	0.04 (1.20)	-0.05 (-1.15)	0.08 (1.24)	0.05 (3.97)	0.05 (1.00)	0.13 (2.39)	1.28% (0.86)	0.329
eps_growth_yoy	0.04	-0.01	-0.05	0.26	0.06	0.02	1.64%	0.334

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
	(0.97)	(-0.13)	(-0.82)	(4.36)	(1.06)	(0.41)	(0.90)	
equity_issuance_yoy_sign_flip	0.96	0.36	0.12	-0.07	0.16	-0.19	-1.57%	0.952
	(45.73)	(9.02)	(4.57)	(-2.39)	(3.09)	(-4.24)	(-1.57)	
momentum_21_252	-0.05	0.01	-0.02	0.88	0.17	0.00	2.13%	0.920
	(-2.02)	(0.37)	(-0.79)	(24.79)	(4.87)	(0.02)	(2.28)	
fcf_yield	0.10	-0.04	0.32	0.02	-0.06	0.52	3.79%	0.625
	(4.11)	(-0.58)	(4.71)	(0.41)	(-1.27)	(9.13)	(2.37)	
momentum_21_126	0.02	0.12	0.11	0.64	0.11	0.06	1.99%	0.491
	(0.52)	(1.39)	(1.26)	(6.60)	(0.94)	(0.52)	(0.95)	
accruals_total	-0.02	0.14	-0.30	-0.12	-0.01	-0.14	4.14%	0.469
	(-0.60)	(3.41)	(-5.29)	(-1.87)	(-0.21)	(-2.90)	(3.51)	
debt_to_equity	0.09	-0.06	0.31	-0.03	0.03	0.14	2.98%	0.408
	(2.68)	(-0.86)	(4.76)	(-0.50)	(0.35)	(1.89)	(1.76)	
momentum_126_252	-0.07	-0.08	-0.12	0.60	0.15	-0.03	1.10%	0.691
	(-1.46)	(-1.20)	(-2.01)	(9.15)	(1.85)	(-0.33)	(0.64)	
resid_momentum_21_252	-0.15	-0.11	-0.10	0.75	0.29	-0.00	1.22%	0.865
	(-4.81)	(-2.42)	(-1.92)	(13.29)	(5.33)	(-0.04)	(0.86)	
gpoa	0.07	-0.03	-0.46	-0.01	-0.05	0.37	0.66%	0.631
	(2.78)	(-0.72)	(-12.11)	(-0.31)	(-0.79)	(7.48)	(0.48)	
corr_1260d_FRENCH_CRSP	0.37	0.07	0.08	-0.03	-0.42	0.32	1.12%	0.465
	(8.10)	(0.96)	(1.33)	(-0.49)	(-5.34)	(3.82)	(0.66)	
corr_1260d_SPY	0.34	0.01	0.12	-0.01	-0.39	0.34	1.03%	0.434
	(7.18)	(0.18)	(1.99)	(-0.16)	(-5.14)	(3.97)	(0.63)	
corr_1260d_VTI	0.36	0.06	0.12	-0.02	-0.41	0.33	1.09%	0.449
	(7.67)	(0.82)	(1.82)	(-0.26)	(-5.21)	(3.86)	(0.65)	
high_52w_252d	-0.15	-0.17	0.07	0.66	0.15	0.55	1.93%	0.725
	(-2.44)	(-2.14)	(0.99)	(10.18)	(1.57)	(4.45)	(1.21)	
info_discreteness_21_252	-0.03	-0.11	-0.11	0.21	0.12	-0.04	0.76%	0.390
	(-0.76)	(-2.18)	(-1.79)	(2.72)	(1.77)	(-0.50)	(0.47)	
roe	0.14	-0.15	0.05	0.10	-0.03	0.64	-0.37%	0.635
	(4.35)	(-2.81)	(1.17)	(2.04)	(-0.62)	(9.09)	(-0.25)	
accruals_noa_proxy	0.00	0.02	0.32	0.06	-0.16	0.12	3.46%	0.532
	(0.19)	(0.39)	(6.75)	(1.45)	(-4.01)	(3.02)	(3.17)	
sga_to_revenue	0.02	-0.22	0.34	0.13	-0.01	-0.00	1.97%	0.484
	(0.77)	(-4.96)	(6.99)	(2.59)	(-0.14)	(-0.02)	(1.52)	
equity_issuance_yoy	0.14	-0.03	0.31	0.00	-0.14	0.53	1.68%	0.695
	(5.55)	(-0.70)	(6.39)	(0.06)	(-2.94)	(10.78)	(1.18)	
revenue_growth_yoy_sign_flip	-0.12	0.20	-0.06	-0.29	0.96	-0.34	10.99%	0.071
	(-0.53)	(0.66)	(-0.23)	(-1.82)	(3.60)	(-1.19)	(1.20)	
is_positive_net_income	0.07	-0.21	0.34	0.25	0.08	0.78	0.56%	0.703
	(1.85)	(-2.91)	(6.03)	(4.01)	(1.04)	(7.90)	(0.28)	
eps_growth_yoy_sign_flip	0.01	-0.06	0.07	0.37	0.20	-0.16	2.55%	0.106
	(0.11)	(-0.34)	(0.51)	(2.21)	(1.47)	(-0.90)	(0.58)	
dpo	0.04	0.10	-0.15	-0.01	-0.03	0.12	0.78%	0.229
	(1.49)	(2.34)	(-6.13)	(-0.34)	(-0.82)	(2.75)	(0.78)	
net_income_growth_yoy_sign_flip	0.01	0.09	0.03	0.35	0.09	0.13	0.58%	0.228
	(0.16)	(1.02)	(0.38)	(3.26)	(0.98)	(1.34)	(0.22)	
cash_earnings_ttm	-0.04	-0.34	0.30	-0.02	-0.09	-0.04	3.82%	0.696
	(-1.17)	(-10.36)	(10.69)	(-1.27)	(-2.82)	(-0.87)	(3.67)	
growth_cfoa_5y	-0.01	-0.06	-0.36	-0.03	0.01	0.18	0.97%	0.487
	(-0.18)	(-1.12)	(-13.23)	(-0.94)	(0.14)	(3.91)	(0.75)	
roa	0.11	-0.18	-0.00	0.14	-0.08	0.69	-0.66%	0.623
	(2.94)	(-3.35)	(-0.06)	(2.60)	(-1.41)	(9.22)	(-0.41)	
accruals_wc_proxy	-0.05	-0.03	0.05	-0.04	0.01	0.01	2.30%	0.138
	(-2.27)	(-0.74)	(1.89)	(-1.18)	(0.28)	(0.15)	(2.27)	
accruals_percent	-0.05	-0.09	0.10	0.00	-0.04	-0.18	2.81%	0.188
	(-1.41)	(-2.27)	(2.84)	(0.13)	(-0.93)	(-3.87)	(2.41)	
is_positive_income_tax	0.00	-0.05	0.10	0.12	0.08	0.41	0.39%	0.545
	(0.07)	(-1.20)	(2.67)	(2.98)	(1.35)	(7.08)	(0.32)	
roc	0.10	-0.28	-0.05	0.06	-0.08	0.64	-0.54%	0.645
	(2.87)	(-4.90)	(-0.86)	(1.17)	(-1.82)	(9.25)	(-0.34)	
ret_volume_corr_63d	0.07	-0.11	0.08	-0.02	0.04	0.10	0.39%	0.165
	(2.68)	(-2.07)	(1.91)	(-0.68)	(0.85)	(1.93)	(0.32)	
evs_ttm	0.18	0.02	0.62	0.04	-0.19	0.38	2.26%	0.754
	(6.10)	(0.35)	(12.34)	(1.11)	(-2.97)	(6.67)	(1.31)	
value_proxy_1260d	-0.01	0.04	-0.58	0.29	0.10	0.23	-0.90%	0.692
	(-0.36)	(0.57)	(-5.06)	(3.30)	(1.53)	(3.09)	(-0.40)	
accruals	-0.04	0.02	-0.16	-0.06	-0.06	-0.24	2.91%	0.365
	(-1.27)	(0.42)	(-4.51)	(-1.63)	(-1.51)	(-4.84)	(2.52)	
working_capital_proxy	0.11	0.16	-0.07	0.09	-0.06	0.00	-0.37%	0.392
	(4.13)	(4.58)	(-1.92)	(2.15)	(-1.17)	(0.09)	(-0.39)	
momentum_21_756	0.04	-0.10	-0.59	0.24	0.17	0.13	-2.03%	0.664
	(0.79)	(-1.22)	(-6.90)	(3.67)	(2.22)	(1.68)	(-1.01)	
dio	0.06	0.07	-0.07	0.05	-0.10	-0.02	0.80%	0.162
	(1.52)	(1.37)	(-1.67)	(0.83)	(-1.40)	(-0.31)	(0.59)	
ps_ttm	0.17	-0.06	0.73	-0.01	-0.13	0.27	2.80%	0.796
	(5.04)	(-0.90)	(13.19)	(-0.35)	(-1.85)	(4.43)	(1.47)	
rd_to_revenue	-0.16	0.11	-0.85	-0.26	-0.08	-0.47	6.27%	0.690
	(-2.84)	(1.53)	(-11.74)	(-3.39)	(-0.71)	(-4.02)	(3.17)	
net_margin	0.01	-0.22	0.16	0.14	0.07	0.43	-0.24%	0.648
	(0.19)	(-5.08)	(3.70)	(3.20)	(1.31)	(5.88)	(-0.18)	
BAB composite	0.46	0.42	-0.02	-0.19	-0.70	-0.38	3.97%	0.822
	(10.88)	(5.21)	(-0.40)	(-3.06)	(-8.18)	(-5.45)	(2.00)	
revenue_growth_yoy	-0.03	-0.01	-0.44	0.08	0.17	-0.23	-0.11%	0.657
	(-0.78)	(-0.11)	(-10.46)	(1.84)	(3.65)	(-4.83)	(-0.07)	
growth_roa_5y	0.04	-0.04	-0.28	0.08	-0.04	0.33	-0.88%	0.472
	(1.04)	(-0.87)	(-7.21)	(1.63)	(-0.66)	(5.98)	(-0.58)	
Beta composite	0.45	0.44	-0.02	-0.19	-0.68	-0.40	3.50%	0.821
	(10.82)	(5.20)	(-0.37)	(-2.92)	(-7.88)	(-5.59)	(1.77)	
is_negative_ev	-0.12	-0.25	0.21	-0.23	-0.09	-0.11	6.57%	0.172
	(-1.95)	(-1.92)	(2.13)	(-2.17)	(-0.72)	(-1.07)	(1.84)	
growth_acc_5y	-0.04	-0.02	-0.16	-0.10	0.02	-0.22	1.86%	0.281
	(-1.21)	(-0.58)	(-4.14)	(-2.66)	(0.48)	(-4.70)	(1.77)	
momentum_0_21	-0.16	0.05	0.01	0.21	-0.11	0.12	3.12%	0.158
	(-2.51)	(0.53)	(0.12)	(2.87)	(-1.14)	(1.24)	(1.12)	
debt_issuance_yoy	-0.01	-0.03	0.19	0.10	-0.06	0.13	0.83%	0.332
	(-0.29)	(-0.66)	(4.49)	(2.27)	(-1.53)	(2.95)	(0.87)	
skew_63d	0.02	-0.10	0.06	0.02	0.05	0.11	-0.09%	0.147
	(0.72)	(-1.89)	(1.51)	(0.69)	(0.81)	(2.44)	(-0.06)	
growth_roe_5y	0.06	-0.02	-0.22	0.08	-0.03	0.32	-1.53%	0.430
	(1.51)	(-0.57)	(-6.12)	(1.79)	(-0.65)	(6.36)	(-1.11)	
is_negative_equity	-0.02	0.09	-0.22	-0.04	0.12	-0.04	0.62%	0.135
	(-0.58)	(1.34)	(-3.21)	(-0.58)	(1.62)	(-0.66)	(0.31)	
roe_volatility	0.11	0.10	-0.11	-0.04	-0.11	-0.35	0.66%	0.604
	(2.88)	(2.56)	(-2.29)	(-0.76)	(-1.60)	(-4.92)	(0.54)	
operating_margin	-0.02	-0.29	0.26	0.13	0.11	0.37	-0.60%	0.662
	(-0.56)	(-6.36)	(5.02)	(2.65)	(1.90)	(4.89)	(-0.40)	
debt_to_assets	0.06	-0.11	0.13	0.04	0.16	0.01	-1.27%	0.217
	(1.88)	(-2.00)	(3.58)	(0.80)	(2.52)	(0.10)	(-0.92)	
growth_gpoa_5y	0.02	0.00	-0.54	-0.06	-0.00	0.07	-0.87%	0.712
	(0.75)	(0.02)	(-20.31)	(-2.03)	(-0.04)	(1.92)	(-0.77)	
asset_growth_yoy	-0.01	0.02	-0.49	-0.05	0.15	-0.25	-0.57%	0.730
	(-0.48)	(0.48)	(-12.30)	(-1.72)	(3.06)	(-5.06)	(-0.55)	
growth_gmar_5y	0.04	0.03	0.45	0.12	-0.08	0.04	0.98%	0.607
	(1.58)	(0.72)	(12.53)	(2.98)	(-1.77)	(1.05)	(0.85)	
ev	0.03	0.56	-0.09	-0.01	-0.01	-0.00	1.04%	0.712

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
	(1.69)	(15.71)	(-3.28)	(-0.50)	(-0.27)	(-0.00)	(1.18)	
vol_63d	-0.21	-0.43	0.23	0.25	0.30	0.82	-0.70%	0.776
	(-3.41)	(-4.42)	(3.47)	(3.24)	(2.50)	(6.53)	(-0.38)	
is_negative_invested_capital	-0.02	-0.18	-0.18	0.02	0.11	-0.09	-0.52%	0.107
	(-0.35)	(-1.64)	(-2.12)	(0.26)	(1.20)	(-1.01)	(-0.23)	
dso	0.03	0.02	-0.05	0.02	-0.11	-0.08	0.50%	0.150
	(0.72)	(0.38)	(-1.22)	(0.39)	(-2.69)	(-1.68)	(0.50)	
lottery_max_5_21d	-0.15	-0.41	0.23	0.14	0.28	0.67	-0.61%	0.731
	(-2.52)	(-4.64)	(3.05)	(1.64)	(2.28)	(5.60)	(-0.29)	
book_to_market	-0.01	0.15	-0.76	0.06	0.04	-0.00	-1.71%	0.825
	(-0.45)	(2.87)	(-15.97)	(1.68)	(0.51)	(-0.07)	(-1.05)	
pb	-0.01	0.15	-0.78	0.07	0.02	0.00	-1.78%	0.822
	(-0.44)	(2.81)	(-15.34)	(2.14)	(0.26)	(0.07)	(-1.06)	
gross_margin	0.13	0.13	0.34	0.12	-0.14	0.07	-0.46%	0.475
	(5.34)	(3.35)	(6.14)	(2.36)	(-4.13)	(1.80)	(-0.43)	
ivol_21d	-0.07	-0.26	0.03	0.14	0.20	0.62	-2.01%	0.658
	(-1.48)	(-3.06)	(0.72)	(2.45)	(1.95)	(5.82)	(-1.36)	
log_market_cap	0.06	0.55	0.07	-0.04	0.03	-0.07	0.91%	0.777
	(2.76)	(16.55)	(2.49)	(-1.62)	(0.82)	(-1.97)	(1.05)	
market_cap	0.06	0.55	0.07	-0.04	0.03	-0.07	0.91%	0.777
	(2.76)	(16.55)	(2.49)	(-1.62)	(0.82)	(-1.97)	(1.05)	
altman_z	-0.00	-0.13	0.69	-0.04	0.02	-0.05	2.39%	0.713
	(-0.00)	(-2.42)	(9.73)	(-1.07)	(0.19)	(-0.62)	(1.18)	
earnings_yield	0.11	-0.17	0.57	0.03	-0.01	0.51	-0.88%	0.759
	(3.45)	(-2.51)	(10.92)	(0.68)	(-0.16)	(7.82)	(-0.50)	
earnings_to_price	0.16	-0.11	0.61	-0.10	-0.06	0.42	-0.19%	0.749
	(5.21)	(-1.76)	(7.99)	(-2.05)	(-0.82)	(6.35)	(-0.09)	
pe_ttm	-0.13	-0.07	-0.55	0.23	0.03	-0.10	-1.20%	0.710
	(-3.39)	(-1.11)	(-7.96)	(4.02)	(0.43)	(-1.52)	(-0.65)	
cash_ratio	-0.01	-0.10	0.41	0.12	0.13	0.30	-0.76%	0.704
	(-0.25)	(-2.12)	(9.50)	(3.13)	(1.89)	(4.86)	(-0.59)	
current_ratio	-0.05	-0.20	0.32	0.02	0.13	0.19	0.14%	0.678
	(-1.67)	(-4.84)	(6.45)	(0.35)	(2.16)	(3.59)	(0.11)	
net_operating_assets_proxy	0.01	-0.54	0.58	0.01	-0.01	0.07	0.51%	0.806
	(0.42)	(-12.00)	(12.65)	(0.35)	(-0.17)	(1.15)	(0.37)	

Return Contributions

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
surprisePercentage	0.90%	-0.14%	0.28%	0.48%	-0.42%	0.16%	3.73%	4.99%
dollar_volume_21d	1.66%	0.37%	0.46%	0.46%	-0.30%	-0.11%	3.48%	6.03%
is_positive_revenue	-0.24%	-0.13%	-1.11%	-0.14%	1.02%	1.45%	28.69%	29.55%
amihud_illiq_21d	0.95%	0.57%	0.23%	0.81%	-0.03%	0.39%	3.09%	6.02%
surprise	1.21%	-0.04%	-0.02%	0.64%	-0.50%	0.40%	1.90%	3.59%
resid_ret_252v_1260c_SPY	0.08%	-0.24%	0.15%	-0.55%	-0.12%	0.28%	6.37%	5.98%
fcf_margin	0.32%	0.29%	0.09%	0.22%	0.15%	0.55%	2.77%	4.39%
fcf	0.55%	0.58%	-0.57%	0.41%	-0.05%	0.60%	3.37%	4.89%
cfoa	1.11%	0.36%	0.09%	0.66%	-0.72%	0.89%	2.89%	5.27%
asset_growth_yoy_sign_flip	13.79%	-0.67%	-0.40%	-0.44%	0.92%	-0.34%	-1.42%	11.45%
debt_issuance_yoy_sign_flip	13.83%	-0.65%	-0.46%	-0.46%	0.99%	-0.35%	-1.48%	11.42%
net_income_growth_yoy	0.53%	0.09%	-0.23%	1.51%	0.30%	0.25%	1.28%	3.72%
eps_growth_yoy	0.52%	0.01%	0.16%	1.63%	0.35%	0.04%	1.64%	4.35%
equity_issuance_yoy_sign_flip	13.74%	-0.68%	-0.37%	-0.46%	0.93%	-0.37%	-1.57%	11.23%
momentum_21_252	-0.71%	-0.02%	0.07%	5.61%	0.99%	0.00%	2.13%	8.08%
fcf_yield	1.47%	0.07%	-0.97%	0.15%	-0.35%	1.00%	3.79%	5.16%
momentum_21_126	0.35%	-0.22%	-0.33%	4.04%	0.62%	0.11%	1.99%	6.55%
accruals_total	-0.27%	-0.27%	0.89%	-0.77%	-0.04%	-0.27%	4.14%	3.41%
debt_to_equity	1.34%	0.11%	-0.92%	-0.16%	0.17%	0.27%	2.98%	3.79%
momentum_126_252	-0.95%	0.16%	0.37%	3.79%	0.84%	-0.05%	1.10%	5.26%
resid_momentum_21_252	-2.16%	0.20%	0.29%	4.74%	1.65%	-0.00%	1.22%	5.93%
gpoa	1.07%	0.06%	1.39%	-0.09%	-0.27%	0.71%	0.66%	3.52%
corr_1260d_FRENCH_CRSP	5.26%	-0.13%	-0.25%	-0.19%	-2.41%	0.62%	1.12%	4.02%
corr_1260d_SPY	4.83%	-0.02%	-0.37%	-0.06%	-2.25%	0.65%	1.03%	3.81%
corr_1260d_VTI	5.10%	-0.11%	-0.35%	-0.10%	-2.34%	0.64%	1.09%	3.92%
high_52w_252d	-2.21%	0.32%	-0.20%	4.17%	0.85%	1.05%	1.93%	5.92%
info_discreteness_21_252	-0.44%	0.20%	0.32%	1.30%	0.69%	-0.08%	0.76%	2.76%
roe	1.99%	0.29%	-0.15%	0.64%	-0.16%	1.24%	-0.37%	3.47%
accruals_noa_proxy	0.07%	-0.03%	-0.95%	0.37%	-0.91%	0.24%	3.46%	2.25%
sga_to_revenue	0.32%	0.41%	-1.03%	0.83%	-0.04%	-0.00%	1.97%	2.47%
equity_issuance_yoy	2.03%	0.06%	-0.93%	0.01%	-0.81%	1.01%	1.68%	3.05%
revenue_growth_yoy_sign_flip	-1.73%	-0.37%	0.17%	-1.85%	5.49%	-0.65%	10.99%	12.04%
is_positive_net_income	1.07%	0.38%	-1.03%	1.60%	0.44%	1.51%	0.56%	4.54%
eps_growth_yoy_sign_flip	0.13%	0.10%	-0.22%	2.37%	1.14%	-0.31%	2.55%	5.77%
dpo	0.57%	-0.18%	0.45%	-0.08%	-0.19%	0.22%	0.78%	1.57%
net_income_growth_yoy_sign_flip	0.12%	-0.17%	-0.10%	2.24%	0.52%	0.24%	0.58%	3.42%
cash_earnings_ttm	-0.52%	0.63%	-0.92%	-0.16%	-0.53%	-0.07%	3.82%	2.24%
growth_cfoa_5y	-0.08%	0.11%	1.08%	-0.19%	0.04%	0.34%	0.97%	2.29%
roa	1.56%	0.34%	0.01%	0.92%	-0.43%	1.32%	-0.66%	3.06%
accruals_wc_proxy	-0.72%	0.06%	-0.16%	-0.28%	0.06%	0.01%	2.30%	1.26%
accruals_percent	-0.70%	0.17%	-0.30%	0.03%	-0.23%	-0.35%	2.81%	1.43%
is_positive_income_tax	0.03%	0.10%	-0.31%	0.74%	0.45%	0.79%	0.39%	2.19%
roc	1.41%	0.53%	0.15%	0.41%	-0.47%	1.24%	-0.54%	2.72%
ret_volume_corr_63d	0.99%	0.21%	-0.23%	-0.15%	0.22%	0.20%	0.39%	1.62%
evs_ttm	2.62%	-0.04%	-1.85%	0.28%	-1.07%	0.73%	2.26%	2.93%
value_proxy_1260d	-0.20%	-0.08%	1.76%	1.83%	0.60%	0.45%	-0.90%	3.46%
accruals	-0.56%	-0.04%	0.48%	-0.38%	-0.35%	-0.47%	2.91%	1.58%
working_capital_proxy	1.59%	-0.29%	0.21%	0.54%	-0.37%	0.01%	-0.37%	1.31%
momentum_21_756	0.52%	0.19%	1.78%	1.49%	0.98%	0.24%	-2.03%	3.17%
dio	0.82%	-0.13%	0.21%	0.29%	-0.58%	-0.04%	0.80%	1.36%
ps_ttm	2.42%	0.11%	-2.18%	-0.09%	-0.75%	0.53%	2.80%	2.83%
rd_to_revenue	-2.25%	-0.21%	2.55%	-1.67%	-0.47%	-0.90%	6.27%	3.33%
net_margin	0.10%	0.41%	-0.49%	0.89%	0.40%	0.83%	-0.24%	1.90%
BAB composite	6.51%	-0.79%	0.06%	-1.23%	-4.02%	-0.73%	3.97%	3.76%
revenue_growth_yoy	-0.40%	0.01%	1.32%	0.54%	0.98%	-0.44%	-0.11%	1.89%
growth_roa_5y	0.59%	0.07%	0.83%	0.50%	-0.21%	0.64%	-0.88%	1.53%
Beta composite	6.43%	-0.81%	0.06%	-1.19%	-3.89%	-0.77%	3.50%	3.32%
is_negative_ev	-1.66%	0.46%	-0.63%	-1.49%	-0.51%	-0.22%	6.57%	2.52%
growth_acc_5y	-0.62%	0.04%	0.48%	-0.61%	0.13%	-0.42%	1.86%	0.85%
momentum_0_21	-2.26%	-0.10%	-0.03%	1.34%	-0.65%	0.23%	3.12%	1.66%
debt_issuance_yoy	-0.10%	0.05%	-0.57%	0.61%	-0.34%	0.25%	0.83%	0.73%
skew_63d	0.27%	0.19%	-0.19%	0.14%	0.26%	0.20%	-0.09%	0.80%
growth_roe_5y	0.82%	0.04%	0.66%	0.49%	-0.18%	0.62%	-1.53%	0.92%
is_negative_equity	-0.34%	-0.16%	0.67%	-0.27%	0.70%	-0.09%	1.15%	1.15%
roe_volatility	1.62%	-0.18%	0.34%	-0.23%	-0.61%	-0.67%	0.66%	0.94%
operating_margin	-0.31%	0.54%	-0.77%	0.81%	0.62%	0.71%	-0.60%	1.00%
debt_to_assets	0.90%	0.20%	-0.40%	0.22%	0.90%	0.01%	-1.27%	0.55%
growth_gpoa_5y	0.29%	-0.00%	1.64%	-0.40%	-0.01%	0.13%	-0.87%	0.78%
asset_growth_yoy	-0.20%	-0.04%	1.47%	-0.32%	0.83%	-0.48%	-0.57%	0.70%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_gmar_5y	0.60%	-0.05%	-1.37%	0.77%	-0.44%	0.08%	0.98%	0.56%
ev	0.42%	-1.04%	0.26%	-0.08%	-0.04%	-0.00%	1.04%	0.55%
vol_63d	-2.99%	0.80%	-0.69%	1.59%	1.74%	1.58%	-0.70%	1.33%
is_negative_invested_capital	-0.22%	0.33%	0.55%	0.13%	0.64%	-0.17%	-0.52%	0.76%
dso	0.38%	-0.03%	0.15%	0.12%	-0.60%	-0.15%	0.50%	0.36%
lottery_max_5_21d	-2.16%	0.77%	-0.69%	0.86%	1.58%	1.30%	-0.61%	1.04%
book_to_market	-0.20%	-0.28%	2.29%	0.38%	0.21%	-0.01%	-1.71%	0.68%
pb	-0.21%	-0.28%	2.36%	0.46%	0.12%	0.01%	-1.78%	0.68%
gross_margin	1.90%	-0.25%	-1.02%	0.79%	-0.79%	0.14%	-0.46%	0.31%
ivol_21d	-1.04%	0.48%	-0.11%	0.92%	1.14%	1.20%	-2.01%	0.60%
log_market_cap	0.83%	-1.02%	-0.20%	-0.25%	0.19%	-0.14%	0.91%	0.32%
market_cap	0.83%	-1.02%	-0.20%	-0.25%	0.19%	-0.14%	0.91%	0.32%
altman_z	-0.00%	0.25%	-2.06%	-0.27%	0.12%	-0.09%	2.39%	0.34%
earnings_yield	1.52%	0.31%	-1.71%	0.18%	-0.05%	0.98%	-0.88%	0.35%
earnings_to_price	2.30%	0.20%	-1.85%	-0.66%	-0.34%	0.80%	-0.19%	0.27%
pe_ttm	-1.82%	0.13%	1.66%	1.45%	0.20%	-0.19%	-1.20%	0.22%
cash_ratio	-0.11%	0.18%	-1.25%	0.73%	0.72%	0.57%	-0.76%	0.09%
current_ratio	-0.68%	0.38%	-0.97%	0.10%	0.75%	0.36%	0.14%	0.07%
net_operating_assets_proxy	0.17%	1.01%	-1.75%	0.07%	-0.06%	0.14%	0.51%	0.09%

1. surprisePercentage

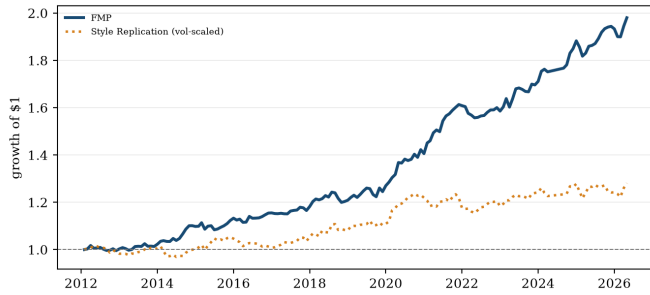


Figure: Common-window cumulative growth of \$1.

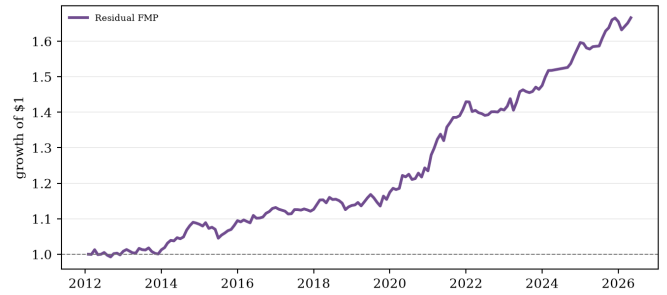


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
surprisePercentage	4.99%	3.85%	1.30	4.96	-3.48%	0.258	4.05%	3.85%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
surprisePercentage	0.06 (3.28)	0.07 (2.43)	-0.09 (-2.82)	0.08 (2.55)	-0.07 (-1.85)	0.08 (1.92)	3.73% (4.49)	0.266

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
surprisePercentage	0.90%	-0.14%	0.28%	0.48%	-0.42%	0.16%	3.73%	4.99%

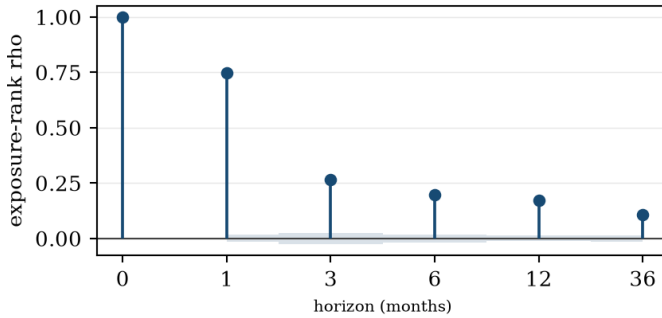


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

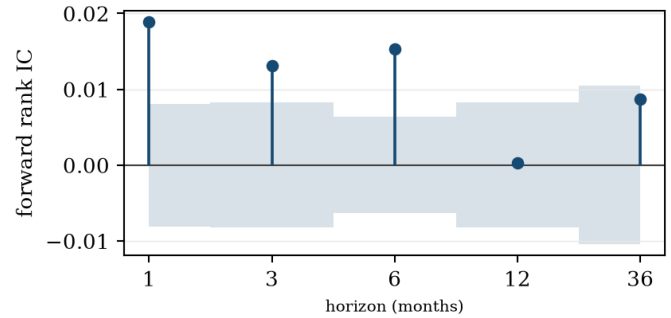


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	surprisePercentage
Component signs	surprisePercentage(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	242 / 167
Available window	2006-06-01 to 2026-07-01
Dropped: low breadth / common	43 / 75
Stocks, min / median / max	684 / 1018 / 1607
Available-history Sharpe	0.71

2. dollar_volume_21d

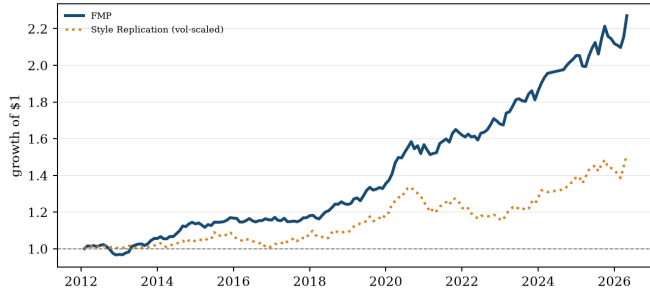


Figure: Common-window cumulative growth of \$1.

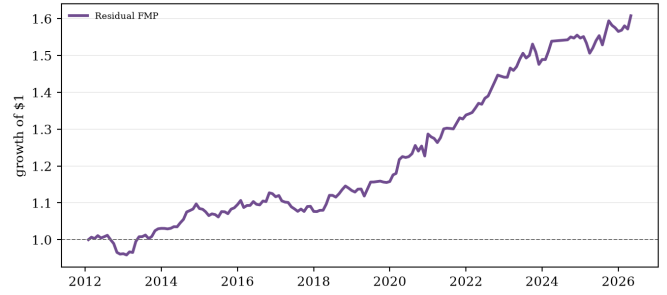


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dollar_volume_21d	6.03%	5.00%	1.21	4.47	-5.54%	0.451	3.94%	5.00%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dollar_volume_21d	0.12 (4.10)	-0.20 (-4.23)	-0.15 (-6.86)	0.07 (2.87)	-0.05 (-0.87)	-0.06 (-1.00)	3.48% (3.33)	0.463

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dollar_volume_21d	1.66%	0.37%	0.46%	0.46%	-0.30%	-0.11%	3.48%	6.03%

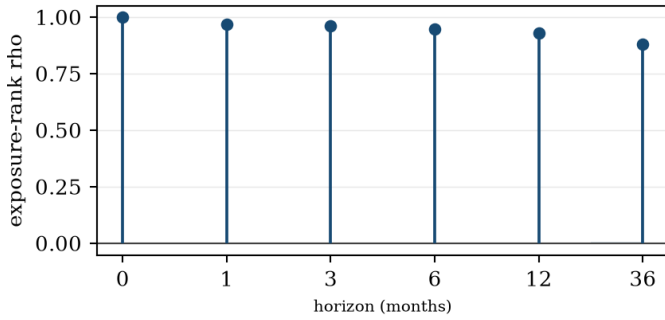


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

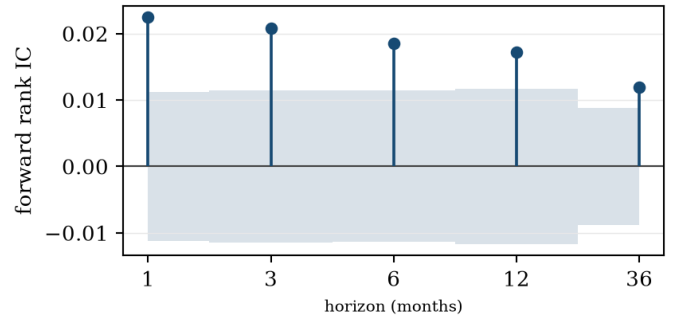


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	dollar_volume_21d
Component signs	dollar_volume_21d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	749 / 1061 / 1676
Available-history Sharpe	0.59

3. is_positive_revenue

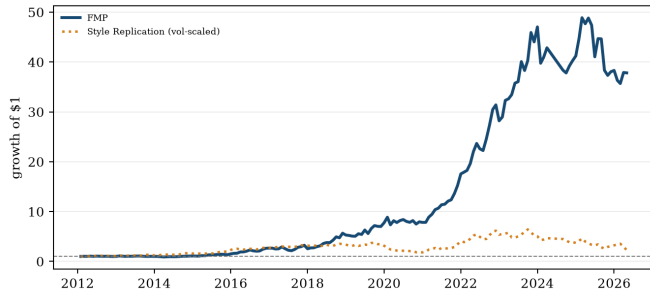


Figure: Common-window cumulative growth of \$1.

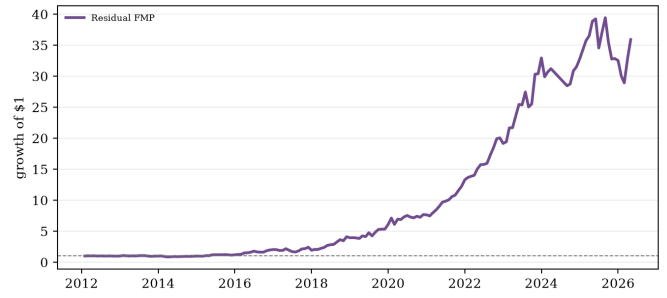


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_positive_revenue	29.55%	25.32%	1.17	4.59	-27.00%	0.136	26.21%	25.32%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_positive_revenue	-0.02 (-0.09)	0.07 (0.30)	0.37 (2.26)	-0.02 (-0.11)	0.18 (0.70)	0.75 (2.59)	28.69% (4.03)	0.144

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_positive_revenue	-0.24%	-0.13%	-1.11%	-0.14%	1.02%	1.45%	28.69%	29.55%

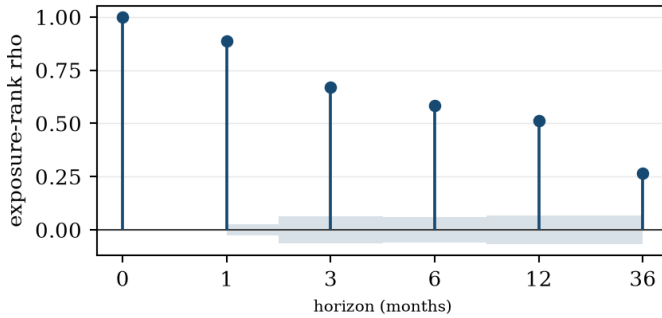


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

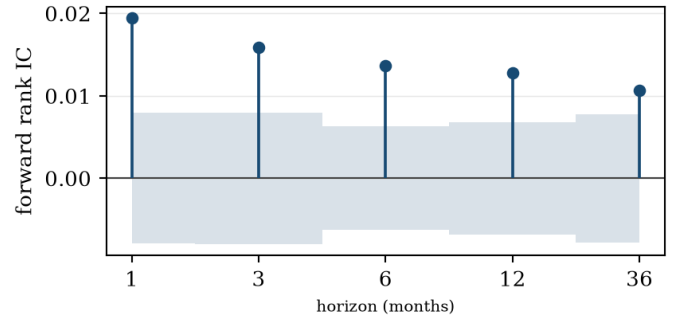


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	is_positive_revenue
Component signs	is_positive_revenue(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	747 / 1057 / 1677
Available-history Sharpe	1.01

4. amihud_illiq_21d

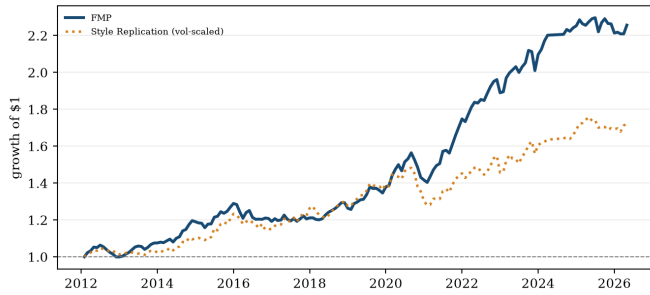


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
amihud_illiq_21d	6.02%	5.66%	1.06	3.73	-10.27%	0.581	2.92%	5.66%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
amihud_illiq_21d	0.07 (2.96)	-0.31 (-5.75)	-0.08 (-2.00)	0.13 (3.41)	-0.00 (-0.15)	0.20 (4.76)	3.09% (2.81)	0.581

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
amihud_illiq_21d	0.95%	0.57%	0.23%	0.81%	-0.03%	0.39%	3.09%	6.02%

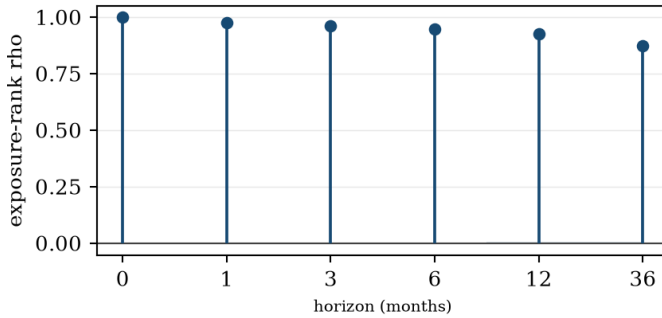


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

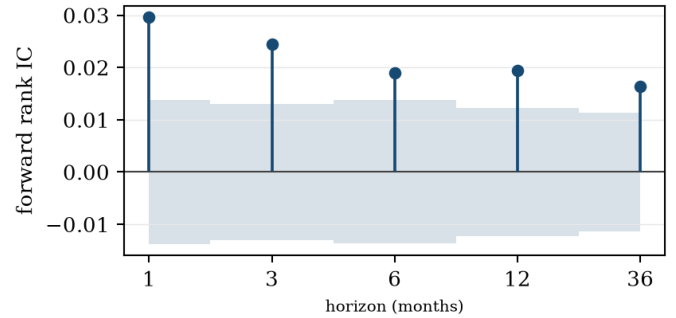


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	amihud_illiq_21d
Component signs	amihud_illiq_21d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	746 / 1055 / 1673
Available-history Sharpe	0.52

5. surprise

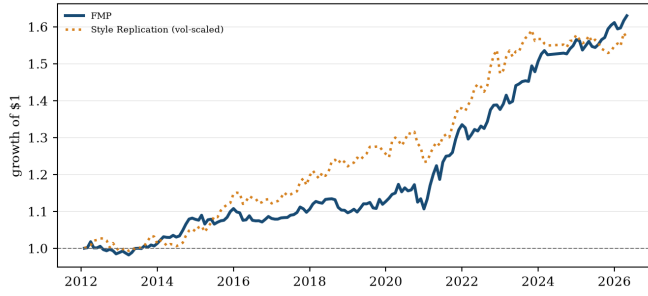


Figure: Common-window cumulative growth of \$1.

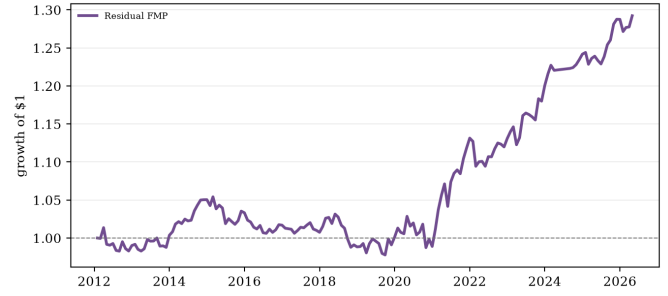


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
surprise	3.59%	3.94%	0.91	3.71	-5.67%	0.217	2.04%	3.94%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
surprise	0.08 (3.32)	0.02 (0.70)	0.01 (0.19)	0.10 (3.24)	-0.09 (-2.15)	0.21 (3.70)	1.90% (2.30)	0.218

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
surprise	1.21%	-0.04%	-0.02%	0.64%	-0.50%	0.40%	1.90%	3.59%

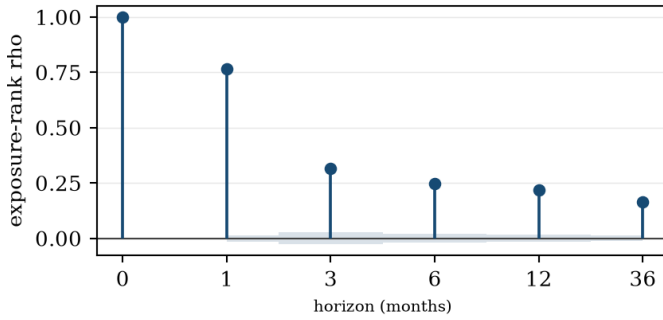


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

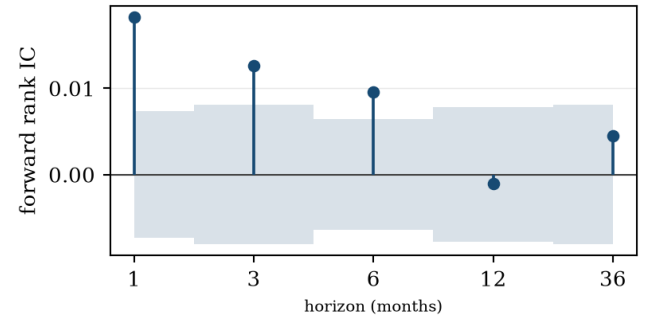


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	surprise
Component signs	surprise(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	242 / 167
Available window	2006-06-01 to 2026-07-01
Dropped: low breadth / common	43 / 75
Stocks, min / median / max	696 / 1041 / 1620
Available-history Sharpe	0.39

6. resid_ret_252v_1260c_SPY

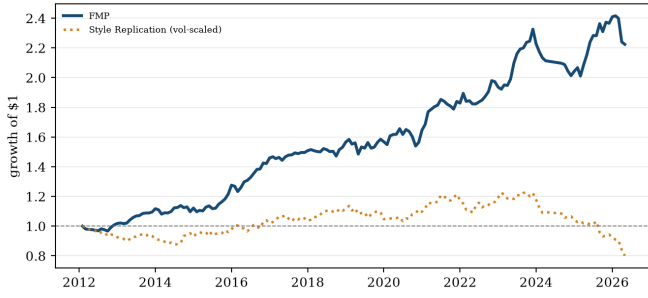


Figure: Common-window cumulative growth of \$1.

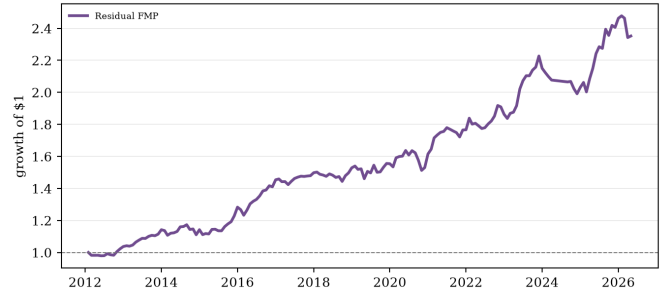


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
resid_ret_252v_1260c_SPY	5.98%	6.72%	0.89	3.07	-13.56%	0.056	6.30%	6.72%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
resid_ret_252v_1260c_SPY	0.01	0.13	-0.05	-0.09	-0.02	0.15	6.37%	0.056
	(0.12)	(1.58)	(-0.93)	(-1.20)	(-0.20)	(1.60)	(3.26)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
resid_ret_252v_1260c_SPY	0.08%	-0.24%	0.15%	-0.55%	-0.12%	0.28%	6.37%	5.98%

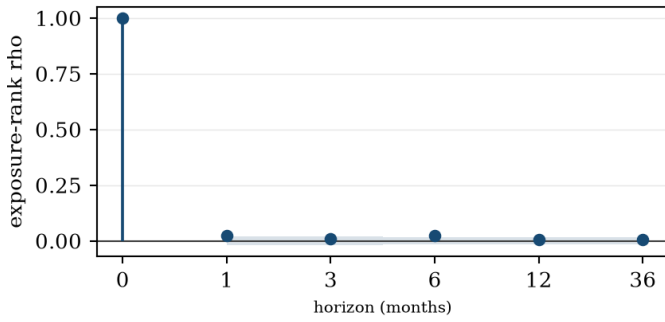


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

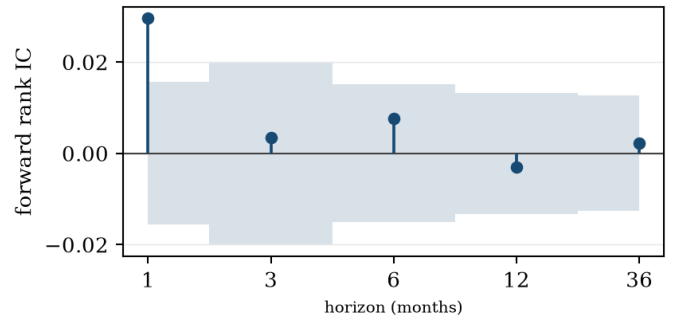


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	resid_ret_252v_1260c_SPY
Component signs	resid_ret_252v_1260c_SPY(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	707 / 982 / 1565
Available-history Sharpe	0.82

7. fcf_margin

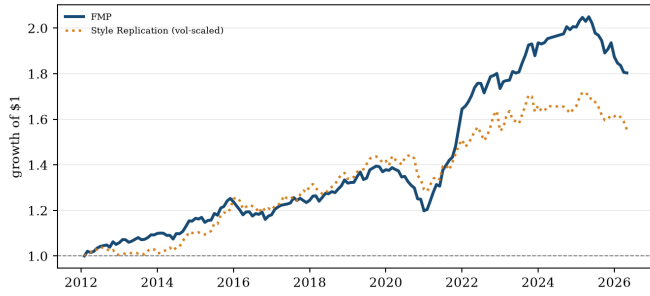


Figure: Common-window cumulative growth of \$1.

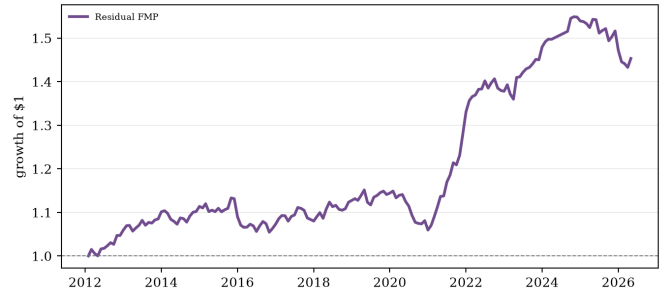


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_margin	4.39%	5.44%	0.81	2.34	-14.08%	0.414	2.25%	5.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_margin	0.02 (0.83)	-0.16 (-3.44)	-0.03 (-0.46)	0.04 (0.78)	0.03 (0.54)	0.28 (5.22)	2.77% (1.91)	0.421

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_margin	0.32%	0.29%	0.09%	0.22%	0.15%	0.55%	2.77%	4.39%

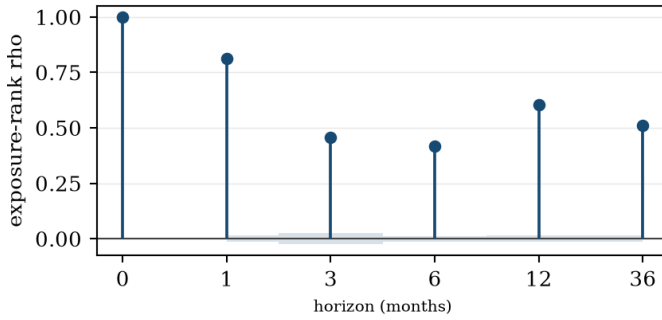


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

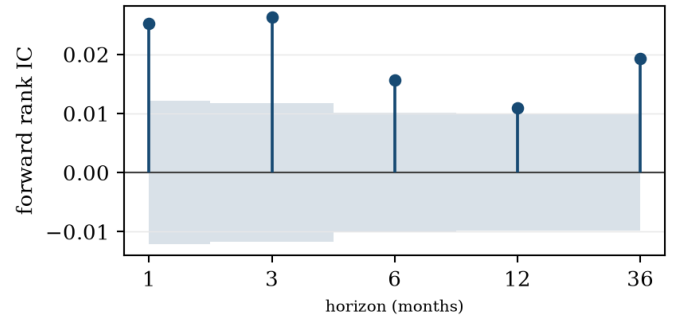


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	fcf_margin
Component signs	fcf_margin(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	731 / 1045 / 1625
Available-history Sharpe	0.88

8. fcf

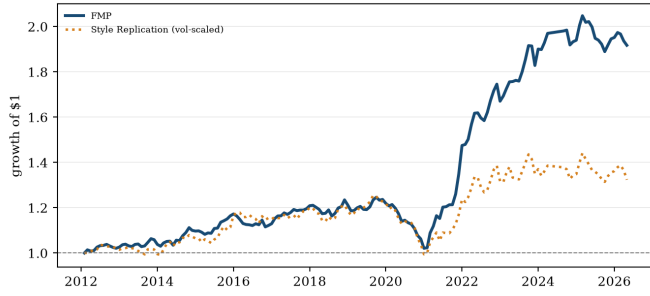


Figure: Common-window cumulative growth of \$1.

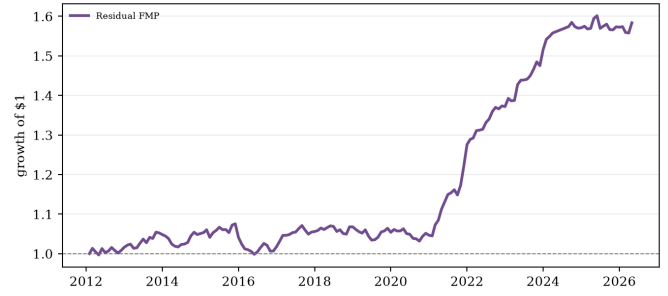


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf	4.89%	6.50%	0.75	2.25	-18.07%	0.684	3.05%	6.50%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf	0.04 (1.82)	-0.31 (-7.22)	0.19 (3.58)	0.06 (2.26)	-0.01 (-0.27)	0.31 (6.38)	3.37% (2.56)	0.686

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf	0.55%	0.58%	-0.57%	0.41%	-0.05%	0.60%	3.37%	4.89%

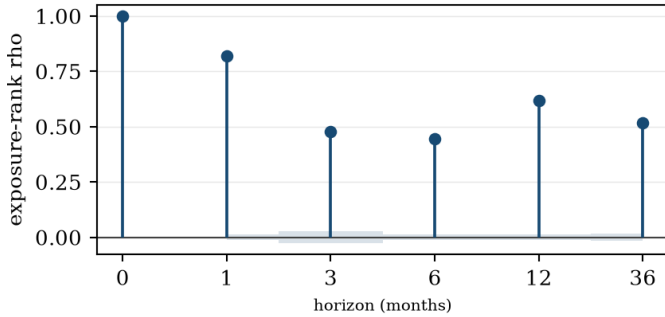


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

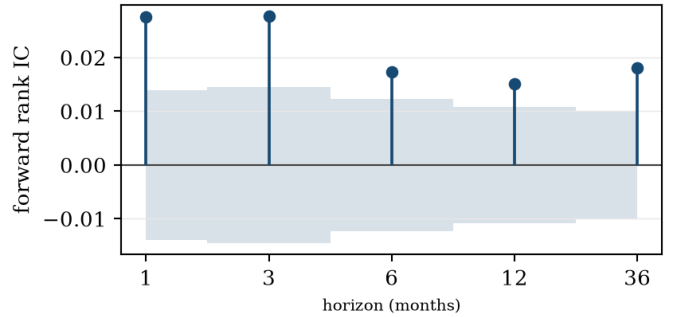


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	fcf
Component signs	fcf(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	733 / 1056 / 1652
Available-history Sharpe	0.78

9. cfoa

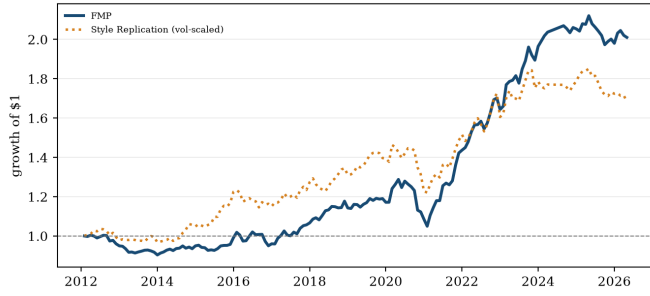


Figure: Common-window cumulative growth of \$1.

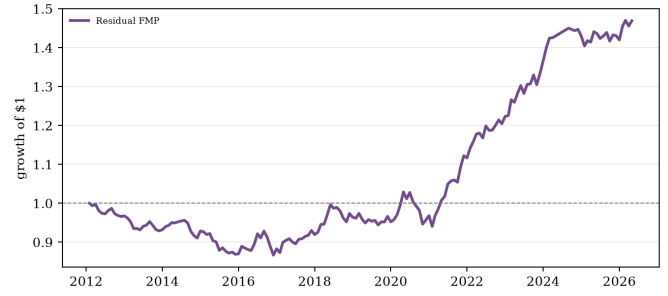


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
cfoa	5.27%	7.03%	0.75	2.42	-18.48%	0.494	2.47%	7.03%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
cfoa	0.08 (2.44)	-0.19 (-3.27)	-0.03 (-0.87)	0.10 (2.71)	-0.13 (-2.18)	0.46 (7.08)	2.89% (1.92)	0.496

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
cfoa	1.11%	0.36%	0.09%	0.66%	-0.72%	0.89%	2.89%	5.27%

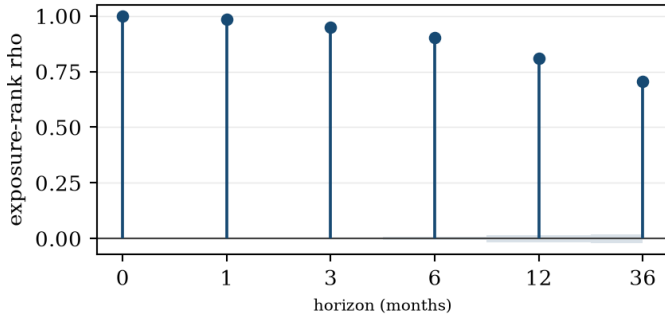


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

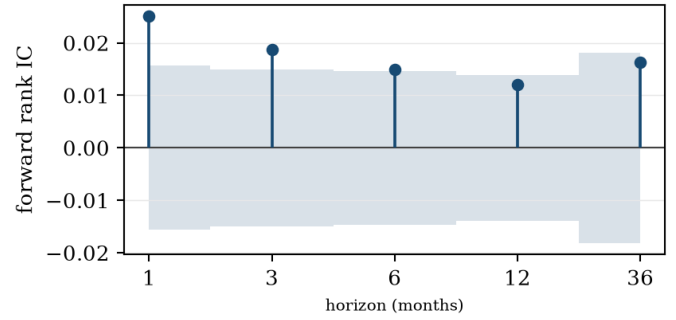


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	cfoa
Component signs	cfoa(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	22 / 66
Stocks, min / median / max	712 / 1047 / 1611
Available-history Sharpe	0.76

10. asset_growth_yoy_sign_flip

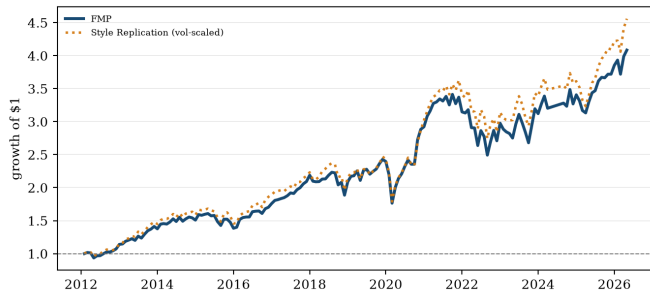


Figure: Common-window cumulative growth of \$1.

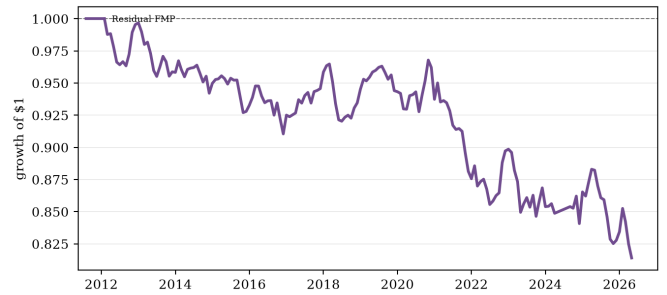


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
asset_growth_yoy_sign_flip	11.45%	16.05%	0.71	3.22	-27.35%	0.948	-0.48%	16.05%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
asset_growth_yoy_sign_flip	0.96	0.36	0.13	-0.07	0.16	-0.18	-1.42%	0.953
	(45.85)	(8.92)	(5.39)	(-2.36)	(2.94)	(-4.12)	(-1.44)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
asset_growth_yoy_sign_flip	13.79%	-0.67%	-0.40%	-0.44%	0.92%	-0.34%	-1.42%	11.45%

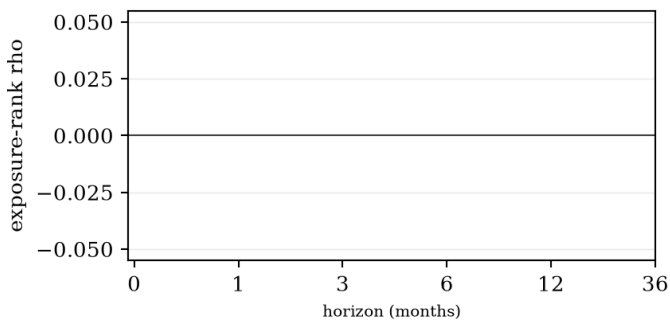


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

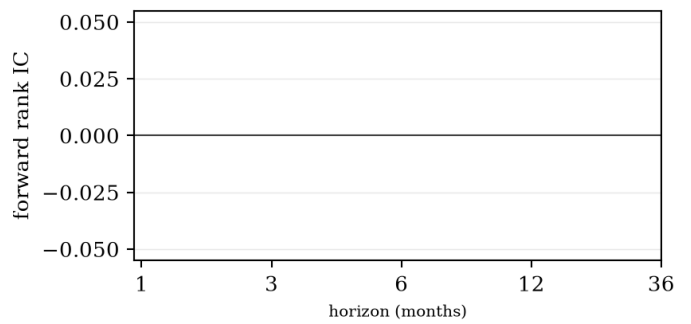


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	asset_growth_yoy_sign_flip
Component signs	asset_growth_yoy_sign_flip(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	237 / 167
Available window	2006-11-01 to 2026-07-01
Dropped: low breadth / common	54 / 70
Stocks, min / median / max	741 / 1047 / 1664
Available-history Sharpe	0.49

11. debt_issuance_yoy_sign_flip

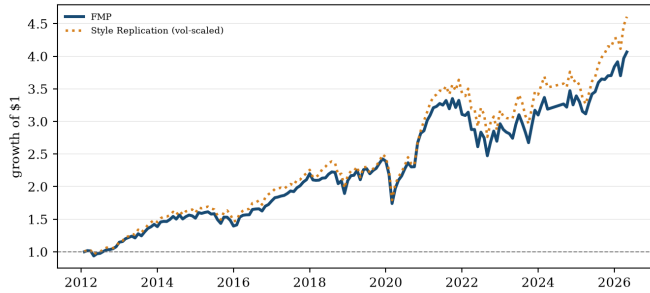


Figure: Common-window cumulative growth of \$1.

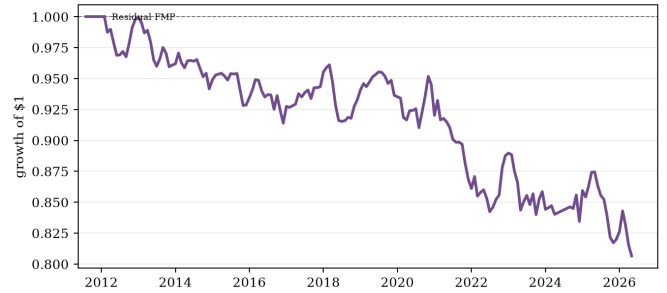


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_issuance_yoy_sign_flip	11.42%	16.06%	0.71	3.22	-28.26%	0.950	-0.59%	16.06%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_issuance_yoy_sign_flip	0.97	0.35	0.15	-0.07	0.17	-0.18	-1.48%	0.954
	(46.16)	(8.81)	(6.46)	(-2.47)	(3.09)	(-4.25)	(-1.49)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_issuance_yoy_sign_flip	13.83%	-0.65%	-0.46%	-0.46%	0.99%	-0.35%	-1.48%	11.42%

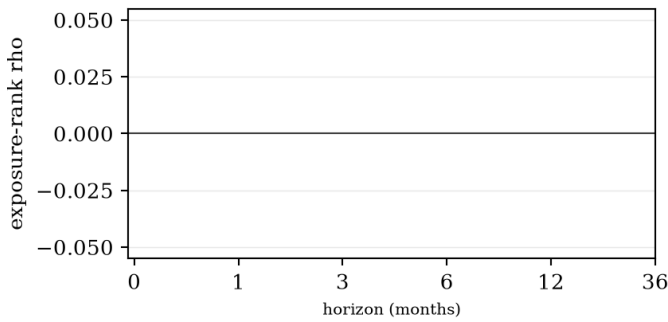


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

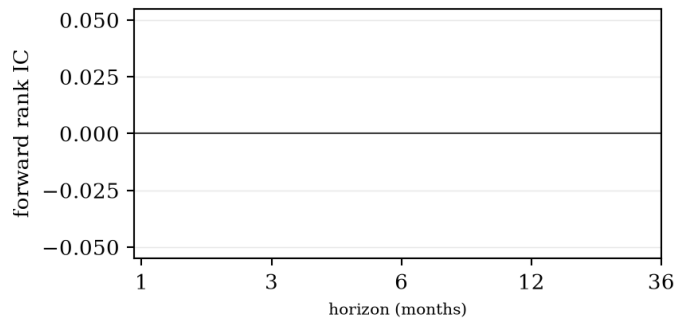


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	debt_issuance_yoy_sign_flip
Component signs	debt_issuance_yoy_sign_flip(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	236 / 167
Available window	2006-12-01 to 2026-07-01
Dropped: low breadth / common	54 / 69
Stocks, min / median / max	683 / 953 / 1616
Available-history Sharpe	0.60

12. net_income_growth_yoy

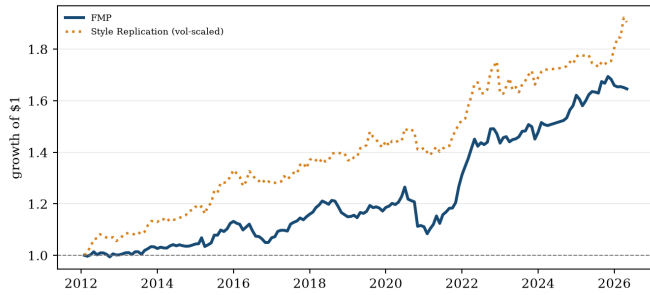


Figure: Common-window cumulative growth of \$1.

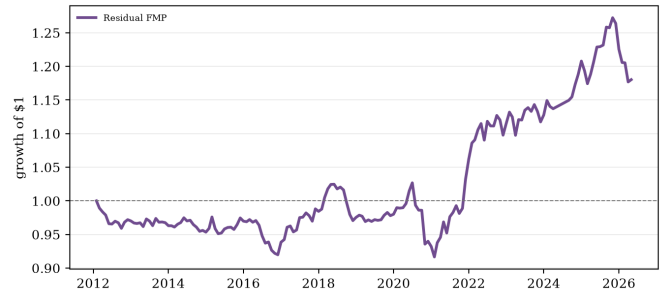


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_growth_yoy	3.72%	5.25%	0.71	2.27	-14.30%	0.327	0.99%	5.25%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_growth_yoy	0.04	-0.05	0.08	0.24	0.05	0.13	1.28%	0.329
	(1.20)	(-1.15)	(1.24)	(3.97)	(1.00)	(2.39)	(0.86)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_growth_yoy	0.53%	0.09%	-0.23%	1.51%	0.30%	0.25%	1.28%	3.72%

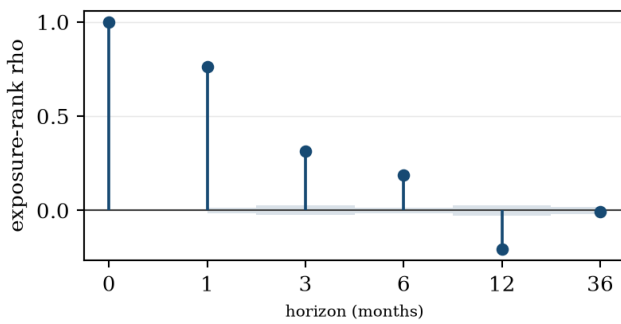


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

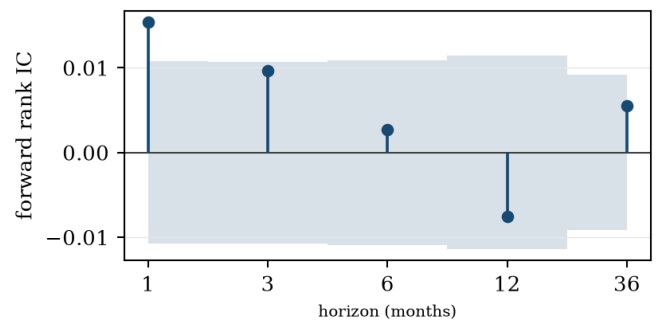


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	net_income_growth_yoy
Component signs	net_income_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	237 / 167
Available window	2006-11-01 to 2026-07-01
Dropped: low breadth / common	54 / 70
Stocks, min / median / max	743 / 1054 / 1665
Available-history Sharpe	0.33

13. eps_growth_yoy

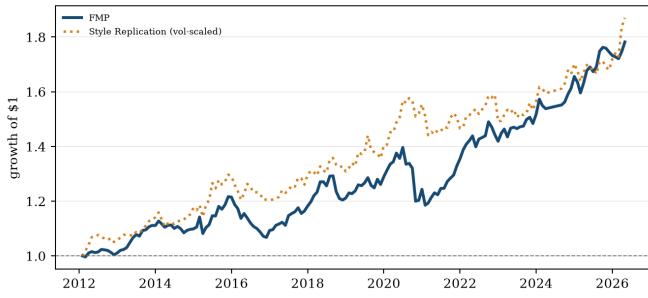


Figure: Common-window cumulative growth of \$1.

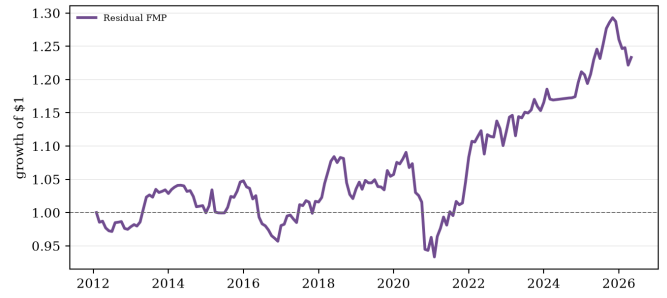


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
eps_growth_yoy	4.35%	6.23%	0.70	2.55	-15.12%	0.334	1.64%	6.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
eps_growth_yoy	0.04 (0.97)	-0.01 (-0.13)	-0.05 (-0.82)	0.26 (4.36)	0.06 (1.06)	0.02 (0.41)	1.64% (0.90)	0.334

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
eps_growth_yoy	0.52%	0.01%	0.16%	1.63%	0.35%	0.04%	1.64%	4.35%

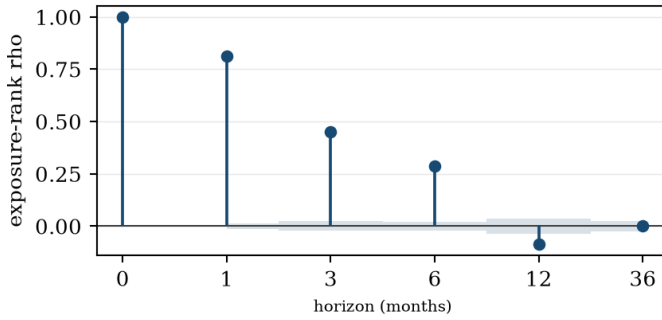


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

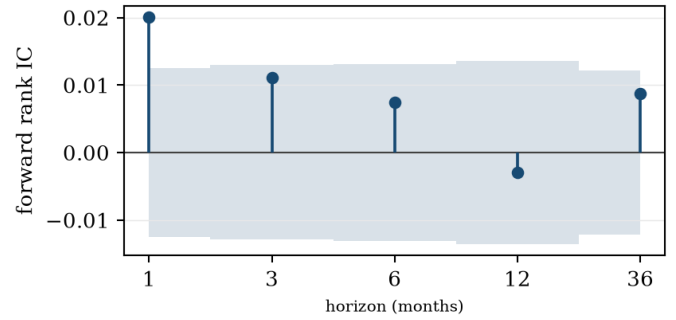


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	eps_growth_yoy
Component signs	eps_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	230 / 167
Available window	2007-06-01 to 2026-07-01
Dropped: low breadth / common	49 / 63
Stocks, min / median / max	708 / 1026 / 1592
Available-history Sharpe	0.28

14. equity_issuance_yoy_sign_flip

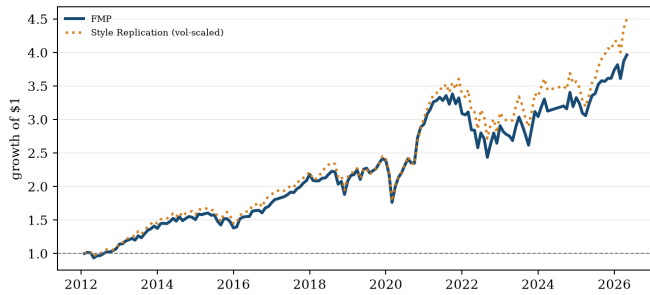


Figure: Common-window cumulative growth of \$1.

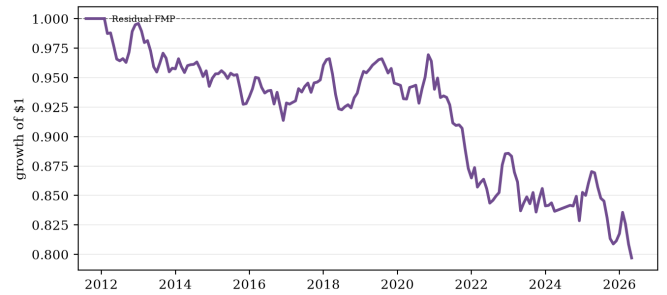


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
equity_issuance_yoy_ sign_flip	11.23%	16.09%	0.70	3.15	-28.01%	0.947	-0.60%	16.09%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
equity_issuance_yoy_ sign_flip	0.96 (45.73)	0.36 (9.02)	0.12 (4.57)	-0.07 (-2.39)	0.16 (3.09)	-0.19 (-4.24)	-1.57% (-1.57)	0.952

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
equity_issuance_yoy_ sign_flip	13.74%	-0.68%	-0.37%	-0.46%	0.93%	-0.37%	-1.57%	11.23%

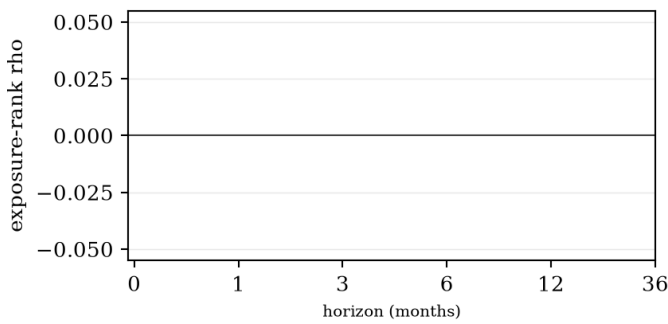


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

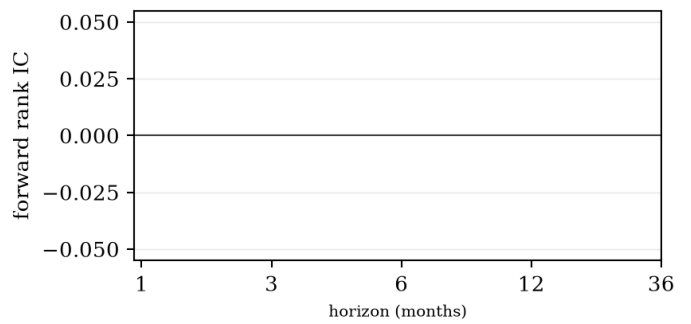


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	equity_issuance_yoy_sign_flip
Component signs	equity_issuance_yoy_sign_flip(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	237 / 167
Available window	2006-11-01 to 2026-07-01
Dropped: low breadth / common	54 / 70
Stocks, min / median / max	745 / 1057 / 1665
Available-history Sharpe	0.61

15. momentum_21_252

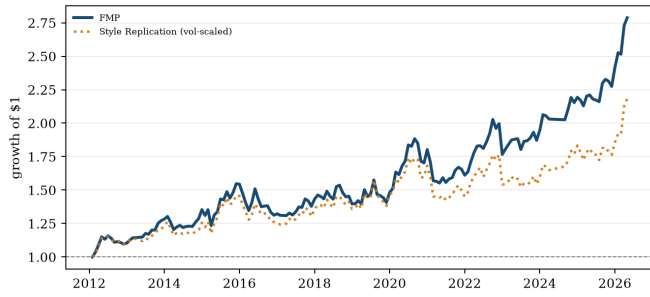


Figure: Common-window cumulative growth of \$1.

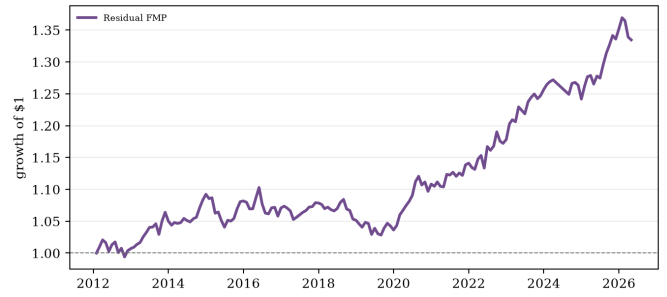


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252	8.08%	11.79%	0.69	2.82	-17.59%	0.920	2.03%	11.79%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252	-0.05 (-2.02)	0.01 (0.37)	-0.02 (-0.79)	0.88 (24.79)	0.17 (4.87)	0.00 (0.02)	2.13% (2.28)	0.920

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252	-0.71%	-0.02%	0.07%	5.61%	0.99%	0.00%	2.13%	8.08%

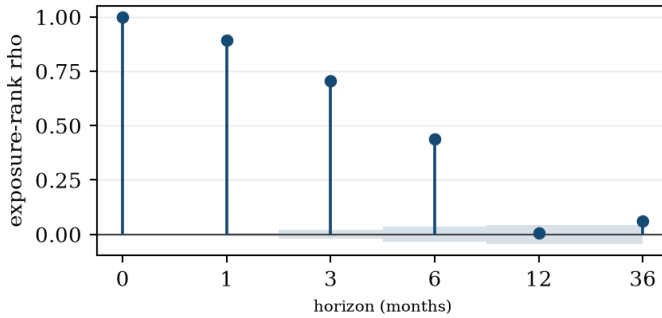


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

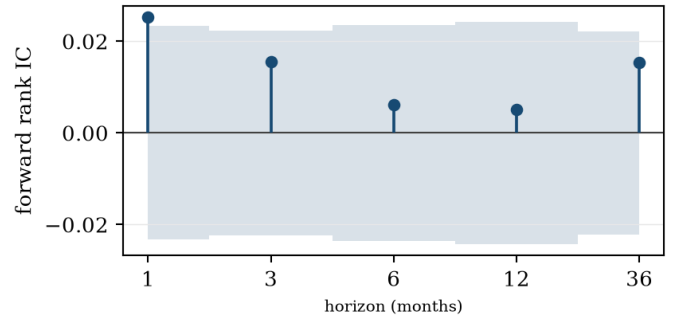


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	momentum_21_252
Component signs	momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	738 / 1032 / 1631
Available-history Sharpe	0.25

16. fcf_yield

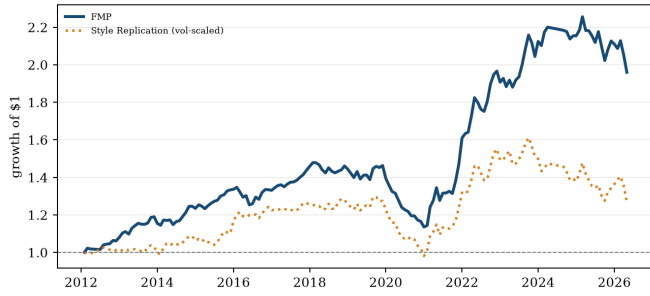


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_yield	5.16%	8.02%	0.64	2.02	-23.23%	0.625	3.56%	8.02%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_yield	0.10 (4.11)	-0.04 (-0.58)	0.32 (4.71)	0.02 (0.41)	-0.06 (-1.27)	0.52 (9.13)	3.79% (2.37)	0.625

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_yield	1.47%	0.07%	-0.97%	0.15%	-0.35%	1.00%	3.79%	5.16%

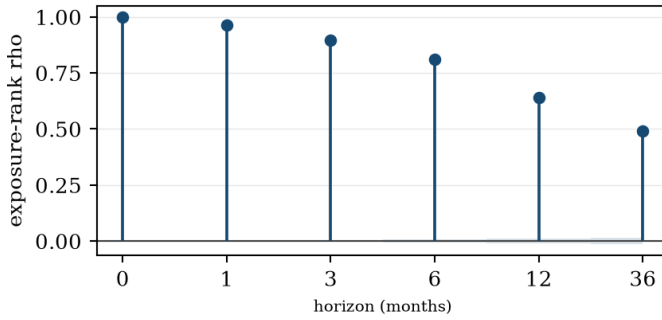


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

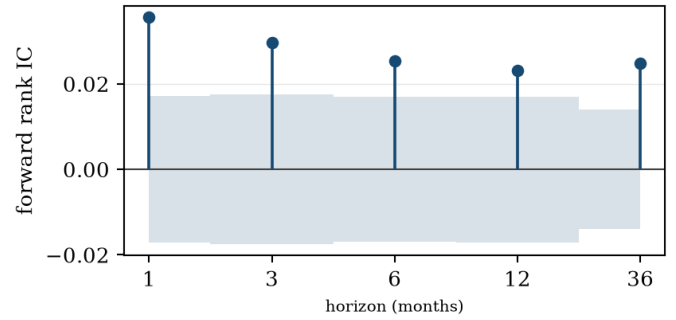


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	fcf_yield
Component signs	fcf_yield(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	15 / 66
Stocks, min / median / max	662 / 959 / 1573
Available-history Sharpe	0.71

17. momentum_21_126

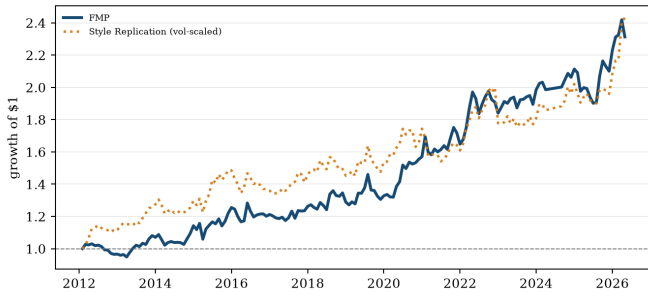


Figure: Common-window cumulative growth of \$1.

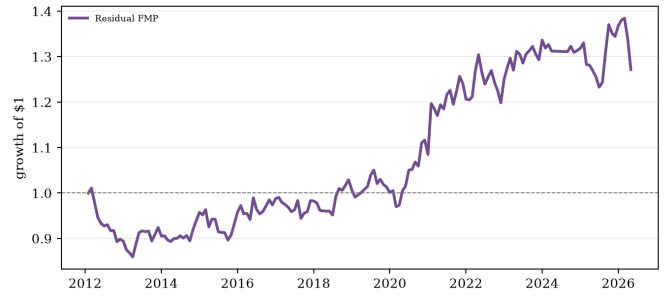


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_126	6.55%	10.19%	0.64	3.07	-10.59%	0.490	1.71%	10.19%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_126	0.02 (0.52)	0.12 (1.39)	0.11 (1.26)	0.64 (6.60)	0.11 (0.94)	0.06 (0.52)	1.99% (0.95)	0.491

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_126	0.35%	-0.22%	-0.33%	4.04%	0.62%	0.11%	1.99%	6.55%

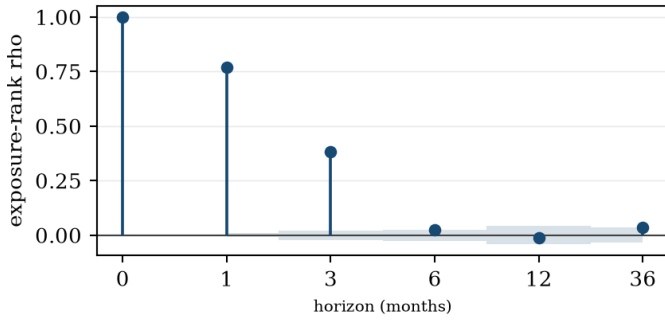


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

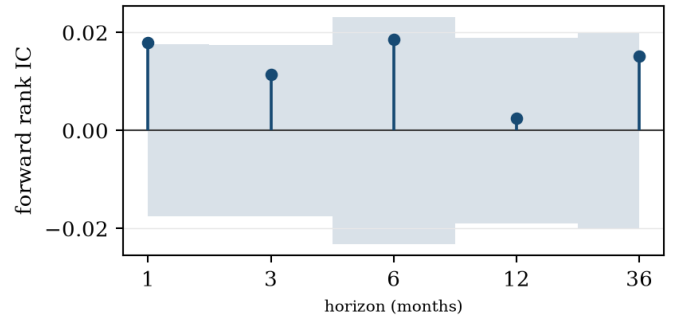


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	momentum_21_126
Component signs	momentum_21_126(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	744 / 1051 / 1657
Available-history Sharpe	0.16

18. accruals_total

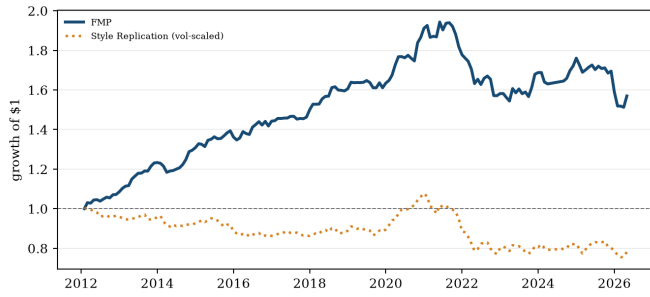


Figure: Common-window cumulative growth of \$1.

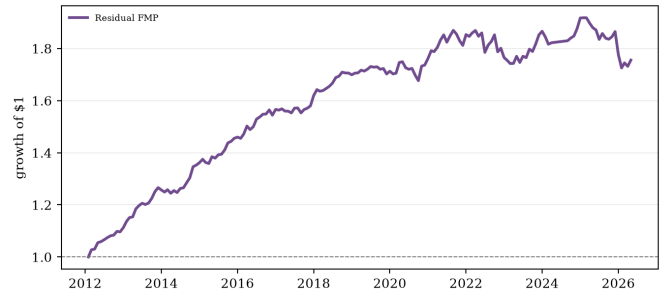


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_total	3.41%	5.86%	0.58	1.90	-22.16%	0.466	4.50%	5.86%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_total	-0.02 (-0.60)	0.14 (3.41)	-0.30 (-5.29)	-0.12 (-1.87)	-0.01 (-0.21)	-0.14 (-2.90)	4.14% (3.51)	0.469

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_total	-0.27%	-0.27%	0.89%	-0.77%	-0.04%	-0.27%	4.14%	3.41%

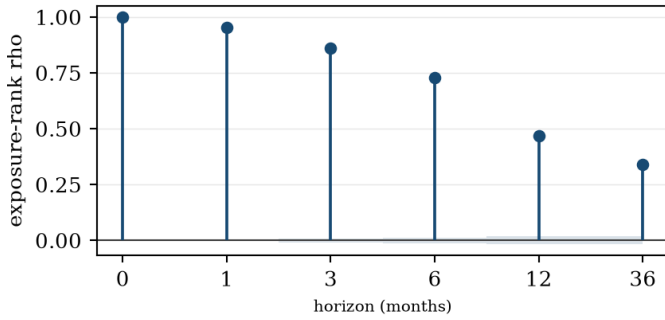


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

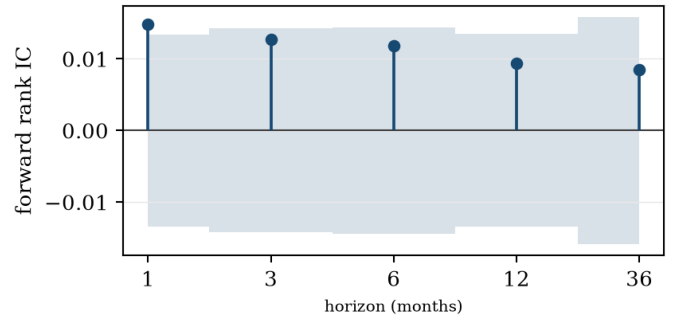


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	accruals_total
Component signs	accruals_total(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	22 / 66
Stocks, min / median / max	711 / 1046 / 1611
Available-history Sharpe	0.53

19. debt_to_equity

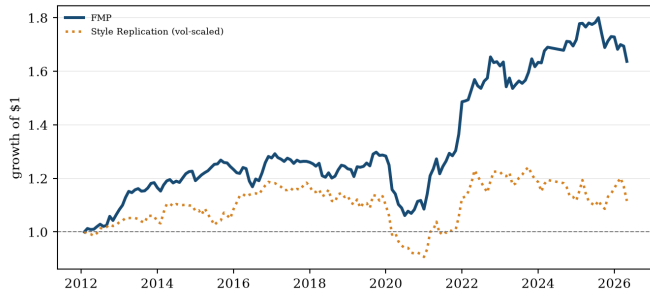


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_equity	3.79%	6.94%	0.55	1.73	-18.24%	0.408	3.13%	6.94%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_equity	0.09 (2.68)	-0.06 (-0.86)	0.31 (4.76)	-0.03 (-0.50)	0.03 (0.35)	0.14 (1.89)	2.98% (1.76)	0.408

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_equity	1.34%	0.11%	-0.92%	-0.16%	0.17%	0.27%	2.98%	3.79%

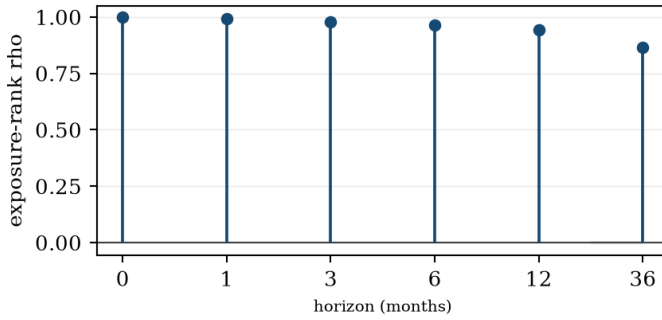


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

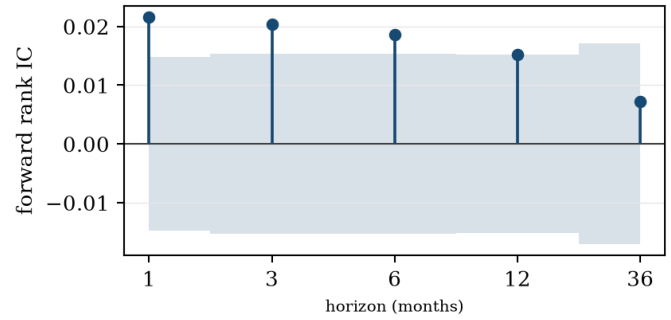


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	debt_to_equity
Component signs	debt_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	728 / 1017 / 1597
Available-history Sharpe	0.29

20. momentum_126_252

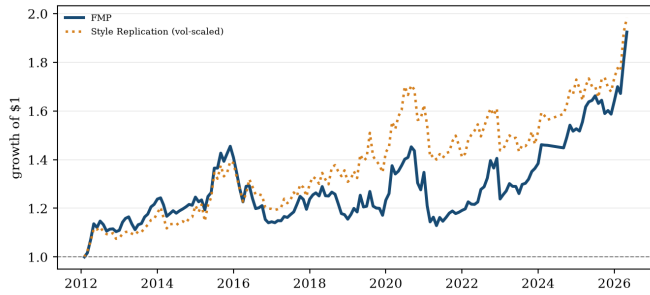


Figure: Common-window cumulative growth of \$1.

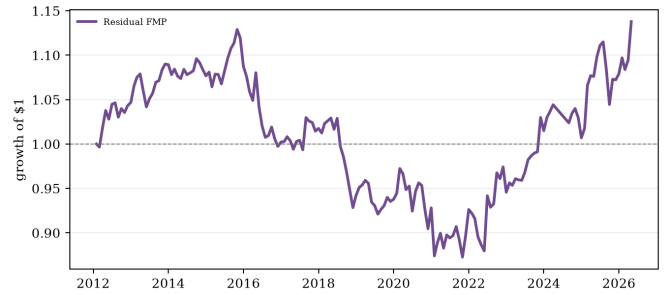


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_126_252	5.26%	10.49%	0.50	1.80	-22.41%	0.690	0.71%	10.49%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_126_252	-0.07 (-1.46)	-0.08 (-1.20)	-0.12 (-2.01)	0.60 (9.15)	0.15 (1.85)	-0.03 (-0.33)	1.10% (0.64)	0.691

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_126_252	-0.95%	0.16%	0.37%	3.79%	0.84%	-0.05%	1.10%	5.26%

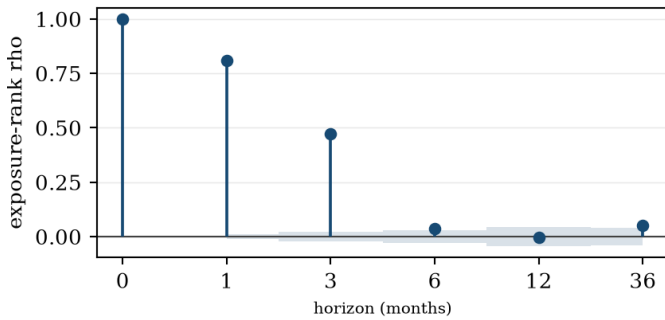


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

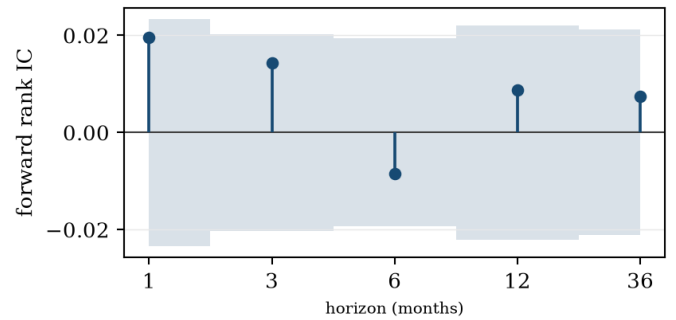


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	momentum_126_252
Component signs	momentum_126_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	738 / 1032 / 1631
Available-history Sharpe	0.28

21. resid_momentum_21_252

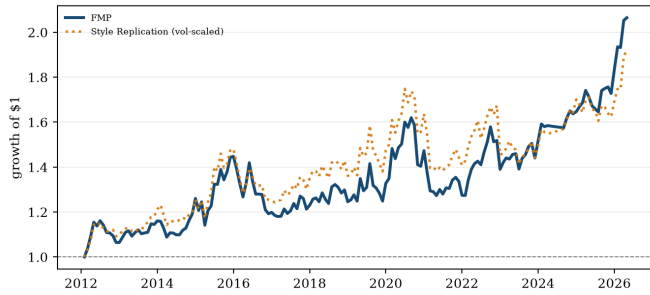


Figure: Common-window cumulative growth of \$1.

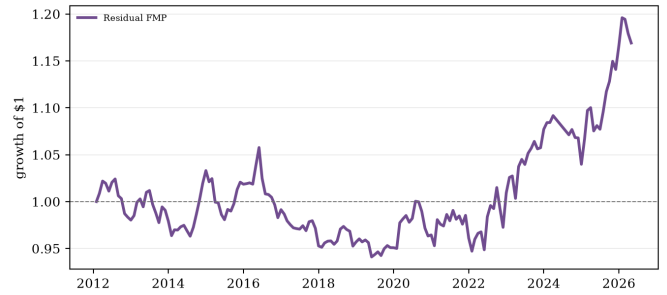


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
resid_momentum_21_252	5.93%	11.94%	0.50	2.03	-21.39%	0.864	0.89%	11.94%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
resid_momentum_21_252	-0.15	-0.11	-0.10	0.75	0.29	-0.00	1.22%	0.865
	(-4.81)	(-2.42)	(-1.92)	(13.29)	(5.33)	(-0.04)	(0.86)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
resid_momentum_21_252	-2.16%	0.20%	0.29%	4.74%	1.65%	-0.00%	1.22%	5.93%

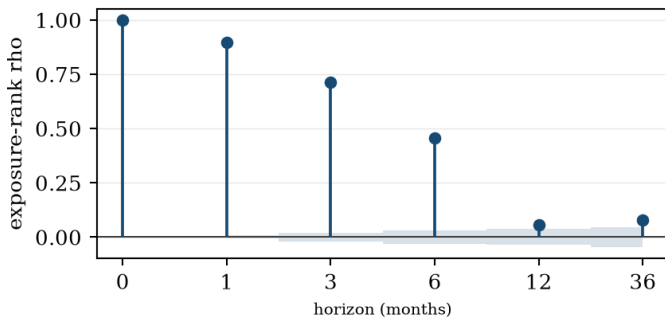


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

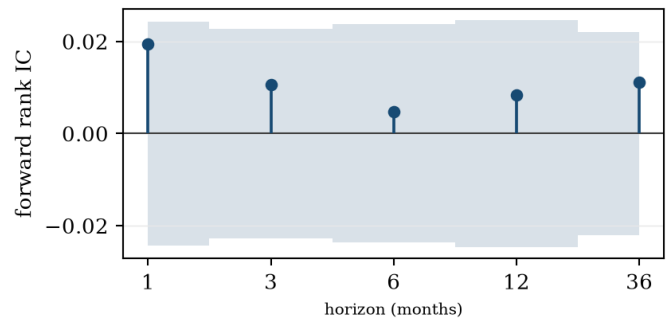


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	resid_momentum_21_252
Component signs	resid_momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	699 / 956 / 1545
Available-history Sharpe	0.17

22. gpoa

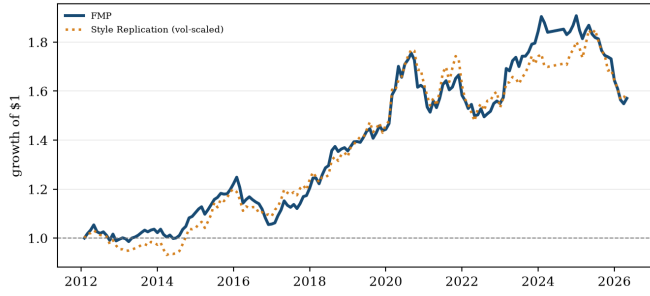


Figure: Common-window cumulative growth of \$1.

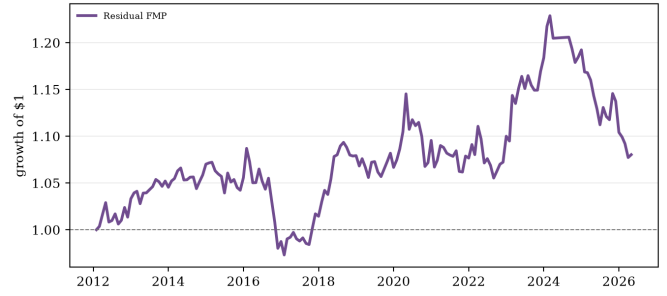


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gpoa	3.52%	7.45%	0.47	1.50	-18.84%	0.631	0.63%	7.45%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gpoa	0.07 (2.78)	-0.03 (-0.72)	-0.46 (-12.11)	-0.01 (-0.31)	-0.05 (-0.79)	0.37 (7.48)	0.66% (0.48)	0.631

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gpoa	1.07%	0.06%	1.39%	-0.09%	-0.27%	0.71%	0.66%	3.52%

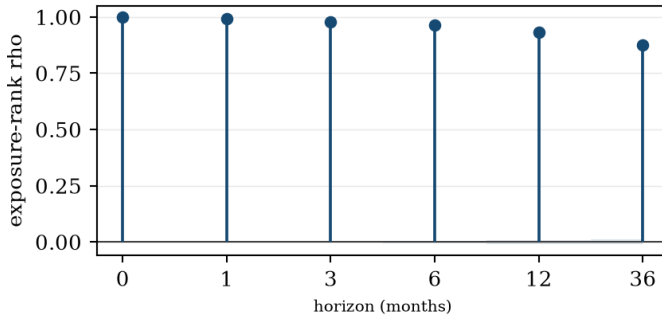


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

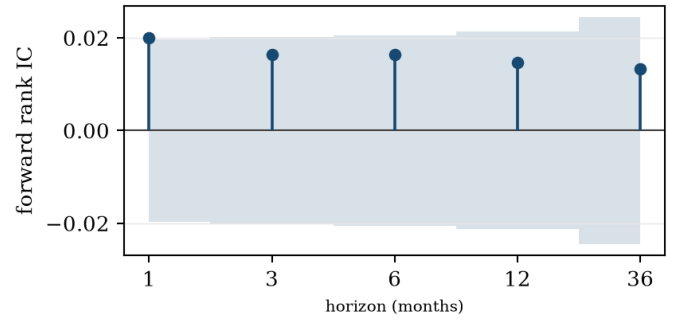


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	gpoa
Component signs	gpoa(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	22 / 66
Stocks, min / median / max	731 / 1053 / 1621
Available-history Sharpe	0.54

23. corr_1260d_FRENCH_CRSP

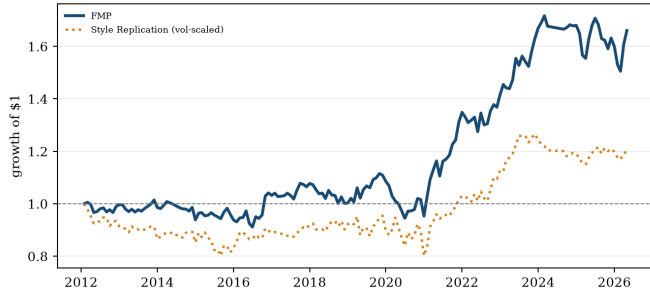


Figure: Common-window cumulative growth of \$1.

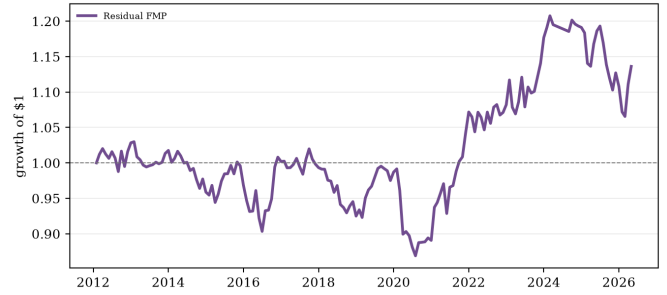


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_FRENCH_CRSP	4.02%	8.67%	0.46	1.82	-15.28%	0.425	2.93%	8.67%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_FRENCH_CRSP	0.37	0.07	0.08	-0.03	-0.42	0.32	1.12%	0.465
	(8.10)	(0.96)	(1.33)	(-0.49)	(-5.34)	(3.82)	(0.66)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_FRENCH_CRSP	5.26%	-0.13%	-0.25%	-0.19%	-2.41%	0.62%	1.12%	4.02%

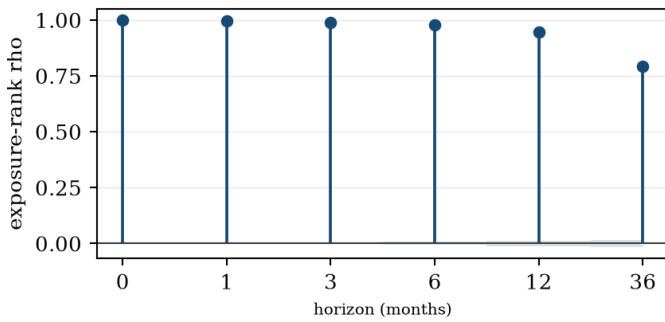


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

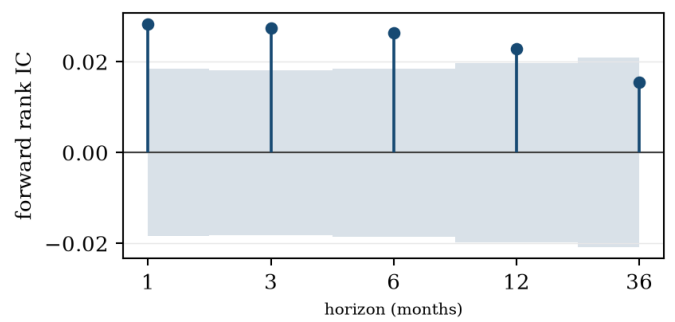


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	corr_1260d_FRENCH_CRSP
Component signs	corr_1260d_FRENCH_CRSP(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	707 / 982 / 1565
Available-history Sharpe	0.27

24. corr_1260d_SPY

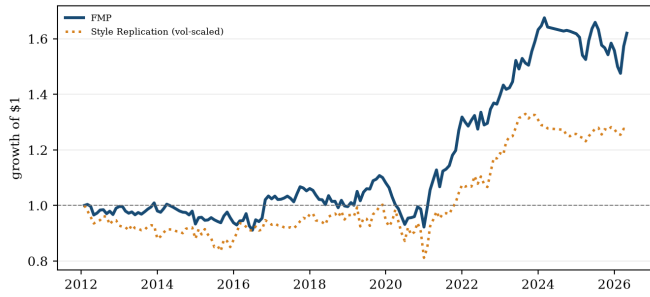


Figure: Common-window cumulative growth of \$1.

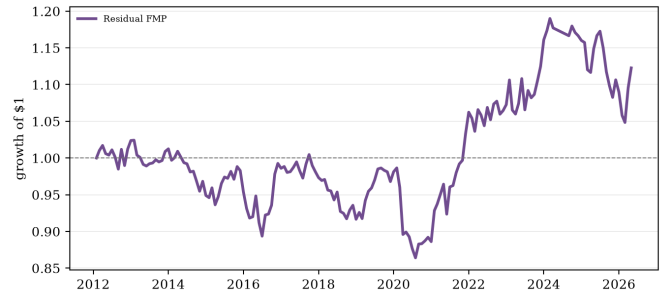


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_SPY	3.81%	8.32%	0.46	1.76	-16.72%	0.403	2.49%	8.32%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_SPY	0.34 (7.18)	0.01 (0.18)	0.12 (1.99)	-0.01 (-0.16)	-0.39 (-5.14)	0.34 (3.97)	1.03% (0.63)	0.434

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_SPY	4.83%	-0.02%	-0.37%	-0.06%	-2.25%	0.65%	1.03%	3.81%

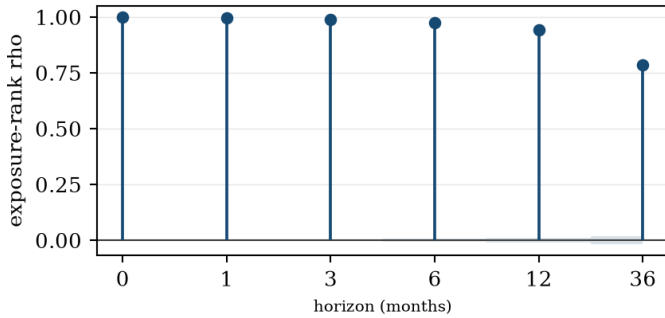


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

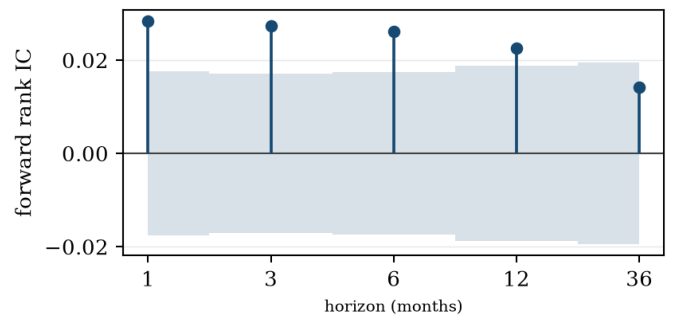


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	corr_1260d_SPY
Component signs	corr_1260d_SPY(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	707 / 982 / 1565
Available-history Sharpe	0.27

25. corr_1260d_VTI

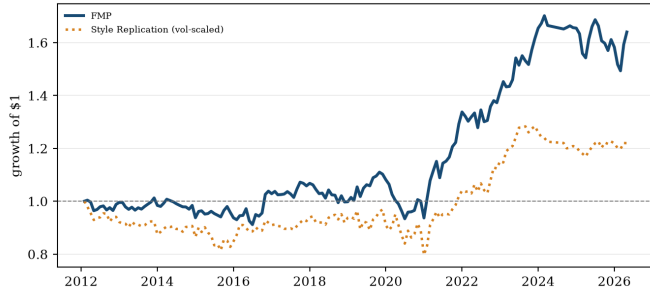


Figure: Common-window cumulative growth of \$1.

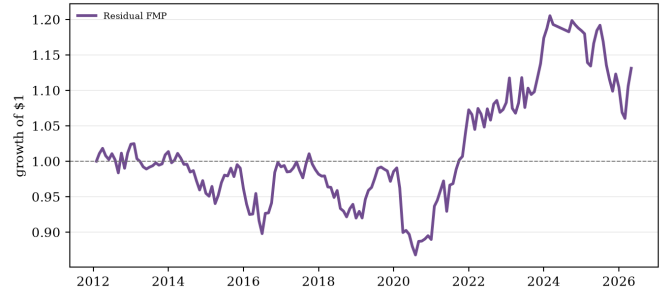


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_VTI	3.92%	8.58%	0.46	1.78	-15.86%	0.413	2.77%	8.58%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_VTI	0.36 (7.67)	0.06 (0.82)	0.12 (1.82)	-0.02 (-0.26)	-0.41 (-5.21)	0.33 (3.86)	1.09% (0.65)	0.449

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_VTI	5.10%	-0.11%	-0.35%	-0.10%	-2.34%	0.64%	1.09%	3.92%

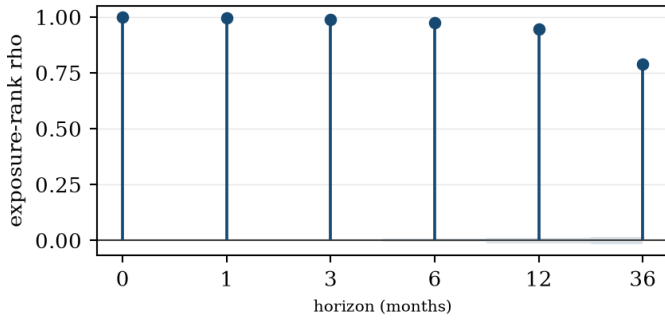


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

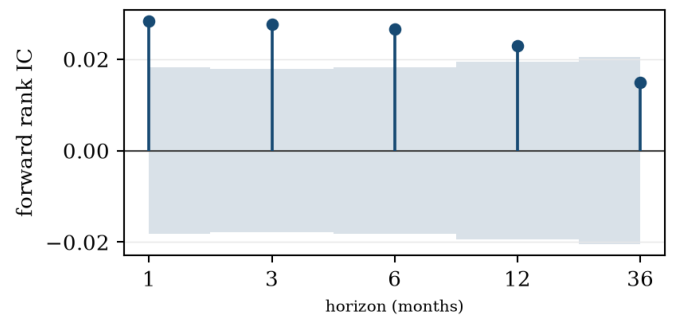


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	corr_1260d_VTI
Component signs	corr_1260d_VTI(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	22 / 76
Stocks, min / median / max	707 / 982 / 1565
Available-history Sharpe	0.27

26. high_52w_252d

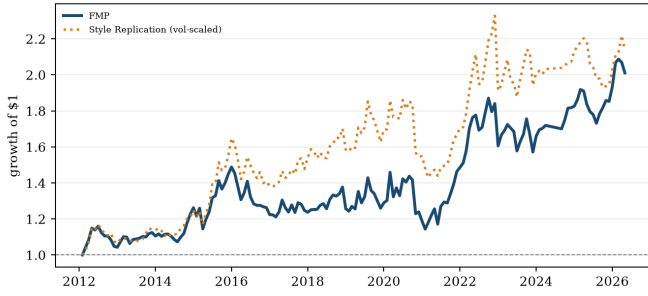


Figure: Common-window cumulative growth of \$1.

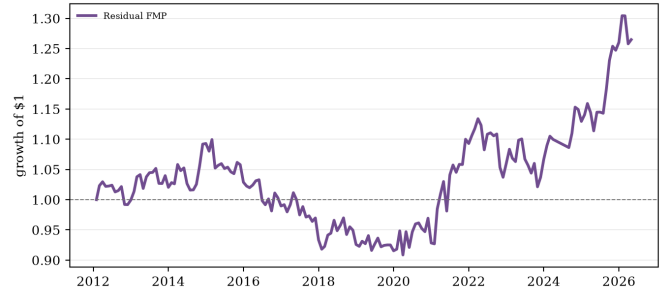


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
high_52w_252d	5.92%	13.30%	0.45	1.87	-23.17%	0.719	0.51%	13.30%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
high_52w_252d	-0.15 (-2.44)	-0.17 (-2.14)	0.07 (0.99)	0.66 (10.18)	0.15 (1.57)	0.55 (4.45)	1.93% (1.21)	0.725

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
high_52w_252d	-2.21%	0.32%	-0.20%	4.17%	0.85%	1.05%	1.93%	5.92%

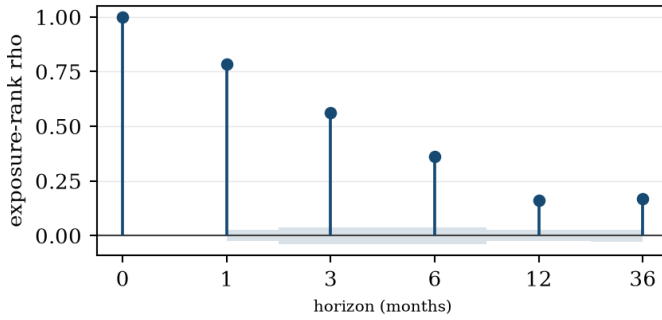


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

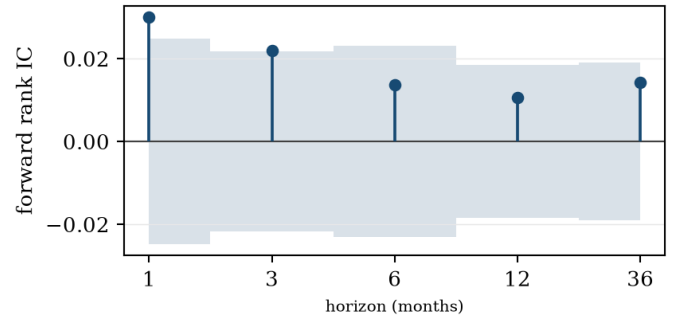


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	high_52w_252d
Component signs	high_52w_252d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	738 / 1032 / 1631
Available-history Sharpe	0.14

27. info_discreteness_21_252

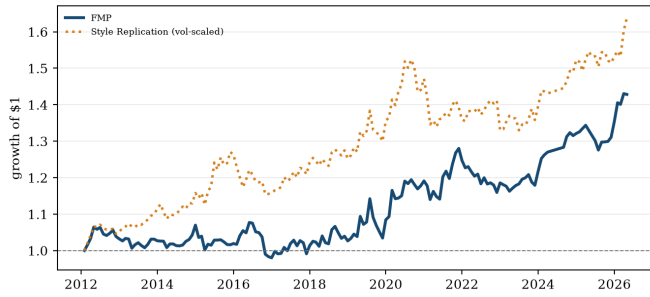


Figure: Common-window cumulative growth of \$1.

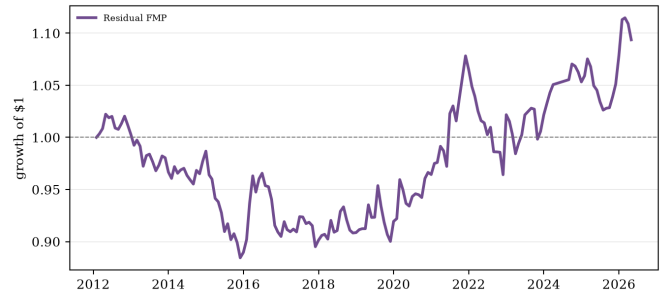


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
info_discreteness_21_252	2.76%	6.25%	0.44	1.76	-9.45%	0.388	0.42%	6.25%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
info_discreteness_21_252	-0.03	-0.11	-0.11	0.21	0.12	-0.04	0.76%	0.390
	(-0.76)	(-2.18)	(-1.79)	(2.72)	(1.77)	(-0.50)	(0.47)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
info_discreteness_21_252	-0.44%	0.20%	0.32%	1.30%	0.69%	-0.08%	0.76%	2.76%

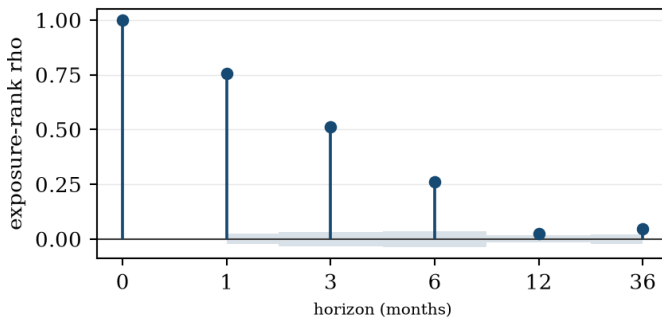


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

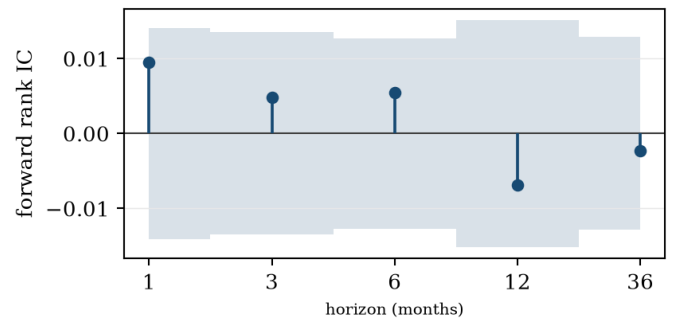


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	info_discreteness_21_252
Component signs	info_discreteness_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	738 / 1032 / 1630
Available-history Sharpe	0.35

28. roe

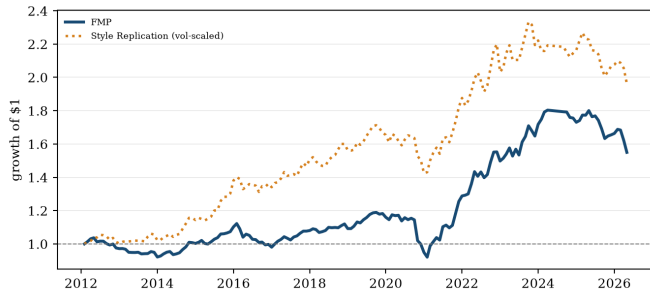


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
roe	3.47%	7.97%	0.44	1.39	-22.59%	0.634	-0.65%	7.97%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
roe	0.14 (4.35)	-0.15 (-2.81)	0.05 (1.17)	0.10 (2.04)	-0.03 (-0.62)	0.64 (9.09)	-0.37% (-0.25)	0.635

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
roe	1.99%	0.29%	-0.15%	0.64%	-0.16%	1.24%	-0.37%	3.47%

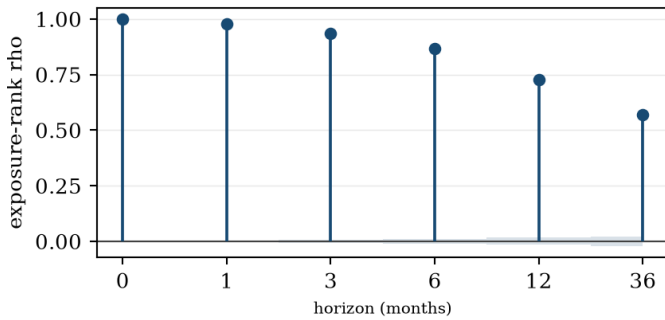


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

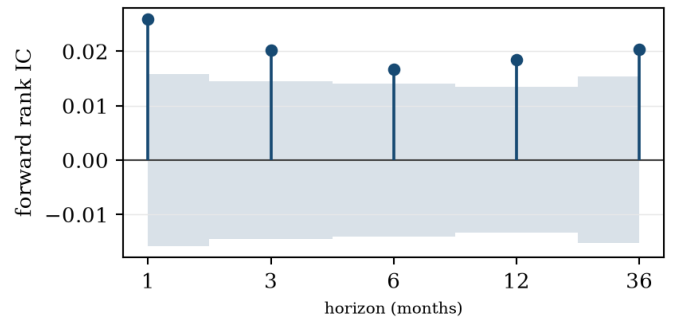


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	roe
Component signs	roe(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	15 / 66
Stocks, min / median / max	713 / 1014 / 1566
Available-history Sharpe	0.49

29. accruals_noa_proxy

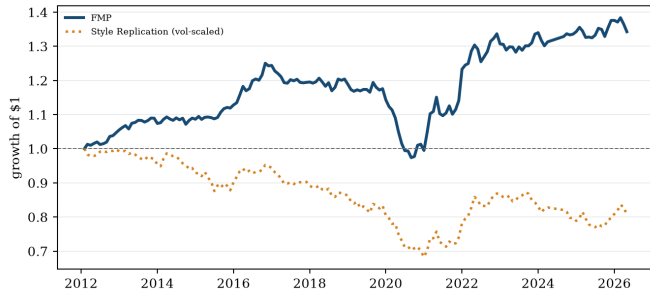


Figure: Common-window cumulative growth of \$1.

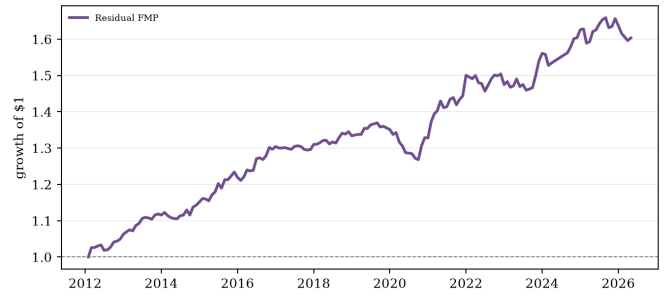


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_noa_proxy	2.25%	5.26%	0.43	1.32	-22.11%	0.531	3.28%	5.26%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_noa_proxy	0.00 (0.19)	0.02 (0.39)	0.32 (6.75)	0.06 (1.45)	-0.16 (-4.01)	0.12 (3.02)	3.46% (3.17)	0.532

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_noa_proxy	0.07%	-0.03%	-0.95%	0.37%	-0.91%	0.24%	3.46%	2.25%

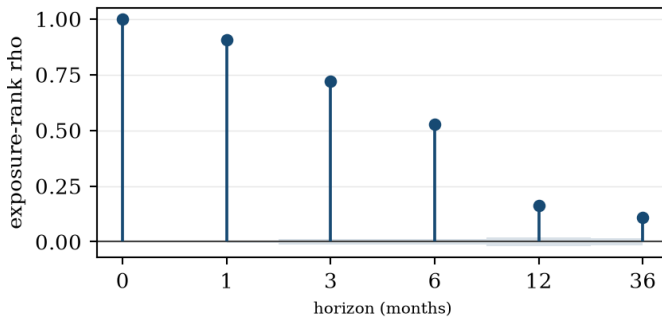


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

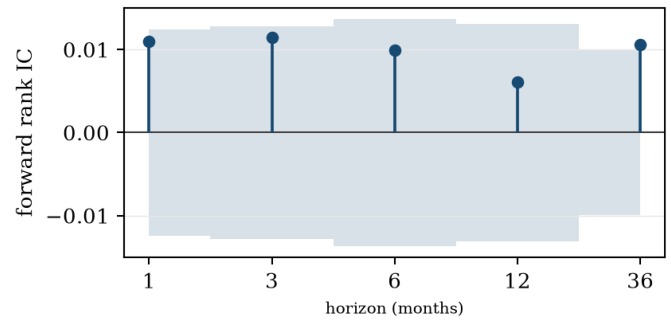


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	accruals_noa_proxy
Component signs	accruals_noa_proxy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	235 / 167
Available window	2007-01-01 to 2026-07-01
Dropped: low breadth / common	55 / 68
Stocks, min / median / max	682 / 952 / 1614
Available-history Sharpe	0.32

30. sga_to_revenue

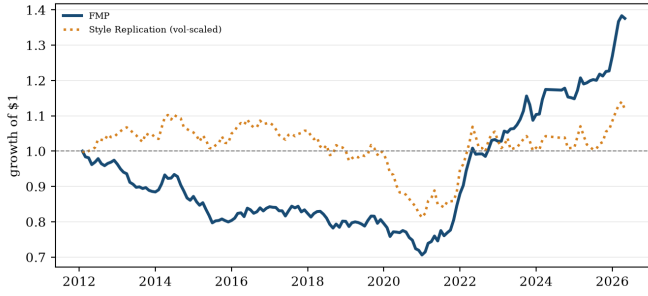


Figure: Common-window cumulative growth of \$1.

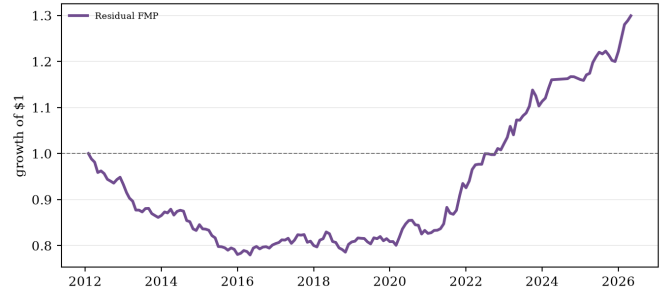


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
sga_to_revenue	2.47%	5.94%	0.42	1.16	-28.22%	0.483	1.79%	5.94%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
sga_to_revenue	0.02 (0.77)	-0.22 (-4.96)	0.34 (6.99)	0.13 (2.59)	-0.01 (-0.14)	-0.00 (-0.02)	1.97% (1.52)	0.484

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
sga_to_revenue	0.32%	0.41%	-1.03%	0.83%	-0.04%	-0.00%	1.97%	2.47%

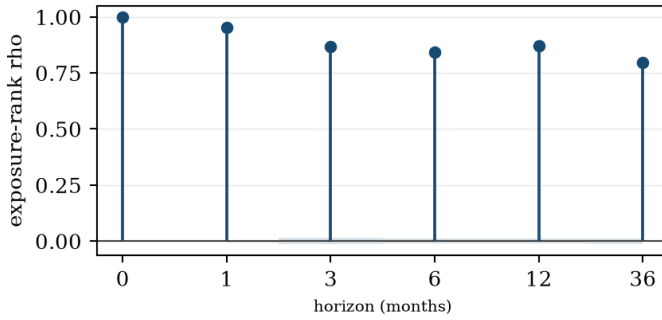


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

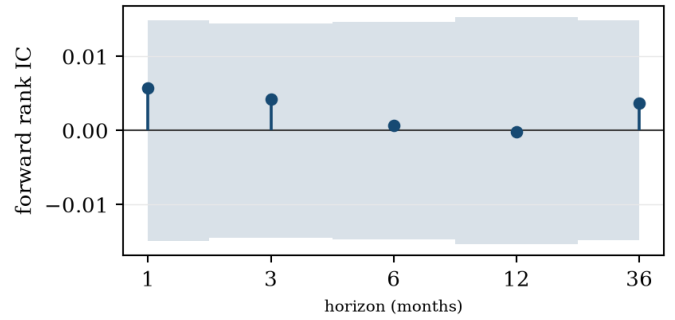


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	sga_to_revenue
Component signs	sga_to_revenue(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	723 / 1029 / 1536
Available-history Sharpe	0.26

31. equity_issuance_yoy

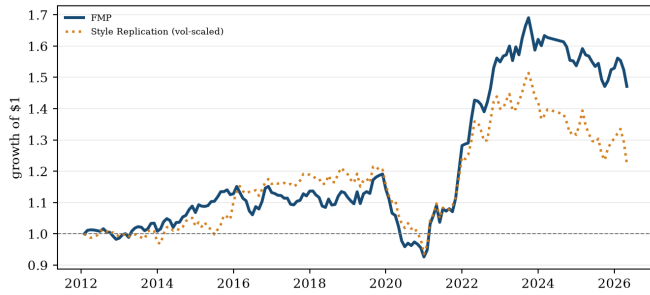


Figure: Common-window cumulative growth of \$1.

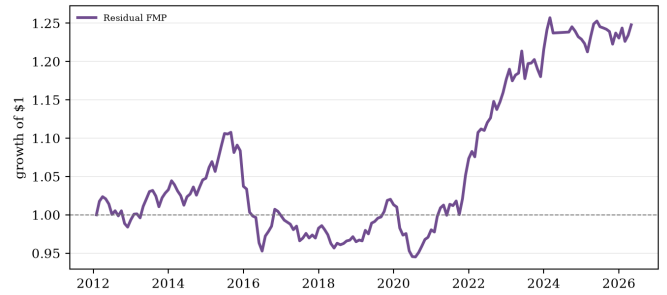


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
equity_issuance_yoy	3.05%	7.49%	0.41	1.27	-22.25%	0.695	1.61%	7.49%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
equity_issuance_yoy	0.14 (5.55)	-0.03 (-0.70)	0.31 (6.39)	0.00 (0.06)	-0.14 (-2.94)	0.53 (10.78)	1.68% (1.18)	0.695

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
equity_issuance_yoy	2.03%	0.06%	-0.93%	0.01%	-0.81%	1.01%	1.68%	3.05%

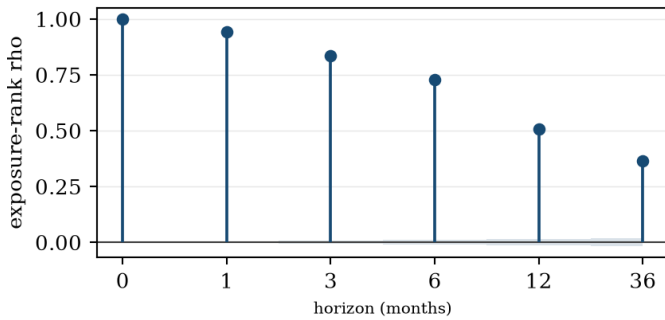


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

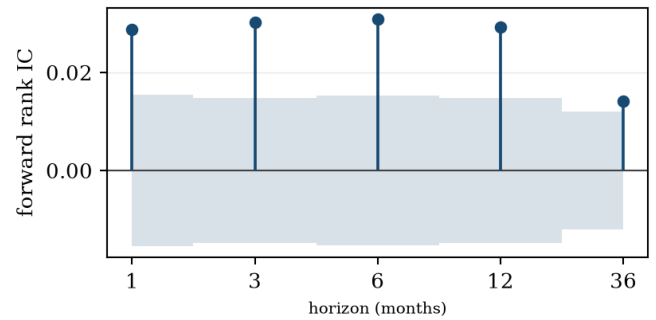


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	equity_issuance_yoy
Component signs	equity_issuance_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	237 / 167
Available window	2006-11-01 to 2026-07-01
Dropped: low breadth / common	54 / 70
Stocks, min / median / max	745 / 1057 / 1665
Available-history Sharpe	0.38

32. revenue_growth_yoy_sign_flip

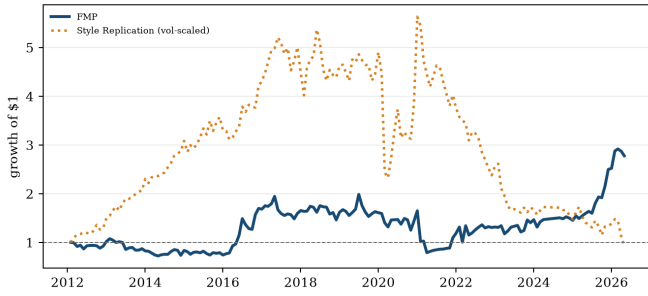


Figure: Common-window cumulative growth of \$1.

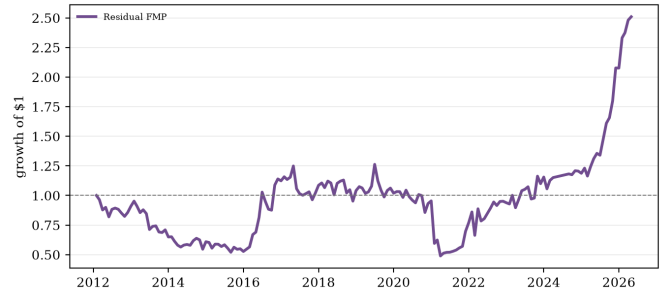


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
revenue_growth_yoy_ sign_flip	12.04%	30.41%	0.40	1.63	-60.31%	0.071	10.88%	30.41%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
revenue_growth_yoy_ sign_flip	-0.12 (-0.53)	0.20 (0.66)	-0.06 (-0.23)	-0.29 (-1.82)	0.96 (3.60)	-0.34 (-1.19)	10.99% (1.20)	0.071

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
revenue_growth_yoy_ sign_flip	-1.73%	-0.37%	0.17%	-1.85%	5.49%	-0.65%	10.99%	12.04%

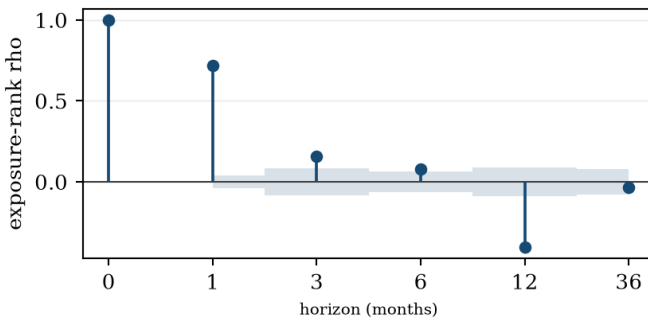


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

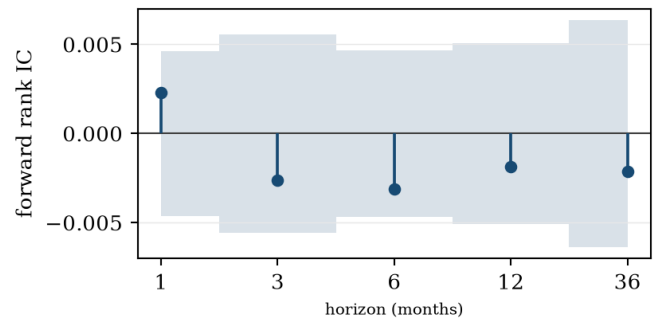


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	revenue_growth_yoy_sign_flip
Component signs	revenue_growth_yoy_sign_flip(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	237 / 167
Available window	2006-11-01 to 2026-07-01
Dropped: low breadth / common	54 / 70
Stocks, min / median / max	744 / 1053 / 1663
Available-history Sharpe	0.40

33. is_positive_net_income

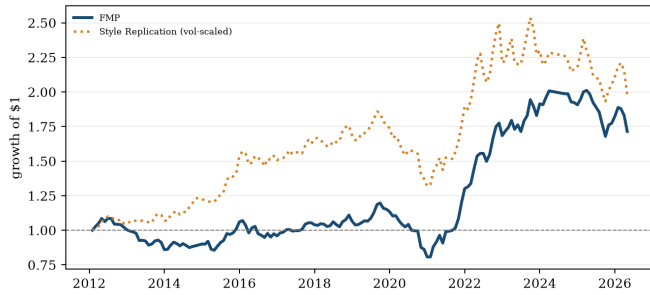


Figure: Common-window cumulative growth of \$1.

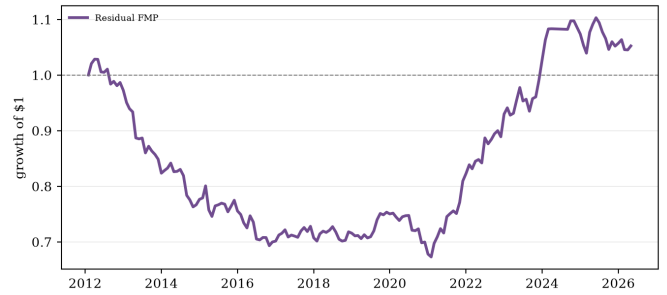


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_positive_net_income	4.54%	11.51%	0.39	1.23	-32.56%	0.700	-0.12%	11.51%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_positive_net_income	0.07	-0.21	0.34	0.25	0.08	0.78	0.56%	0.703
	(1.85)	(-2.91)	(6.03)	(4.01)	(1.04)	(7.90)	(0.28)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_positive_net_income	1.07%	0.38%	-1.03%	1.60%	0.44%	1.51%	0.56%	4.54%

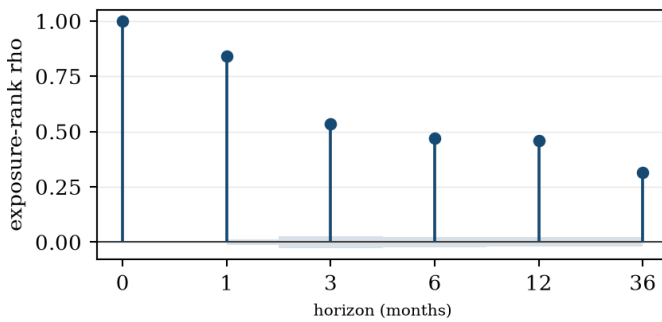


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

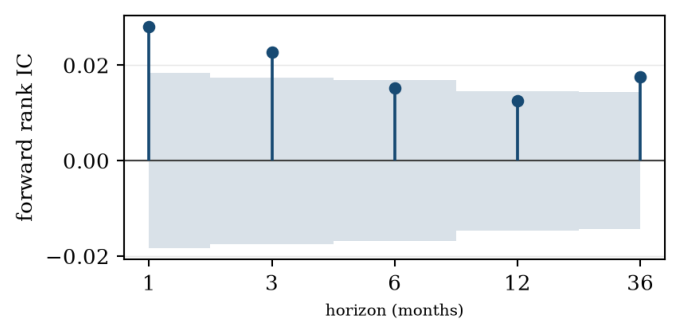


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	is_positive_net_income
Component signs	is_positive_net_income(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	748 / 1060 / 1678
Available-history Sharpe	0.37

34. eps_growth_yoy_sign_flip

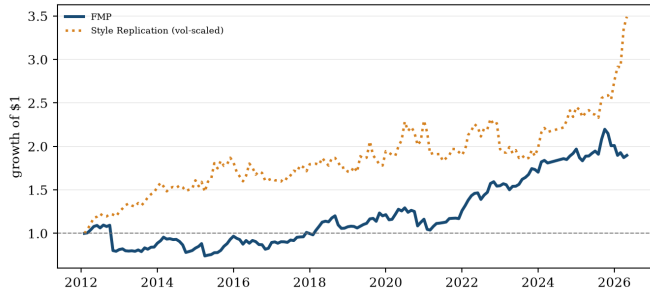


Figure: Common-window cumulative growth of \$1.

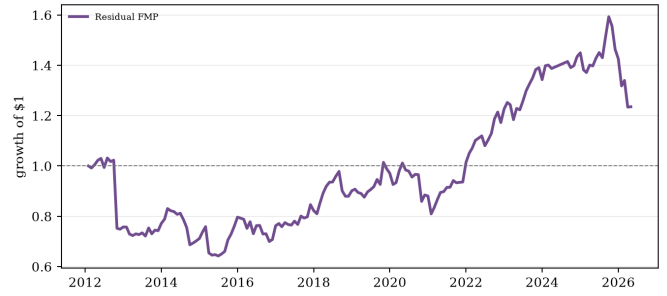


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
eps_growth_yoy_sign_flip	5.77%	14.76%	0.39	1.49	-32.41%	0.106	2.48%	14.76%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
eps_growth_yoy_sign_flip	0.01	-0.06	0.07	0.37	0.20	-0.16	2.55%	0.106
	(0.11)	(-0.34)	(0.51)	(2.21)	(1.47)	(-0.90)	(0.58)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
eps_growth_yoy_sign_flip	0.13%	0.10%	-0.22%	2.37%	1.14%	-0.31%	2.55%	5.77%

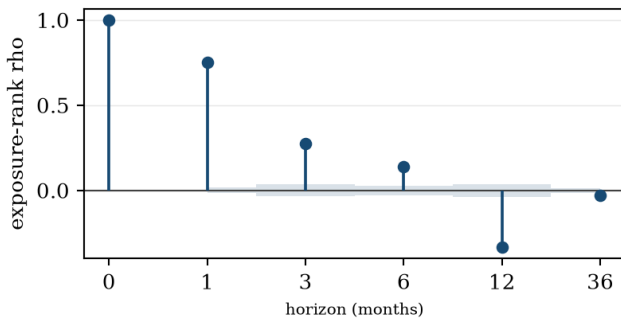


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

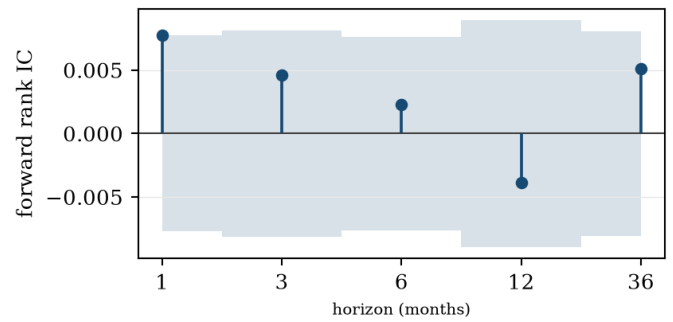


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	eps_growth_yoy_sign_flip
Component signs	eps_growth_yoy_sign_flip(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	230 / 167
Available window	2007-06-01 to 2026-07-01
Dropped: low breadth / common	49 / 63
Stocks, min / median / max	711 / 1032 / 1600
Available-history Sharpe	0.29

35. dpo

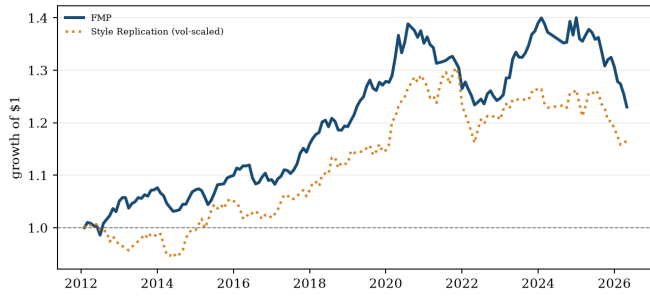


Figure: Common-window cumulative growth of \$1.

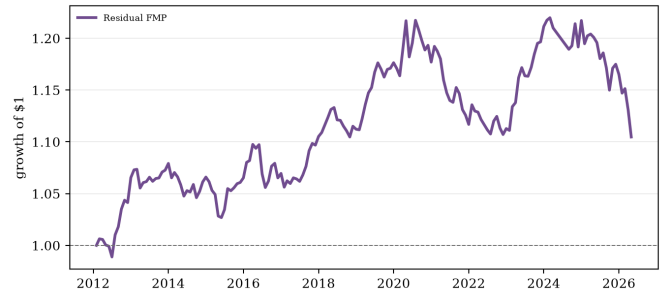


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dpo	1.57%	4.02%	0.39	1.29	-12.15%	0.226	1.00%	4.02%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dpo	0.04 (1.49)	0.10 (2.34)	-0.15 (-6.13)	-0.01 (-0.34)	-0.03 (-0.82)	0.12 (2.75)	0.78% (0.78)	0.229

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dpo	0.57%	-0.18%	0.45%	-0.08%	-0.19%	0.22%	0.78%	1.57%

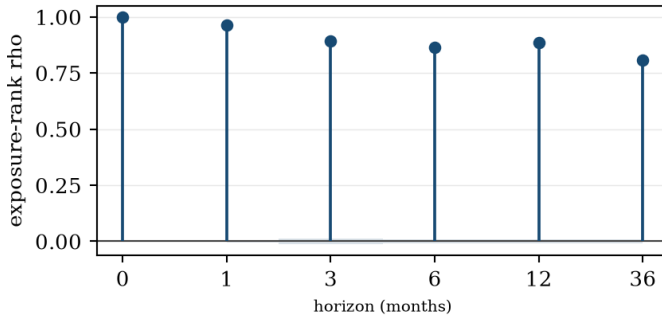


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

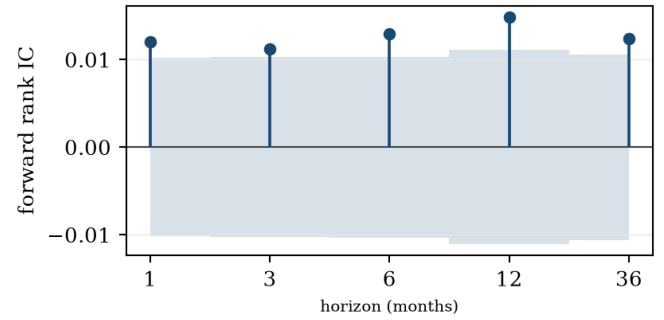


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	dpo
Component signs	dpo(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	12 / 66
Stocks, min / median / max	686 / 997 / 1460
Available-history Sharpe	0.44

36. net_income_growth_yoy_sign_flip

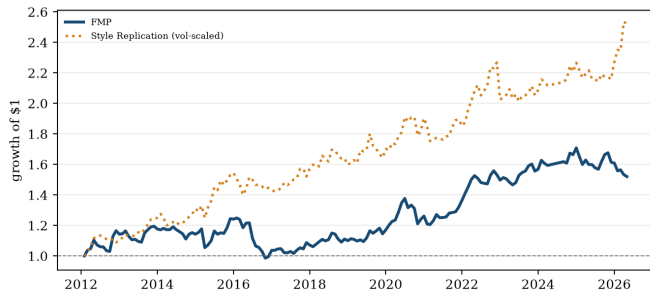


Figure: Common-window cumulative growth of \$1.

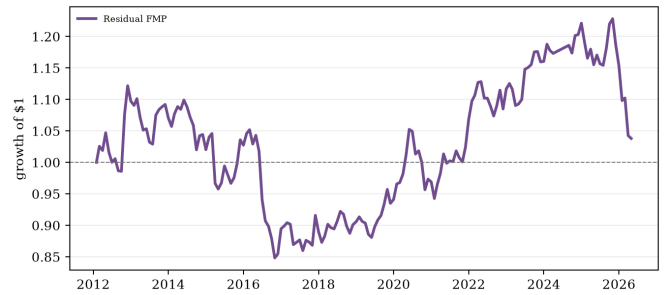


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_growth_yoy_sign_flip	3.42%	9.06%	0.38	1.38	-21.06%	0.226	0.06%	9.06%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_growth_yoy_sign_flip	0.01	0.09	0.03	0.35	0.09	0.13	0.58%	0.228
	(0.16)	(1.02)	(0.38)	(3.26)	(0.98)	(1.34)	(0.22)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_growth_yoy_sign_flip	0.12%	-0.17%	-0.10%	2.24%	0.52%	0.24%	0.58%	3.42%

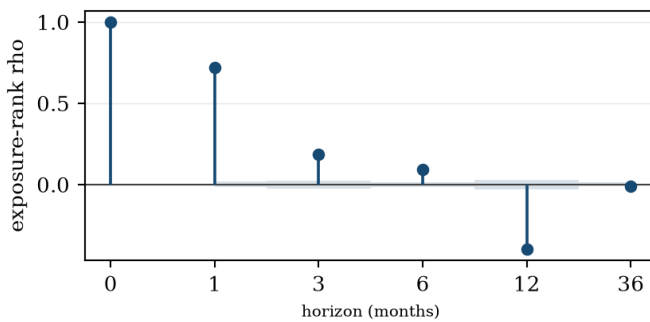


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

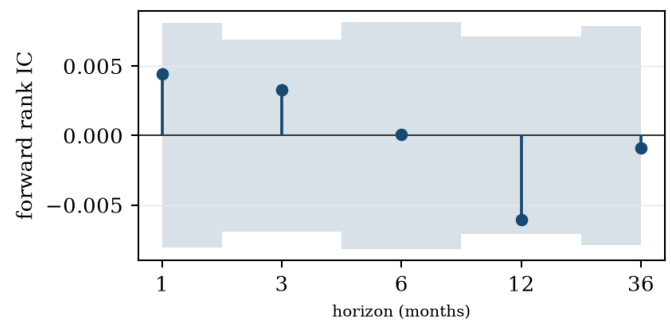


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	net_income_growth_yoy_sign_flip
Component signs	net_income_growth_yoy_sign_flip(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	237 / 167
Available window	2006-11-01 to 2026-07-01
Dropped: low breadth / common	54 / 70
Stocks, min / median / max	744 / 1056 / 1665
Available-history Sharpe	0.09

37. cash_earnings_ttm

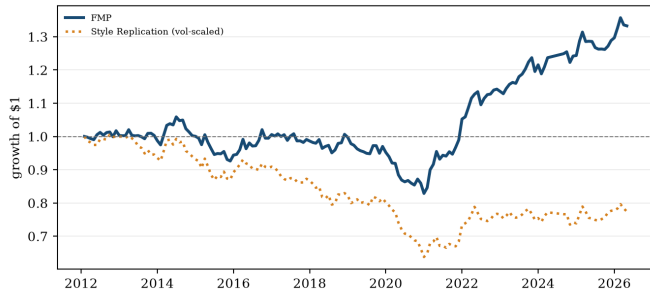


Figure: Common-window cumulative growth of \$1.

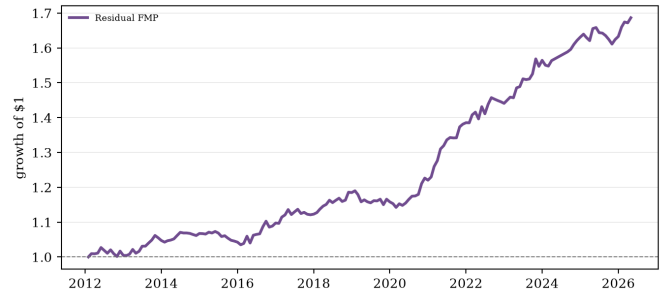


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
cash_earnings_ttm	2.24%	5.98%	0.37	1.31	-21.73%	0.695	3.63%	5.98%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
cash_earnings_ttm	-0.04 (-1.17)	-0.34 (-10.36)	0.30 (10.69)	-0.02 (-1.27)	-0.09 (-2.82)	-0.04 (-0.87)	3.82% (3.67)	0.696

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
cash_earnings_ttm	-0.52%	0.63%	-0.92%	-0.16%	-0.53%	-0.07%	3.82%	2.24%

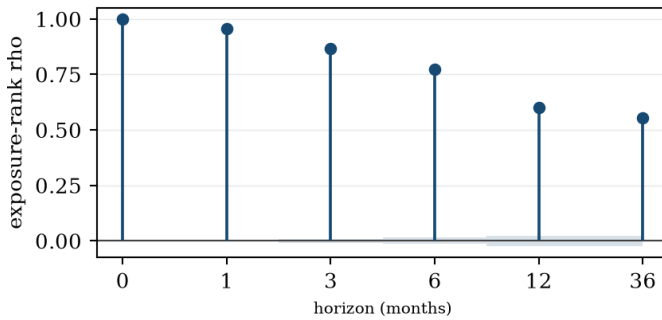


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

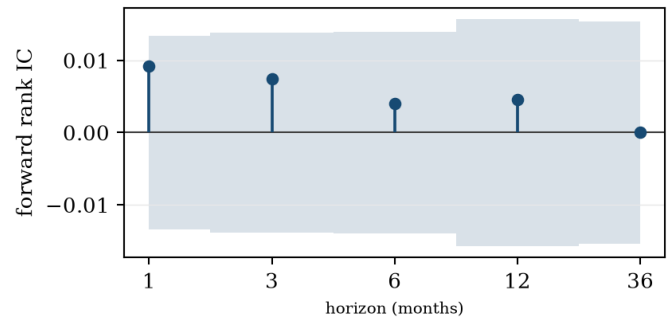


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	cash_earnings_ttm
Component signs	cash_earnings_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	22 / 66
Stocks, min / median / max	711 / 1047 / 1611
Available-history Sharpe	0.38

38. growth_cfoa_5y

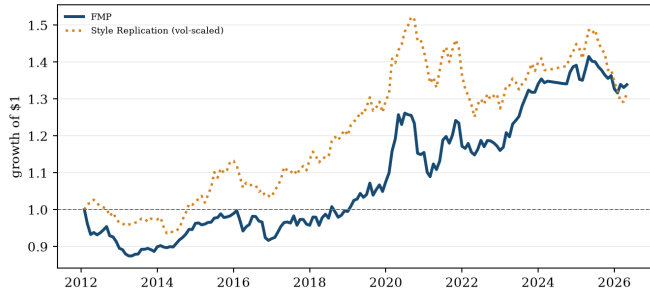


Figure: Common-window cumulative growth of \$1.

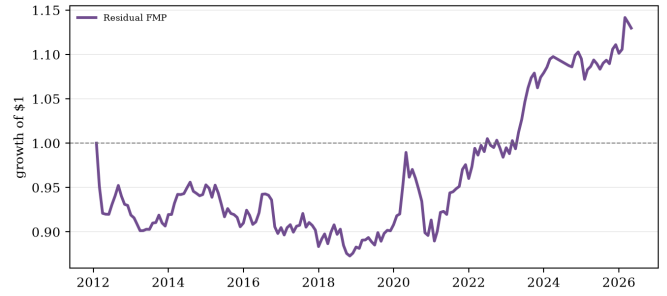


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_cfoa_5y	2.29%	6.18%	0.37	1.21	-13.64%	0.486	0.78%	6.18%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_cfoa_5y	-0.01 (-0.18)	-0.06 (-1.12)	-0.36 (-13.23)	-0.03 (-0.94)	0.01 (0.14)	0.18 (3.91)	0.97% (0.75)	0.487

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_cfoa_5y	-0.08%	0.11%	1.08%	-0.19%	0.04%	0.34%	0.97%	2.29%

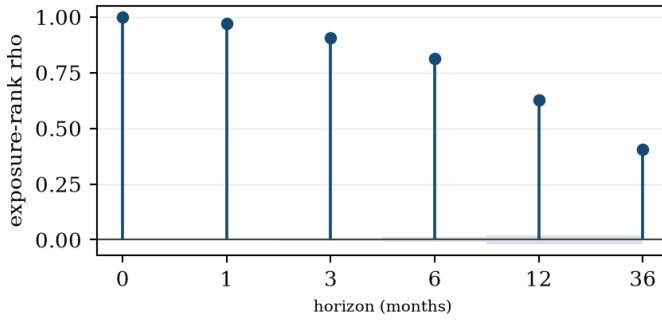


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

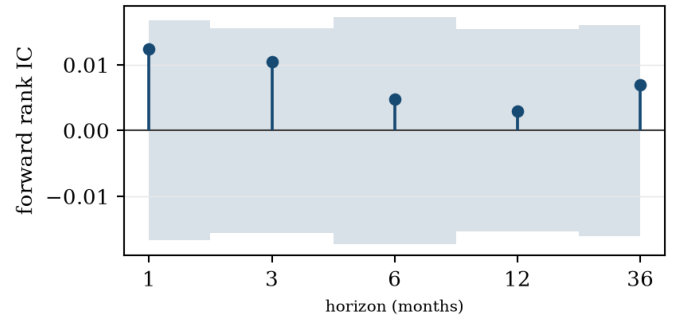


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	growth_cfoa_5y
Component signs	growth_cfoa_5y(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	173 / 167
Available window	2012-03-01 to 2026-07-01
Dropped: low breadth / common	16 / 6
Stocks, min / median / max	52 / 933 / 1484
Available-history Sharpe	0.36

39. roa

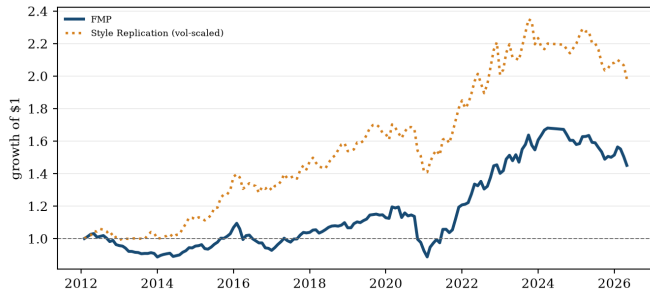


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
roa	3.06%	8.77%	0.35	1.18	-25.75%	0.621	-1.13%	8.77%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
roa	0.11 (2.94)	-0.18 (-3.35)	-0.00 (-0.06)	0.14 (2.60)	-0.08 (-1.41)	0.69 (9.22)	-0.66% (-0.41)	0.623

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
roa	1.56%	0.34%	0.01%	0.92%	-0.43%	1.32%	-0.66%	3.06%

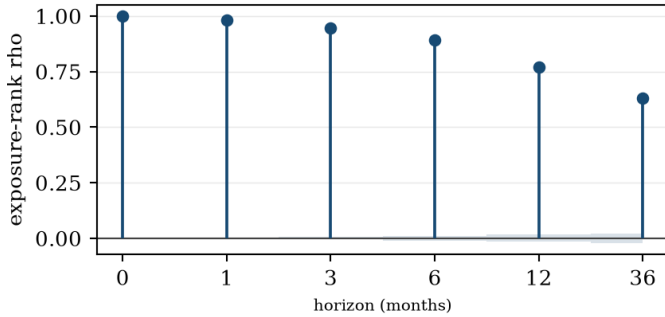


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

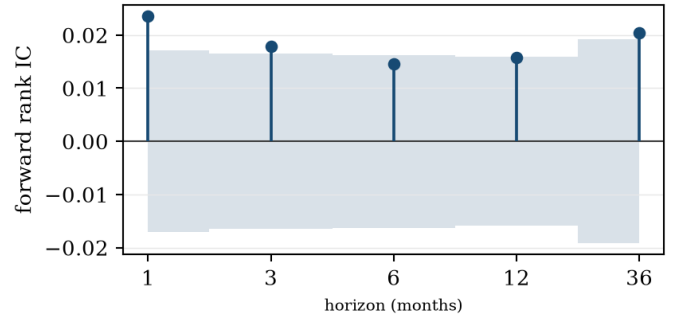


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	roa
Component signs	roa(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	22 / 66
Stocks, min / median / max	732 / 1055 / 1642
Available-history Sharpe	0.43

40. accruals_wc_proxy

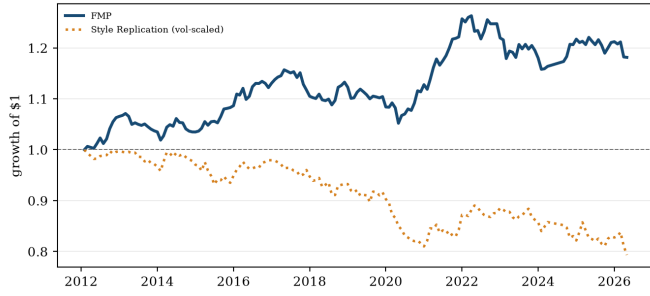


Figure: Common-window cumulative growth of \$1.

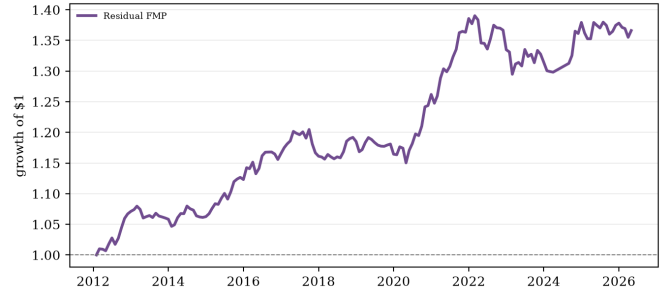


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_wc_proxy	1.26%	3.62%	0.35	1.22	-9.08%	0.118	1.81%	3.62%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_wc_proxy	-0.05 (-2.27)	-0.03 (-0.74)	0.05 (1.89)	-0.04 (-1.18)	0.01 (0.28)	0.01 (0.15)	2.30% (2.27)	0.138

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_wc_proxy	-0.72%	0.06%	-0.16%	-0.28%	0.06%	0.01%	2.30%	1.26%

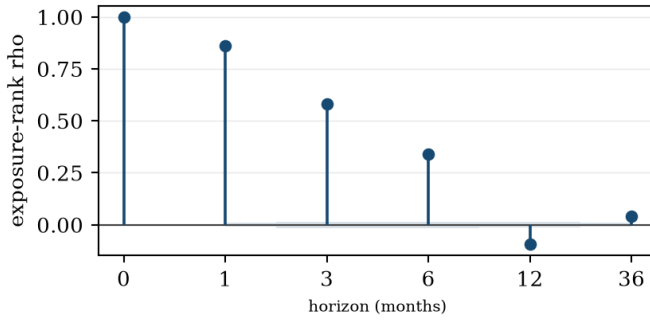


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

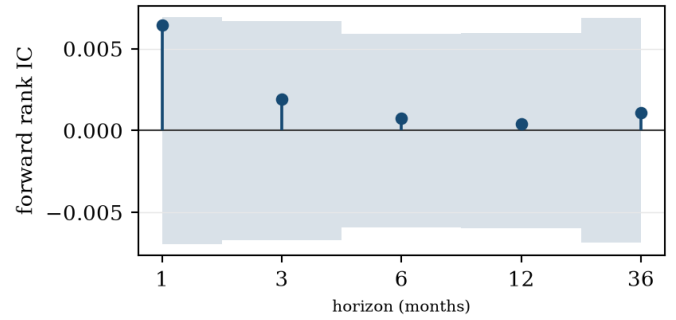


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	accruals_wc_proxy
Component signs	accruals_wc_proxy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	236 / 167
Available window	2006-12-01 to 2026-07-01
Dropped: low breadth / common	55 / 69
Stocks, min / median / max	726 / 1044 / 1633
Available-history Sharpe	0.24

41. accruals_percent

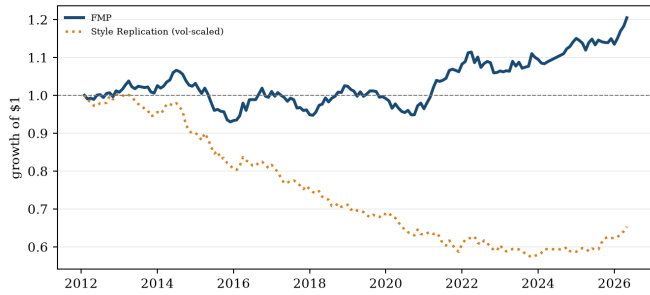


Figure: Common-window cumulative growth of \$1.

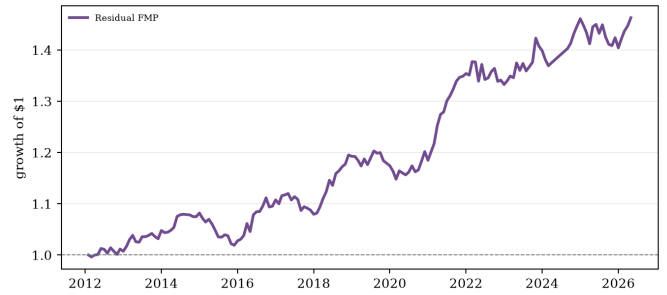


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_percent	1.43%	4.20%	0.34	1.21	-12.77%	0.188	2.71%	4.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_percent	-0.05 (-1.41)	-0.09 (-2.27)	0.10 (2.84)	0.00 (0.13)	-0.04 (-0.93)	-0.18 (-3.87)	2.81% (2.41)	0.188

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_percent	-0.70%	0.17%	-0.30%	0.03%	-0.23%	-0.35%	2.81%	1.43%

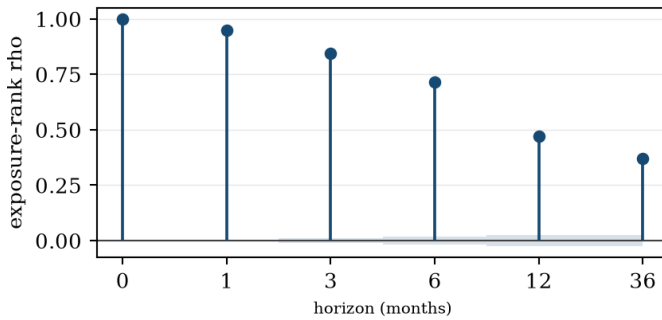


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

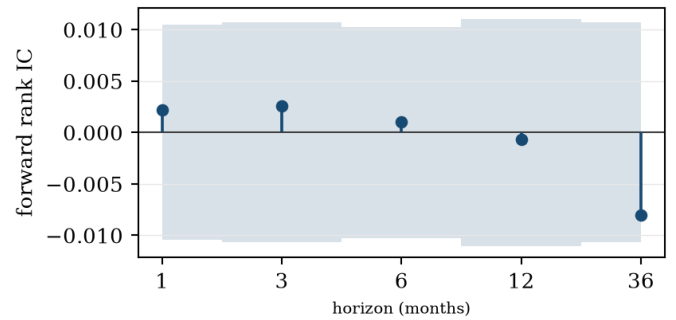


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	accruals_percent
Component signs	accruals_percent(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	22 / 66
Stocks, min / median / max	694 / 1034 / 1585
Available-history Sharpe	0.38

42. is_positive_income_tax

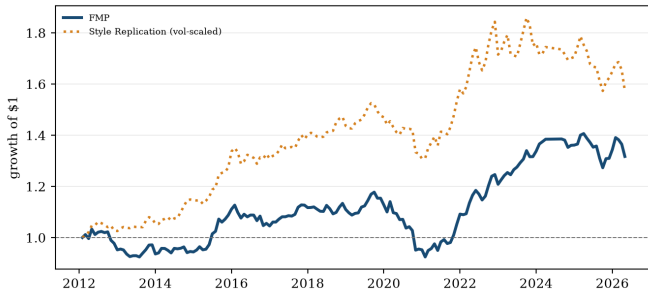


Figure: Common-window cumulative growth of \$1.

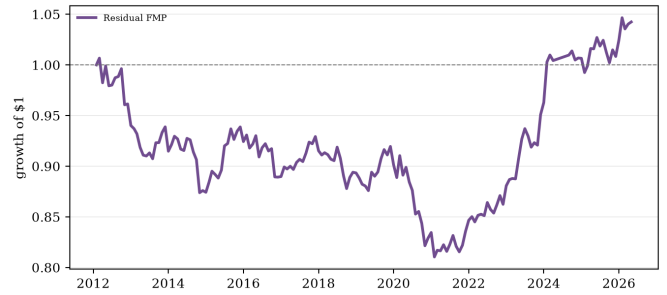


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_positive_income_tax	2.19%	6.52%	0.34	1.13	-21.50%	0.537	-0.35%	6.52%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_positive_income_tax	0.00	-0.05	0.10	0.12	0.08	0.41	0.39%	0.545
	(0.07)	(-1.20)	(2.67)	(2.98)	(1.35)	(7.08)	(0.32)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_positive_income_tax	0.03%	0.10%	-0.31%	0.74%	0.45%	0.79%	0.39%	2.19%

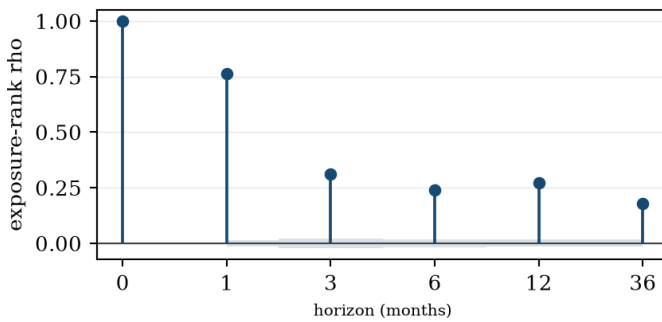


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

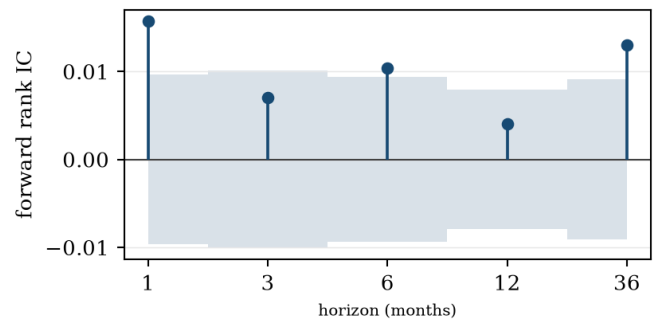


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	is_positive_income_tax
Component signs	is_positive_income_tax(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	747 / 1060 / 1610
Available-history Sharpe	0.36

43. roc

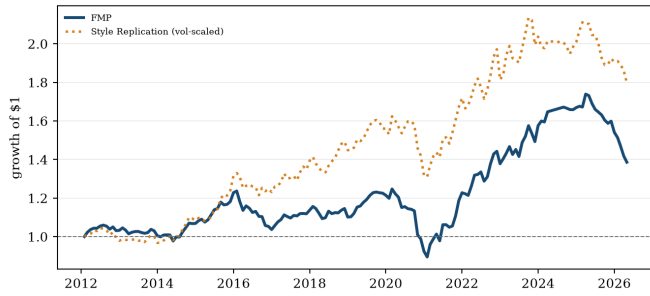


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
roc	2.72%	8.68%	0.31	1.02	-28.26%	0.643	-0.96%	8.68%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
roc	0.10 (2.87)	-0.28 (-4.90)	-0.05 (-0.86)	0.06 (1.17)	-0.08 (-1.82)	0.64 (9.25)	-0.54% (-0.34)	0.645

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
roc	1.41%	0.53%	0.15%	0.41%	-0.47%	1.24%	-0.54%	2.72%

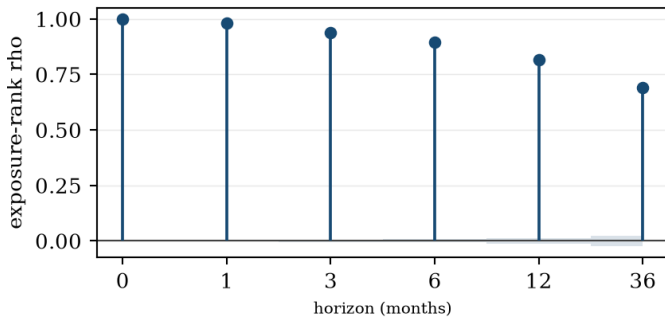


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

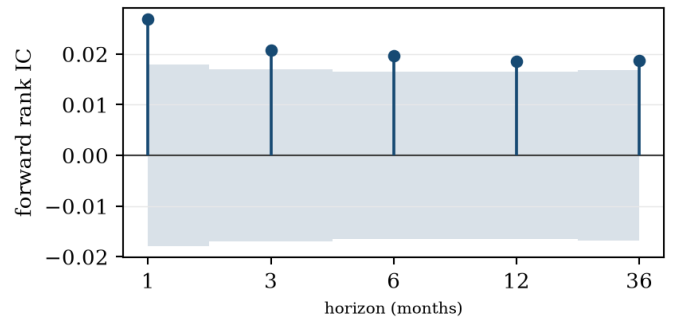


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	roc
Component signs	roc(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	228 / 167
Available window	2007-03-01 to 2026-06-01
Dropped: low breadth / common	17 / 61
Stocks, min / median / max	22 / 777 / 1216
Available-history Sharpe	0.33

44. ret_volume_corr_63d

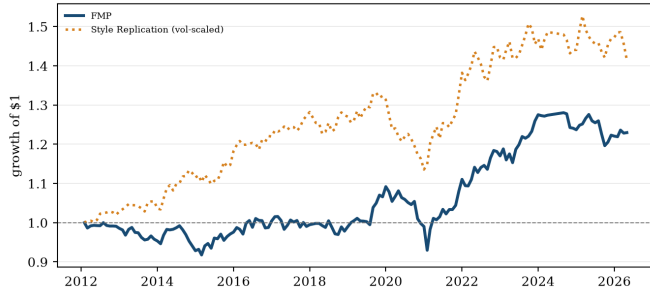


Figure: Common-window cumulative growth of \$1.

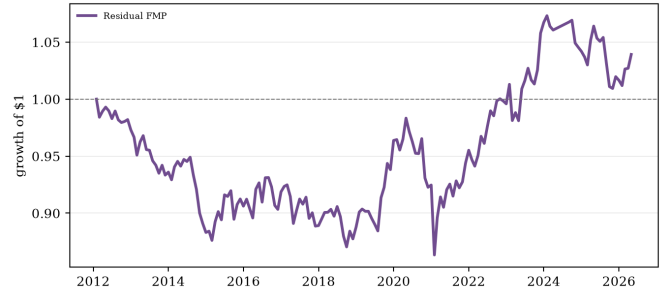


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_volume_corr_63d	1.62%	5.20%	0.31	1.23	-14.85%	0.164	0.55%	5.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_volume_corr_63d	0.07 (2.68)	-0.11 (-2.07)	0.08 (1.91)	-0.02 (-0.68)	0.04 (0.85)	0.10 (1.93)	0.39% (0.32)	0.165

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_volume_corr_63d	0.99%	0.21%	-0.23%	-0.15%	0.22%	0.20%	0.39%	1.62%

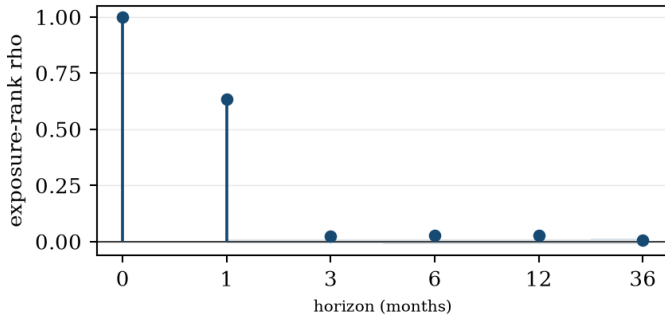


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

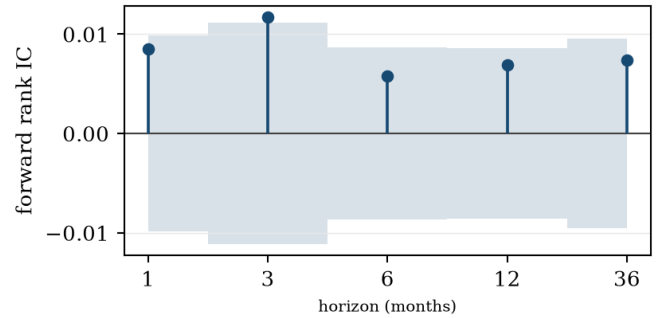


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	ret_volume_corr_63d
Component signs	ret_volume_corr_63d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	749 / 1058 / 1674
Available-history Sharpe	0.31

45. evs_ttm

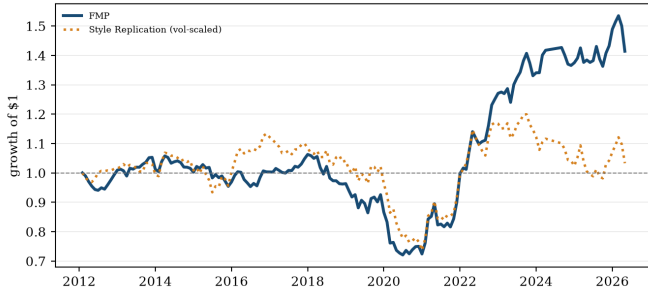


Figure: Common-window cumulative growth of \$1.

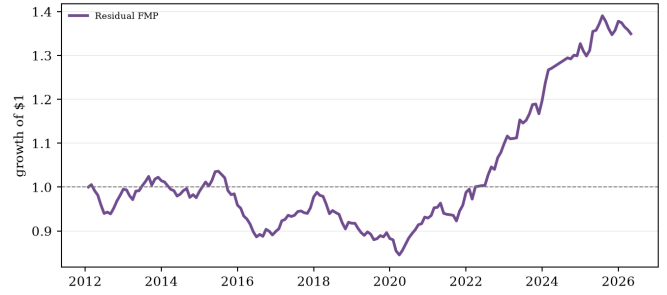


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
evs_ttm	2.93%	9.44%	0.31	0.99	-32.18%	0.754	2.35%	9.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
evs_ttm	0.18 (6.10)	0.02 (0.35)	0.62 (12.34)	0.04 (1.11)	-0.19 (-2.97)	0.38 (6.67)	2.26% (1.31)	0.754

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
evs_ttm	2.62%	-0.04%	-1.85%	0.28%	-1.07%	0.73%	2.26%	2.93%

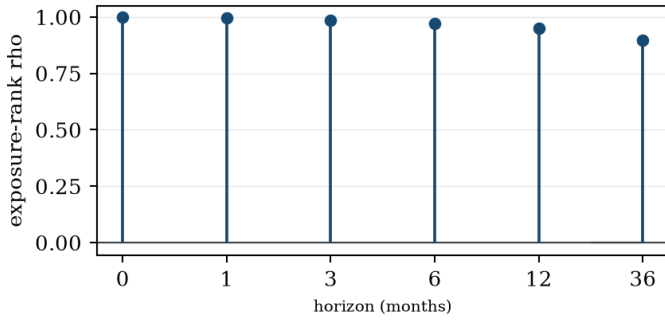


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

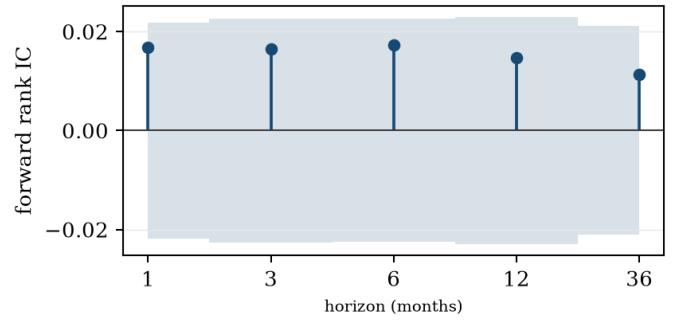


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	evs_ttm
Component signs	evs_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	15 / 66
Stocks, min / median / max	673 / 958 / 1577
Available-history Sharpe	0.46

46. value_proxy_1260d

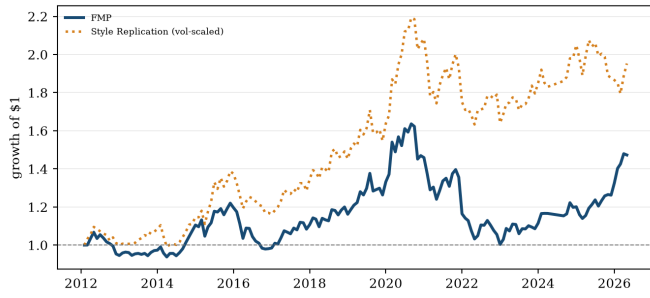


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
value_proxy_1260d	3.46%	11.58%	0.30	1.04	-38.66%	0.692	-1.11%	11.58%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
value_proxy_1260d	-0.01 (-0.36)	0.04 (0.57)	-0.58 (-5.06)	0.29 (3.30)	0.10 (1.53)	0.23 (3.09)	-0.90% (-0.40)	0.692

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
value_proxy_1260d	-0.20%	-0.08%	1.76%	1.83%	0.60%	0.45%	-0.90%	3.46%

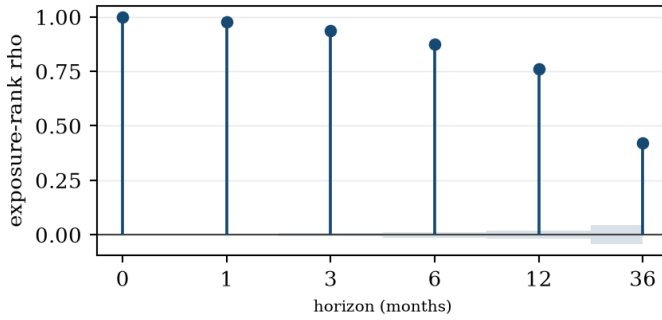


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

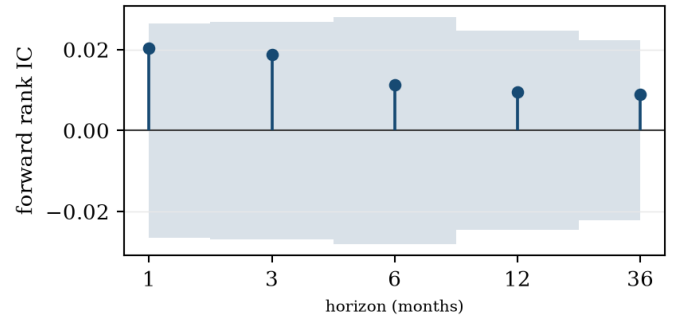


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	value_proxy_1260d
Component signs	value_proxy_1260d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	680 / 933 / 1479
Available-history Sharpe	0.10

47. accruals

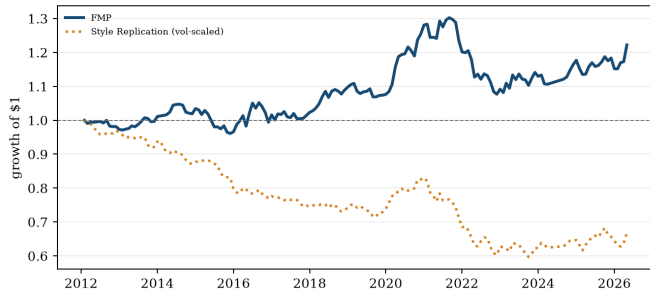


Figure: Common-window cumulative growth of \$1.

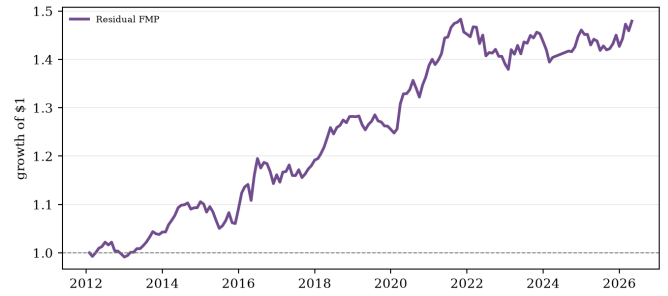


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals	1.58%	5.33%	0.30	1.07	-17.32%	0.362	3.24%	5.33%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals	-0.04 (-1.27)	0.02 (0.42)	-0.16 (-4.51)	-0.06 (-1.63)	-0.06 (-1.51)	-0.24 (-4.84)	2.91% (2.52)	0.365

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals	-0.56%	-0.04%	0.48%	-0.38%	-0.35%	-0.47%	2.91%	1.58%

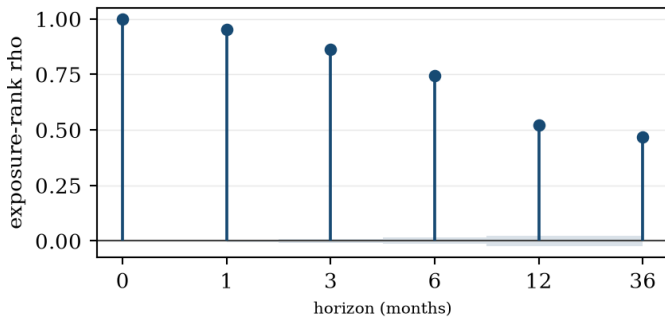


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

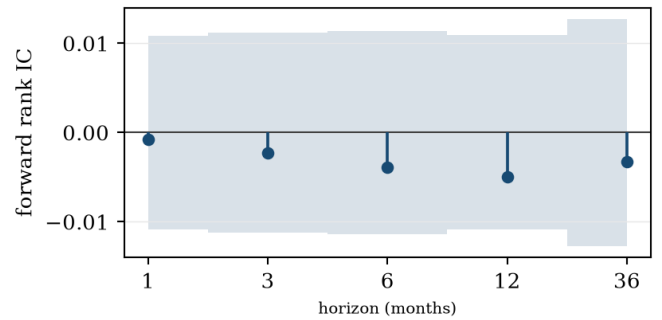


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	accruals
Component signs	accruals(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	22 / 66
Stocks, min / median / max	711 / 1046 / 1611
Available-history Sharpe	0.45

48. working_capital_proxy

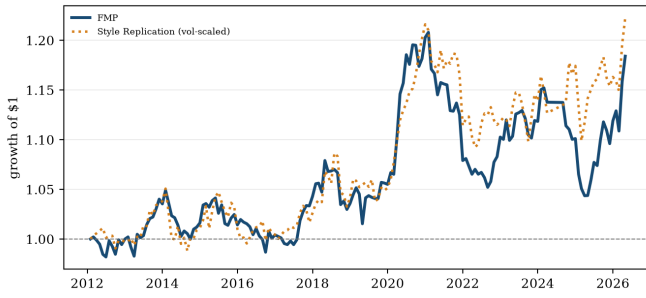


Figure: Common-window cumulative growth of \$1.

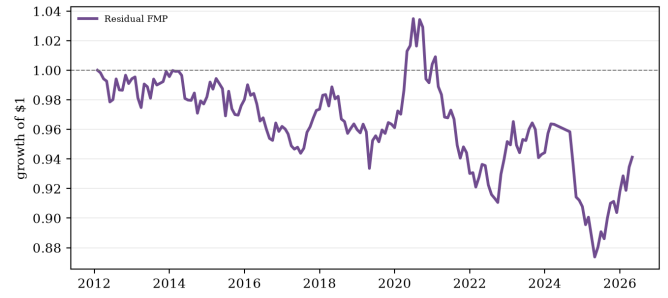


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
working_capital_proxy	1.31%	4.51%	0.29	0.98	-13.61%	0.360	0.39%	4.51%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
working_capital_proxy	0.11 (4.13)	0.16 (4.58)	-0.07 (-1.92)	0.09 (2.15)	-0.06 (-1.17)	0.00 (0.09)	-0.37% (-0.39)	0.392

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
working_capital_proxy	1.59%	-0.29%	0.21%	0.54%	-0.37%	0.01%	-0.37%	1.31%

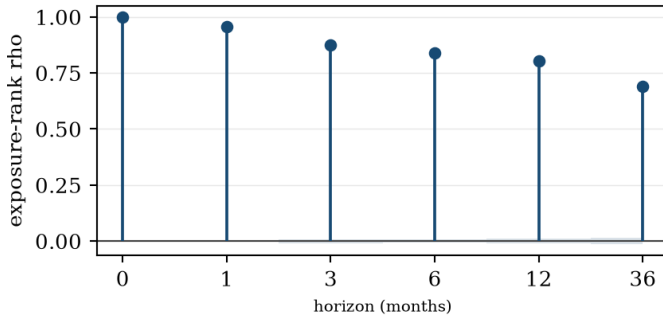


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

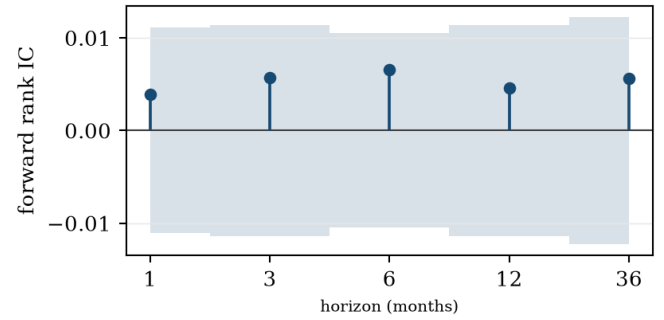


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	working_capital_proxy
Component signs	working_capital_proxy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	737 / 1058 / 1653
Available-history Sharpe	0.25

49. momentum_21_756

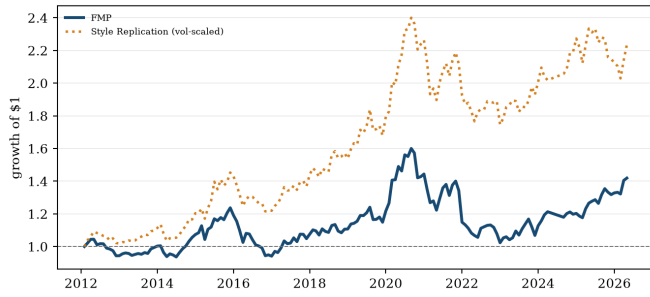


Figure: Common-window cumulative growth of \$1.

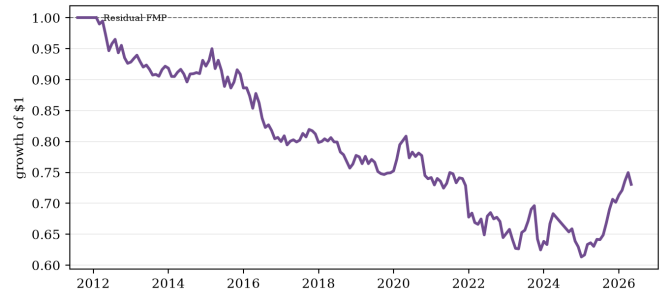


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_756	3.17%	11.41%	0.28	0.95	-36.06%	0.664	-2.08%	11.41%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_756	0.04 (0.79)	-0.10 (-1.22)	-0.59 (-6.90)	0.24 (3.67)	0.17 (2.22)	0.13 (1.68)	-2.03% (-1.01)	0.664

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_756	0.52%	0.19%	1.78%	1.49%	0.98%	0.24%	-2.03%	3.17%

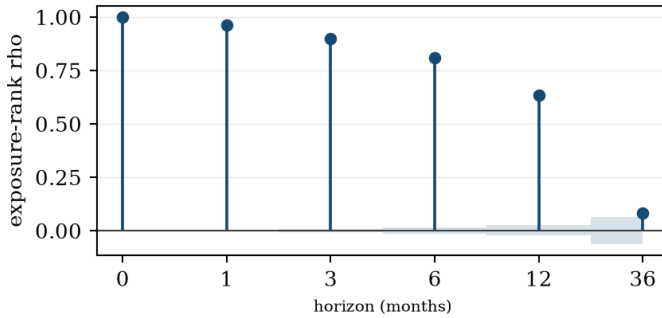


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

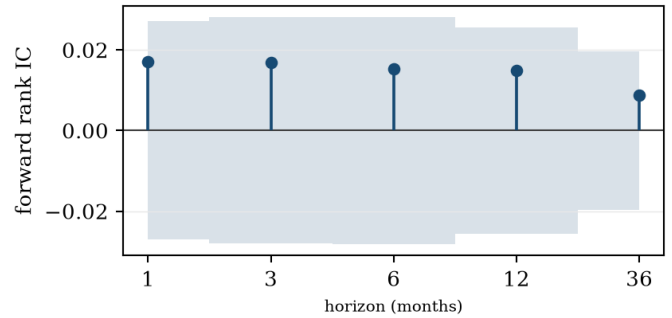


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	momentum_21_756
Component signs	momentum_21_756(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	707 / 982 / 1565
Available-history Sharpe	0.12

50. dio

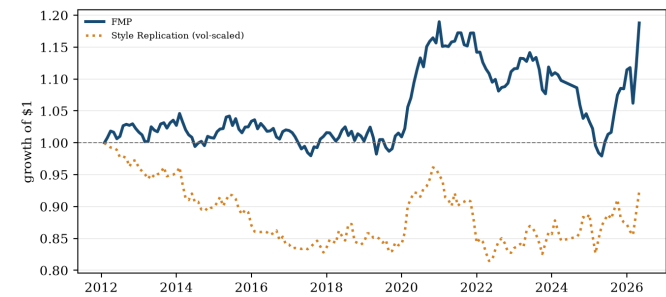


Figure: Common-window cumulative growth of \$1.

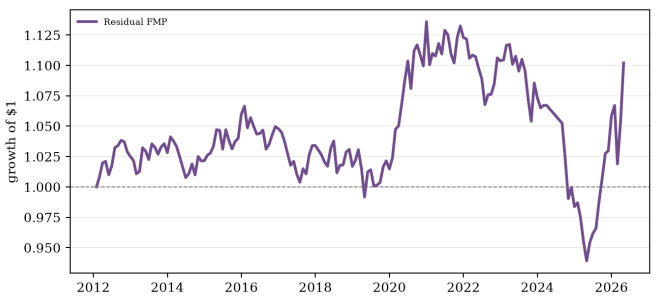


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dio	1.36%	5.00%	0.27	0.93	-17.69%	0.143	1.53%	5.00%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dio	0.06 (1.52)	0.07 (1.37)	-0.07 (-1.67)	0.05 (0.83)	-0.10 (-1.40)	-0.02 (-0.31)	0.80% (0.59)	0.162

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dio	0.82%	-0.13%	0.21%	0.29%	-0.58%	-0.04%	0.80%	1.36%

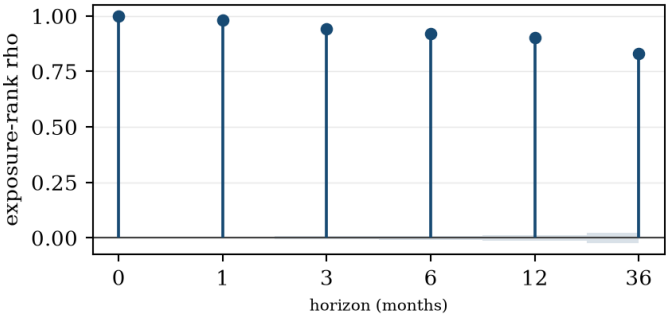


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

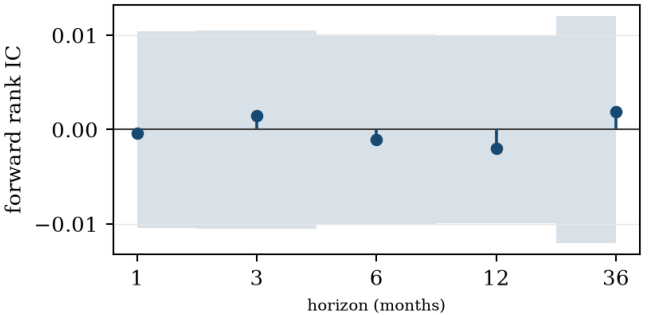


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	dio
Component signs	dio(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	12 / 66
Stocks, min / median / max	668 / 888 / 1126
Available-history Sharpe	0.08

51. ps_ttm

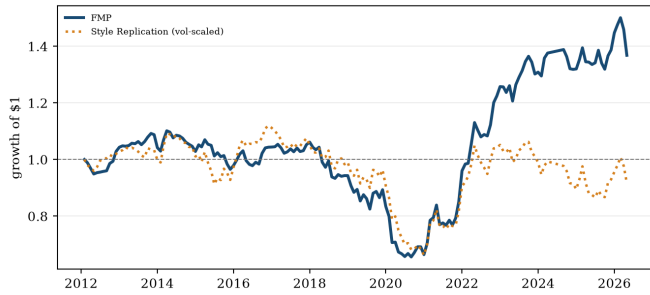


Figure: Common-window cumulative growth of \$1.

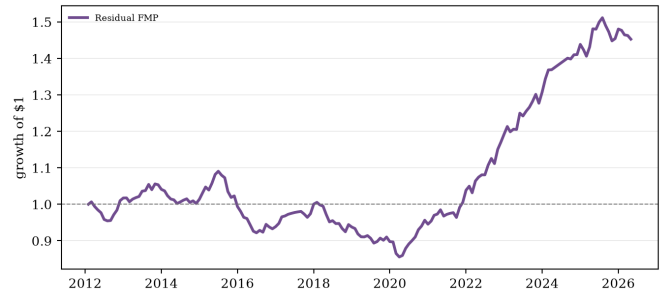


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ps_ttm	2.83%	10.78%	0.26	0.83	-40.45%	0.796	2.88%	10.78%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ps_ttm	0.17 (5.04)	-0.06 (-0.90)	0.73 (13.19)	-0.01 (-0.35)	-0.13 (-1.85)	0.27 (4.43)	2.80% (1.47)	0.796

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ps_ttm	2.42%	0.11%	-2.18%	-0.09%	-0.75%	0.53%	2.80%	2.83%

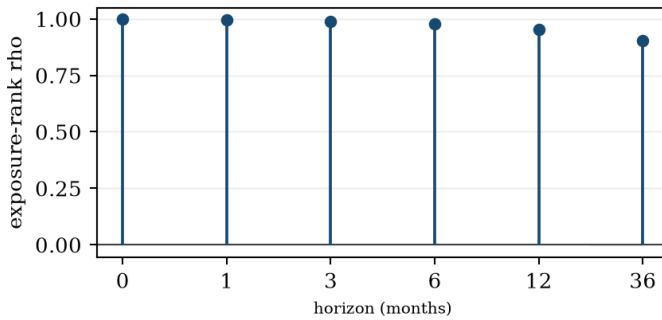


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

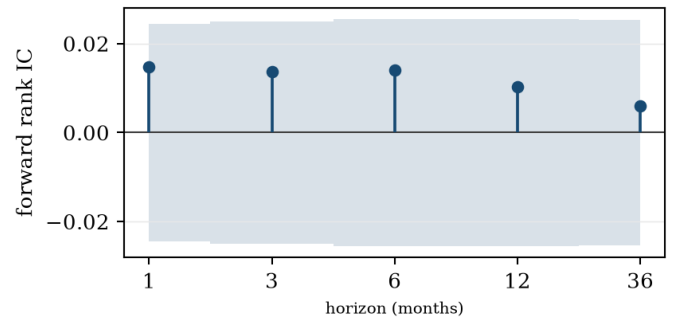


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	ps_ttm
Component signs	ps_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	22 / 66
Stocks, min / median / max	729 / 1048 / 1623
Available-history Sharpe	0.33

52. rd_to_revenue

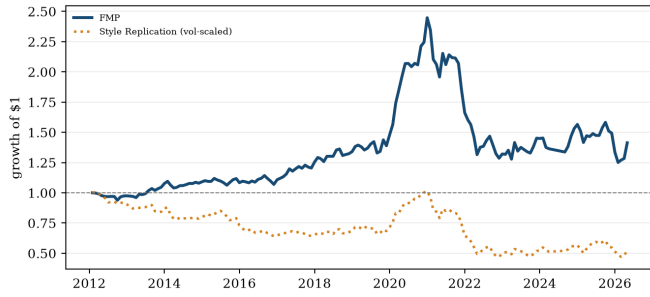


Figure: Common-window cumulative growth of \$1.

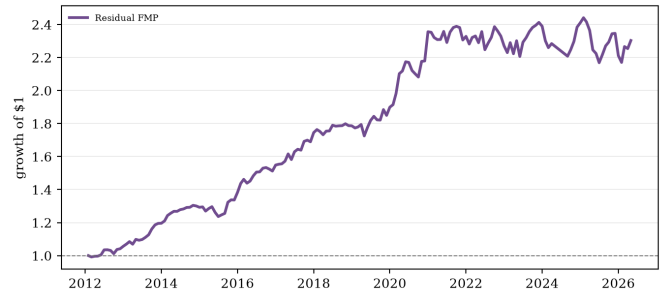


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
rd_to_revenue	3.33%	12.97%	0.26	0.83	-48.91%	0.690	6.46%	12.97%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
rd_to_revenue	-0.16 (-2.84)	0.11 (1.53)	-0.85 (-11.74)	-0.26 (-3.39)	-0.08 (-0.71)	-0.47 (-4.02)	6.27% (3.17)	0.690

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
rd_to_revenue	-2.25%	-0.21%	2.55%	-1.67%	-0.47%	-0.90%	6.27%	3.33%

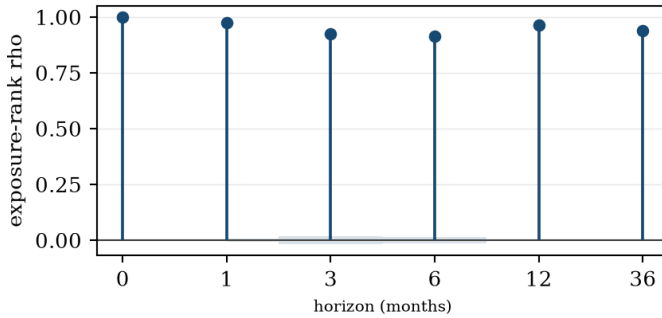


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

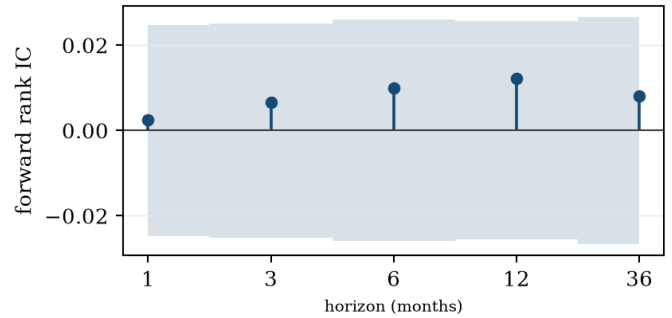


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	rd_to_revenue
Component signs	rd_to_revenue(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	51 / 76
Stocks, min / median / max	430 / 689 / 992
Available-history Sharpe	0.28

53. net_margin

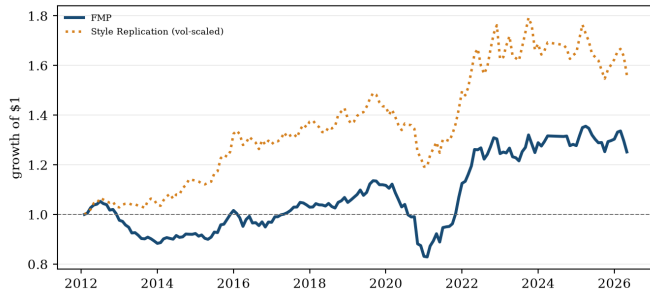


Figure: Common-window cumulative growth of \$1.

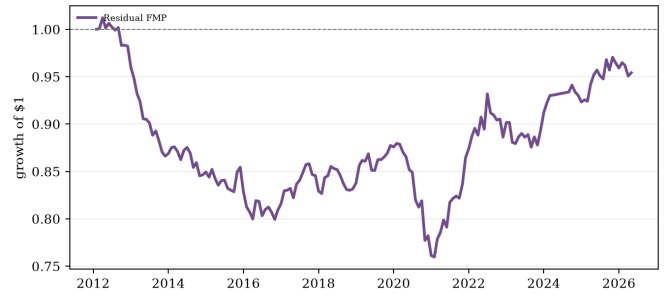


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_margin	1.90%	7.59%	0.25	0.77	-27.02%	0.643	-0.90%	7.59%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_margin	0.01 (0.19)	-0.22 (-5.08)	0.16 (3.70)	0.14 (3.20)	0.07 (1.31)	0.43 (5.88)	-0.24% (-0.18)	0.648

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_margin	0.10%	0.41%	-0.49%	0.89%	0.40%	0.83%	-0.24%	1.90%

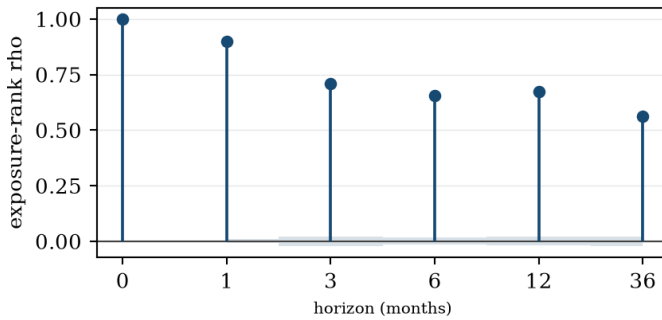


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

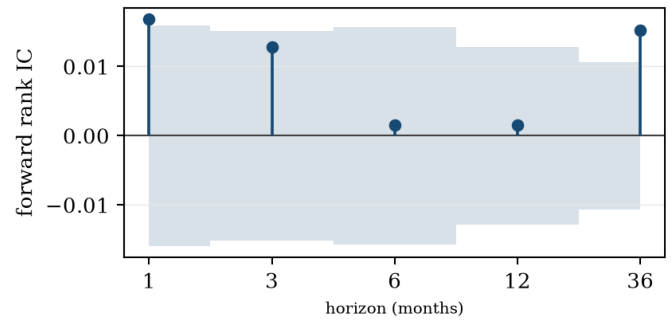


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	net_margin
Component signs	net_margin(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	745 / 1050 / 1650
Available-history Sharpe	0.20

54. BAB composite

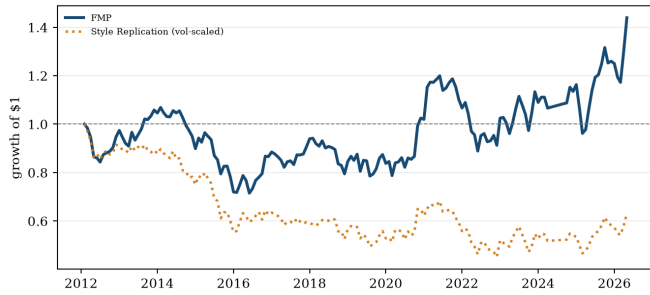


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
BAB composite	3.76%	15.23%	0.25	0.92	-33.10%	0.816	5.71%	15.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
BAB composite	0.46 (10.88)	0.42 (5.21)	-0.02 (-0.40)	-0.19 (-3.06)	-0.70 (-8.18)	-0.38 (-5.45)	3.97% (2.00)	0.822

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
BAB composite	6.51%	-0.79%	0.06%	-1.23%	-4.02%	-0.73%	3.97%	3.76%

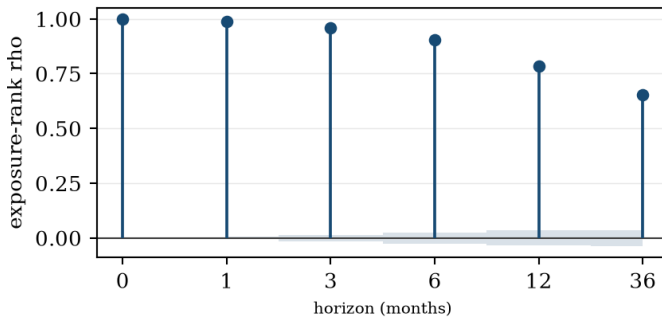


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

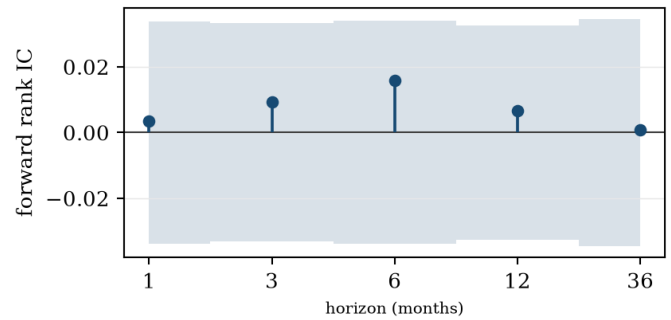


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	bab_252v_1260c_SPY, bab_shrunk_252v_1260c_SPY, bab_252v_1260c_VTI, bab_shrunk_252v_1260c_VTI, bab_252v_1260c_FRENCH_CRSP, bab_shrunk_252v_1260c_FRENCH_CRSP
Component signs	bab_252v_1260c_SPY(+1), bab_shrunk_252v_1260c_SPY(+1), bab_252v_1260c_VTI(+1), bab_shrunk_252v_1260c_VTI(+1), bab_252v_1260c_FRENCH_CRSP(+1), bab_shrunk_252v_1260c_FRENCH_CRSP(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 6
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	707 / 982 / 1565
Available-history Sharpe	0.14

55. revenue_growth_yoy

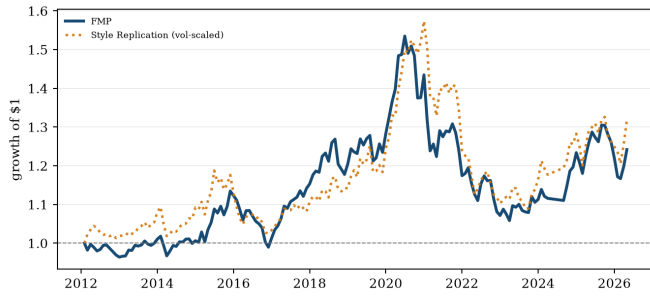


Figure: Common-window cumulative growth of \$1.

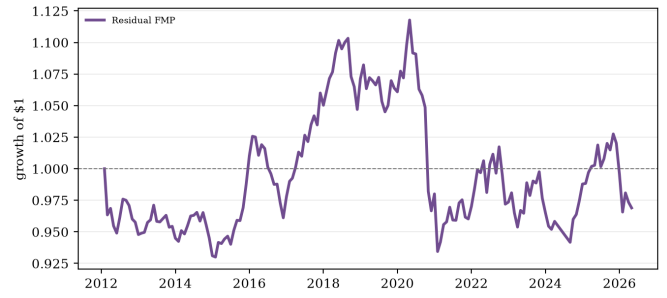


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
revenue_growth_yoy	1.89%	8.15%	0.23	0.81	-31.05%	0.657	0.01%	8.15%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
revenue_growth_yoy	-0.03 (-0.78)	-0.01 (-0.11)	-0.44 (-10.46)	0.08 (1.84)	0.17 (3.65)	-0.23 (-4.83)	-0.11% (-0.07)	0.657

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
revenue_growth_yoy	-0.40%	0.01%	1.32%	0.54%	0.98%	-0.44%	-0.11%	1.89%

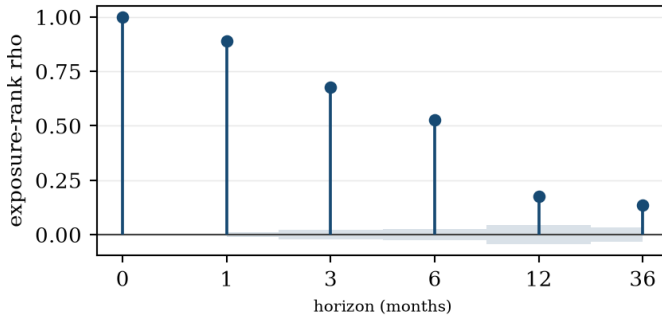


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

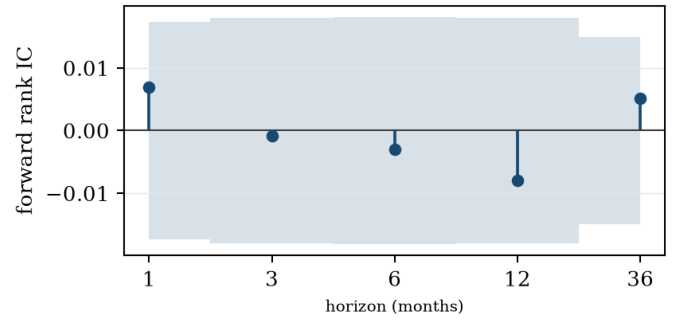


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	revenue_growth_yoy
Component signs	revenue_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	237 / 167
Available window	2006-11-01 to 2026-07-01
Dropped: low breadth / common	54 / 70
Stocks, min / median / max	741 / 1046 / 1646
Available-history Sharpe	0.16

56. growth_roa_5y

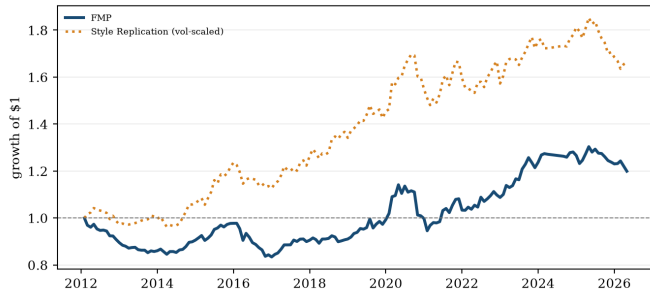


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_roa_5y	1.53%	6.72%	0.23	0.76	-17.14%	0.471	-1.13%	6.72%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_roa_5y	0.04 (1.04)	-0.04 (-0.87)	-0.28 (-7.21)	0.08 (1.63)	-0.04 (-0.66)	0.33 (5.98)	-0.88% (-0.58)	0.472

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_roa_5y	0.59%	0.07%	0.83%	0.50%	-0.21%	0.64%	-0.88%	1.53%

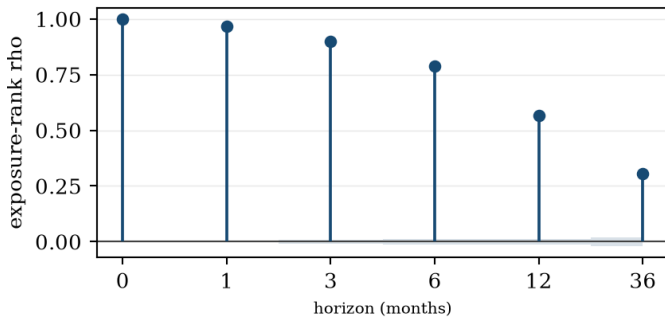


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

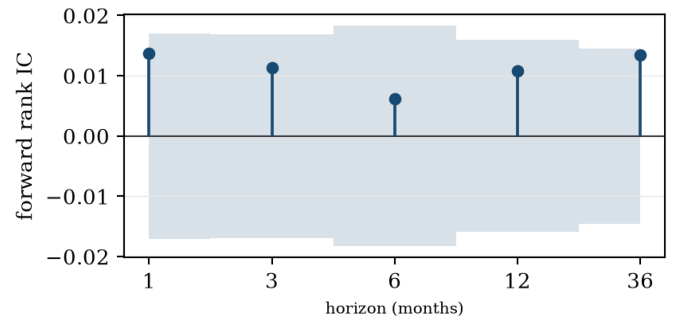


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	growth_roa_5y
Component signs	growth_roa_5y(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	173 / 167
Available window	2012-03-01 to 2026-07-01
Dropped: low breadth / common	16 / 6
Stocks, min / median / max	57 / 953 / 1513
Available-history Sharpe	0.22

57. Beta composite

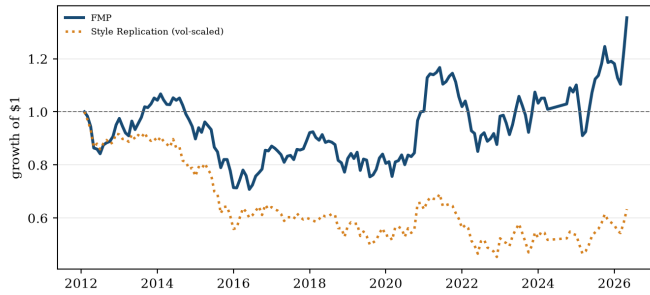


Figure: Common-window cumulative growth of \$1.

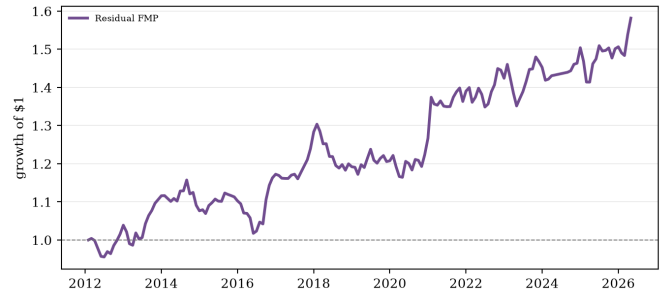


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Beta composite	3.32%	15.25%	0.22	0.81	-33.68%	0.815	5.26%	15.25%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Beta composite	0.45 (10.82)	0.44 (5.20)	-0.02 (-0.37)	-0.19 (-2.92)	-0.68 (-7.88)	-0.40 (-5.59)	3.50% (1.77)	0.821

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Beta composite	6.43%	-0.81%	0.06%	-1.19%	-3.89%	-0.77%	3.50%	3.32%

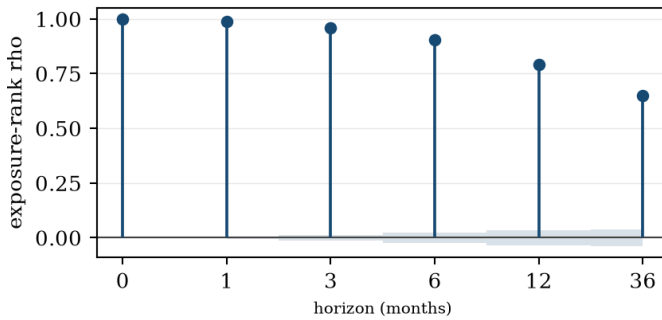


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

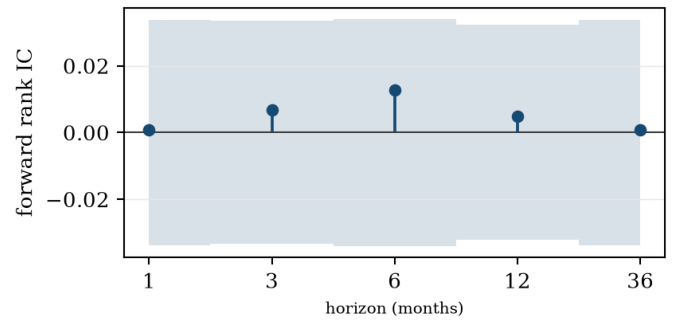


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	beta_252v_1260c_SPY(-1), beta_shrunk_252v_1260c_SPY(-1), beta_stability_252d_252v_1260c_SPY(-1), beta_252v_1260c_VTI(-1), beta_shrunk_252v_1260c_VTI(-1), beta_252v_1260c_FRENCH_CRSP(-1), beta_shrunk_252v_1260c_FRENCH_CRSP(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 7
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	707 / 982 / 1565
Available-history Sharpe	0.12

58. is_negative_ev

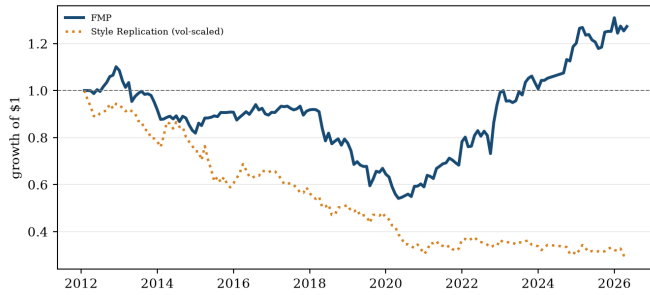


Figure: Common-window cumulative growth of \$1.

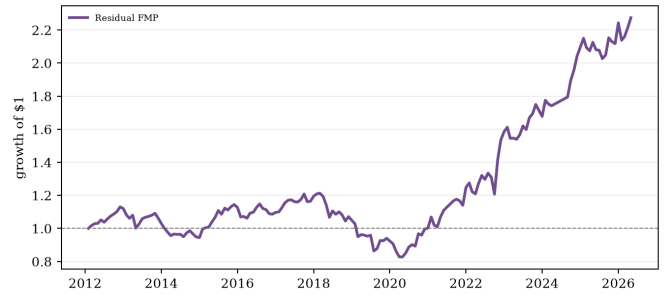


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_negative_ev	2.52%	12.71%	0.20	0.75	-50.87%	0.168	5.81%	12.71%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_negative_ev	-0.12 (-1.95)	-0.25 (-1.92)	0.21 (2.13)	-0.23 (-2.17)	-0.09 (-0.72)	-0.11 (-1.07)	6.57% (1.84)	0.172

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_negative_ev	-1.66%	0.46%	-0.63%	-1.49%	-0.51%	-0.22%	6.57%	2.52%

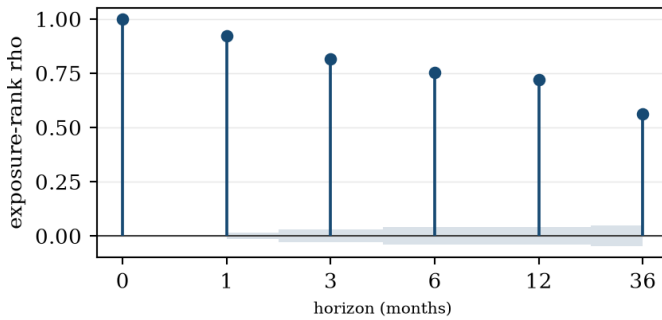


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

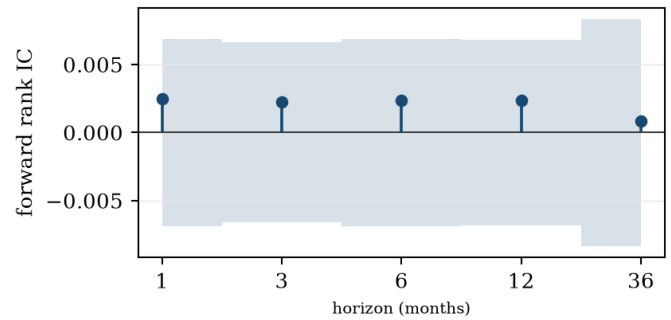


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	is_negative_ev
Component signs	is_negative_ev(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	50 / 76
Stocks, min / median / max	699 / 980 / 1649
Available-history Sharpe	0.27

59. growth_acc_5y

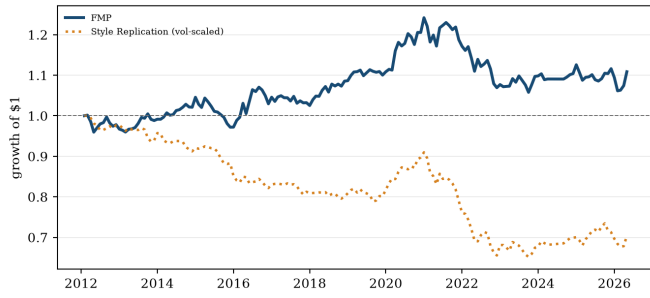


Figure: Common-window cumulative growth of \$1.

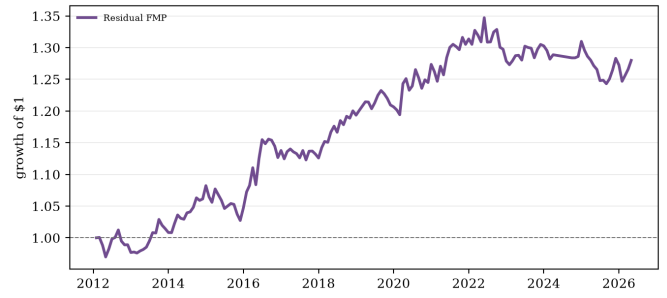


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_acc_5y	0.85%	4.78%	0.18	0.73	-14.82%	0.279	2.12%	4.78%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_acc_5y	-0.04 (-1.21)	-0.02 (-0.58)	-0.16 (-4.14)	-0.10 (-2.66)	0.02 (0.48)	-0.22 (-4.70)	1.86% (1.77)	0.281

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_acc_5y	-0.62%	0.04%	0.48%	-0.61%	0.13%	-0.42%	1.86%	0.85%

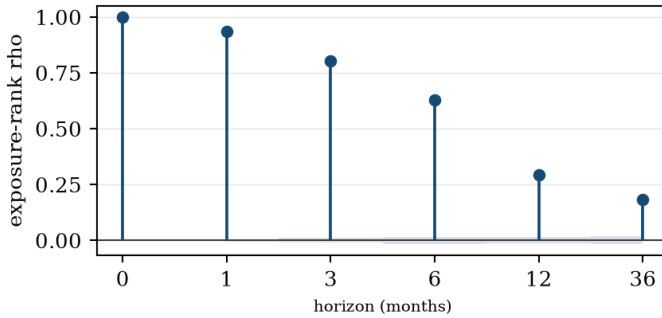


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

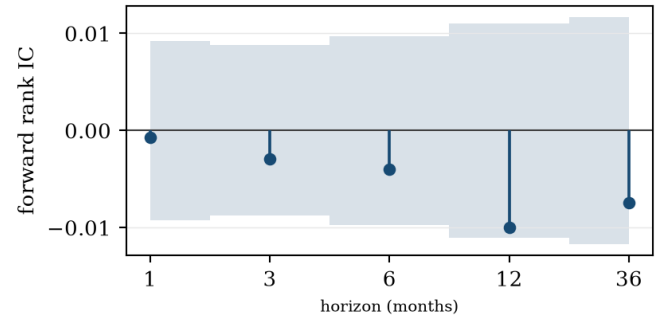


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	growth_acc_5y
Component signs	growth_acc_5y(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	173 / 167
Available window	2012-03-01 to 2026-07-01
Dropped: low breadth / common	16 / 6
Stocks, min / median / max	51 / 932 / 1484
Available-history Sharpe	0.15

60. momentum_0_21

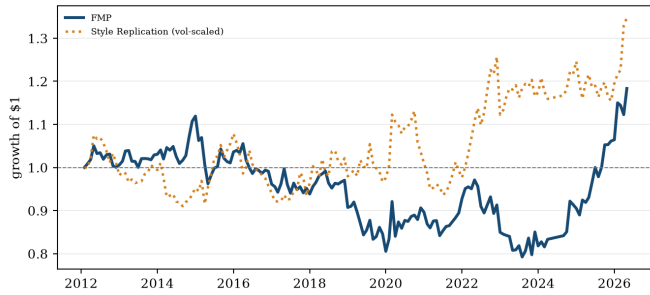


Figure: Common-window cumulative growth of \$1.

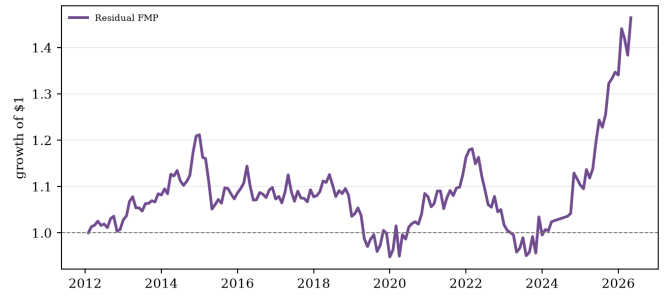


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_0_21	1.66%	9.48%	0.17	0.75	-29.20%	0.112	0.79%	9.48%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_0_21	-0.16 (-2.51)	0.05 (0.53)	0.01 (0.12)	0.21 (2.87)	-0.11 (-1.14)	0.12 (1.24)	3.12% (1.12)	0.158

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_0_21	-2.26%	-0.10%	-0.03%	1.34%	-0.65%	0.23%	3.12%	1.66%

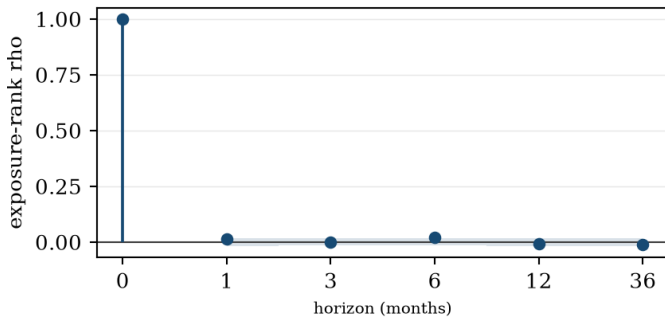


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

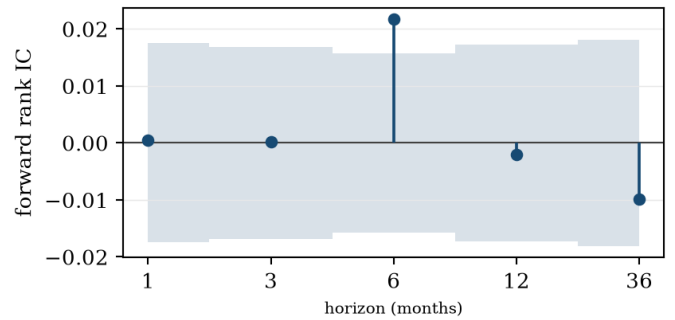


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	momentum_0_21
Component signs	momentum_0_21(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	749 / 1060 / 1676
Available-history Sharpe	0.06

61. debt_issuance_yoy

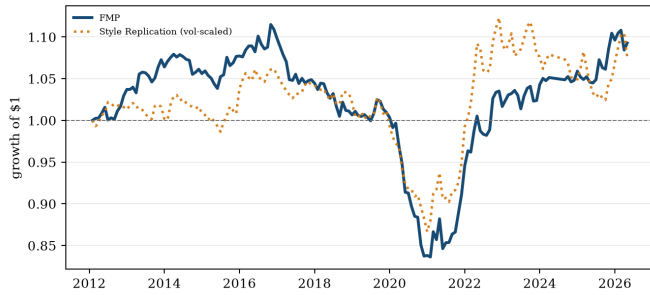


Figure: Common-window cumulative growth of \$1.

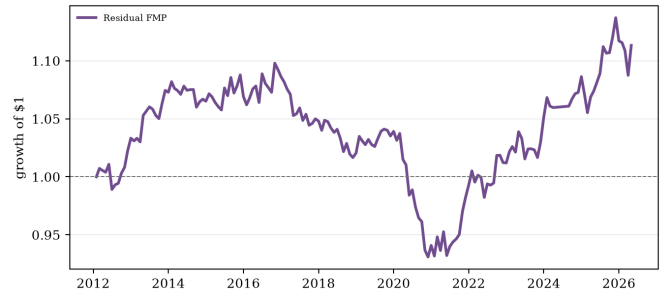


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_issuance_yoy	0.73%	4.30%	0.17	0.53	-24.98%	0.322	0.38%	4.30%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_issuance_yoy	-0.01 (-0.29)	-0.03 (-0.66)	0.19 (4.49)	0.10 (2.27)	-0.06 (-1.53)	0.13 (2.95)	0.83% (0.87)	0.332

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_issuance_yoy	-0.10%	0.05%	-0.57%	0.61%	-0.34%	0.25%	0.83%	0.73%

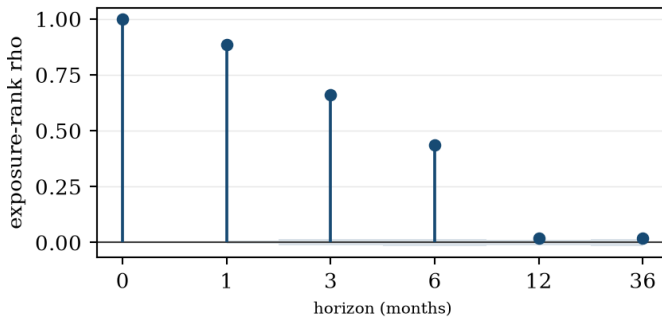


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

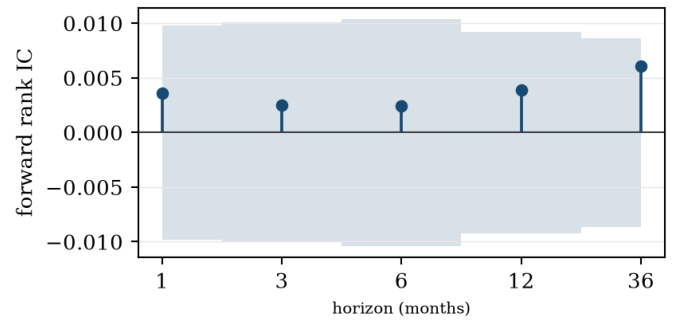


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	debt_issuance_yoy
Component signs	debt_issuance_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	236 / 167
Available window	2006-12-01 to 2026-07-01
Dropped: low breadth / common	54 / 69
Stocks, min / median / max	683 / 953 / 1615
Available-history Sharpe	0.36

62. skew_63d

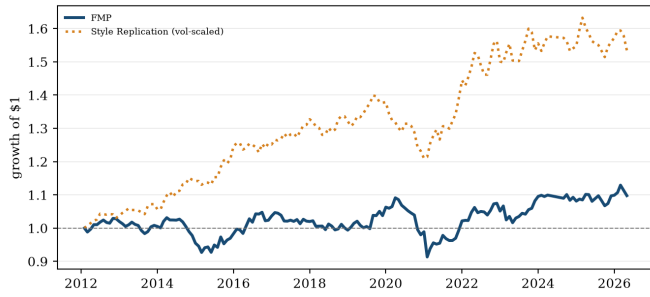


Figure: Common-window cumulative growth of \$1.

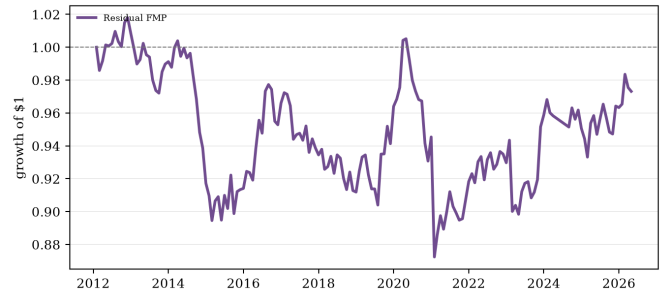


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
skew_63d	0.80%	5.07%	0.16	0.59	-16.29%	0.144	-0.42%	5.07%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
skew_63d	0.02 (0.72)	-0.10 (-1.89)	0.06 (1.51)	0.02 (0.69)	0.05 (0.81)	0.11 (2.44)	-0.09% (-0.06)	0.147

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
skew_63d	0.27%	0.19%	-0.19%	0.14%	0.26%	0.20%	-0.09%	0.80%

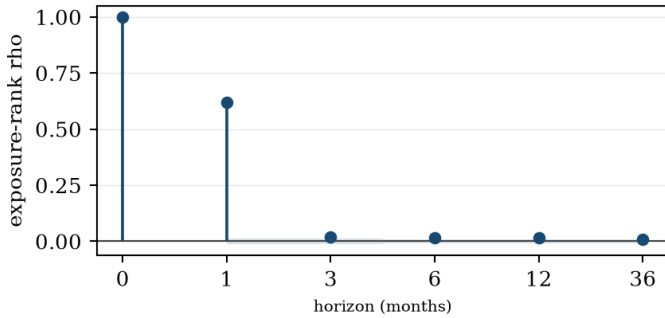


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

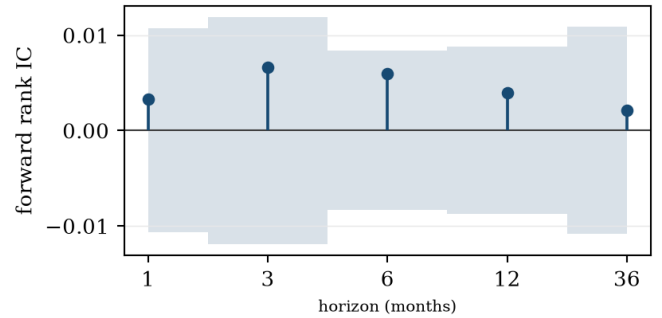


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	skew_63d
Component signs	skew_63d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	749 / 1058 / 1674
Available-history Sharpe	0.11

63. growth_roe_5y

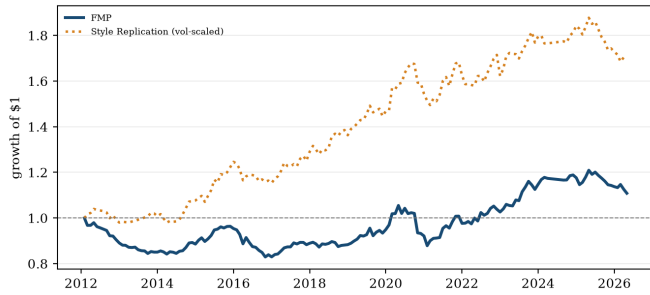


Figure: Common-window cumulative growth of \$1.

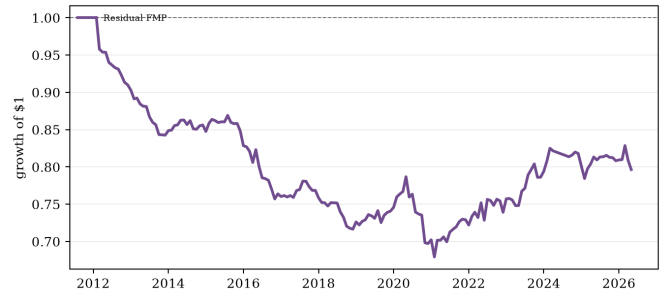


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_roe_5y	0.92%	6.09%	0.15	0.50	-16.70%	0.430	-1.71%	6.09%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_roe_5y	0.06 (1.51)	-0.02 (-0.57)	-0.22 (-6.12)	0.08 (1.79)	-0.03 (-0.65)	0.32 (6.36)	-1.53% (-1.11)	0.430

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_roe_5y	0.82%	0.04%	0.66%	0.49%	-0.18%	0.62%	-1.53%	0.92%

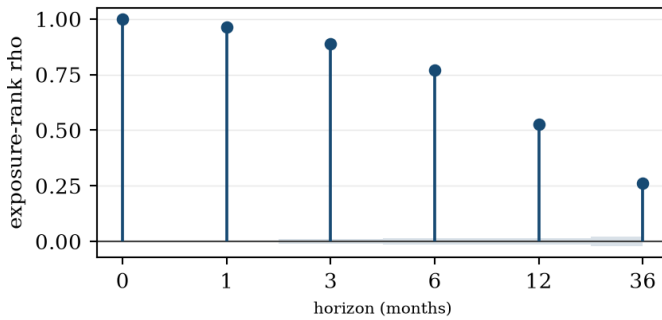


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

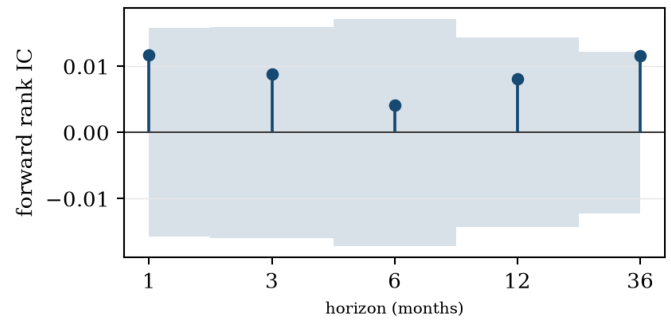


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	growth_roe_5y
Component signs	growth_roe_5y(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	173 / 167
Available window	2012-03-01 to 2026-07-01
Dropped: low breadth / common	16 / 6
Stocks, min / median / max	57 / 952 / 1517
Available-history Sharpe	0.17

64. is_negative_equity

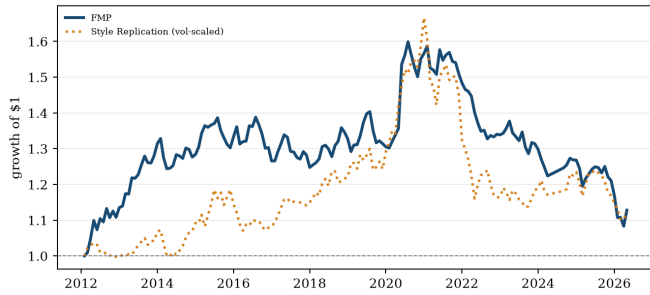


Figure: Common-window cumulative growth of \$1.

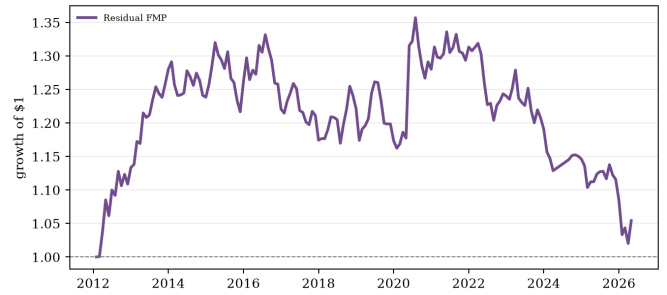


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_negative_equity	1.15%	7.59%	0.15	0.52	-32.27%	0.135	0.73%	7.59%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_negative_equity	-0.02 (-0.58)	0.09 (1.34)	-0.22 (-3.21)	-0.04 (-0.58)	0.12 (1.62)	-0.04 (-0.66)	0.62% (0.31)	0.135

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_negative_equity	-0.34%	-0.16%	0.67%	-0.27%	0.70%	-0.09%	0.62%	1.15%

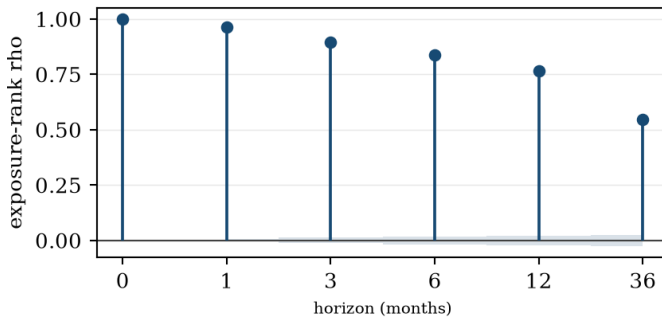


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

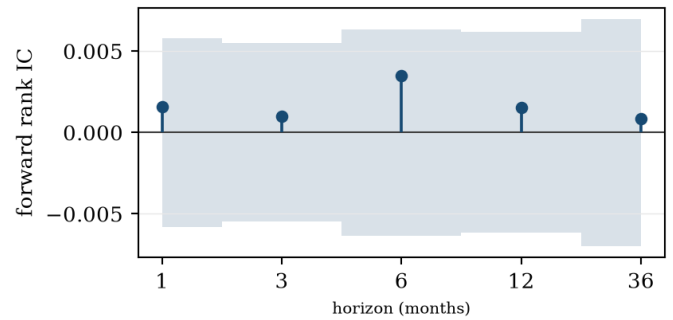


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	is_negative_equity
Component signs	is_negative_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	744 / 1059 / 1677
Available-history Sharpe	0.25

65. roe_volatility

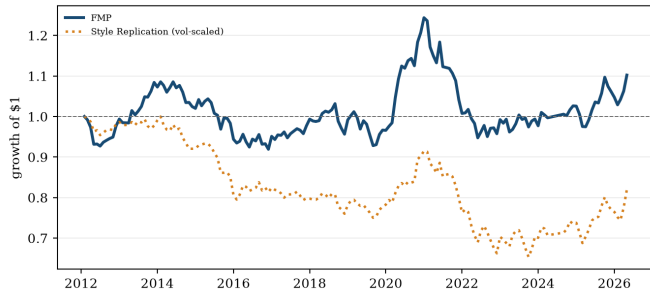


Figure: Common-window cumulative growth of \$1.

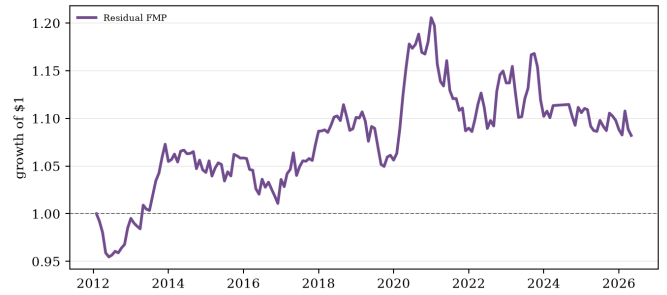


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
roe_volatility	0.94%	6.98%	0.13	0.47	-23.82%	0.586	1.85%	6.98%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
roe_volatility	0.11 (2.88)	0.10 (2.56)	-0.11 (-2.29)	-0.04 (-0.76)	-0.11 (-1.60)	-0.35 (-4.92)	0.66% (0.54)	0.604

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
roe_volatility	1.62%	-0.18%	0.34%	-0.23%	-0.61%	-0.67%	0.66%	0.94%

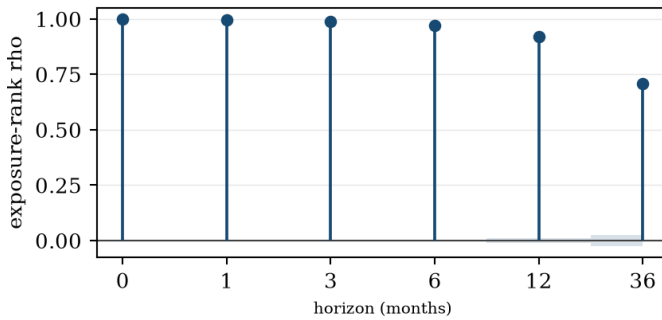


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

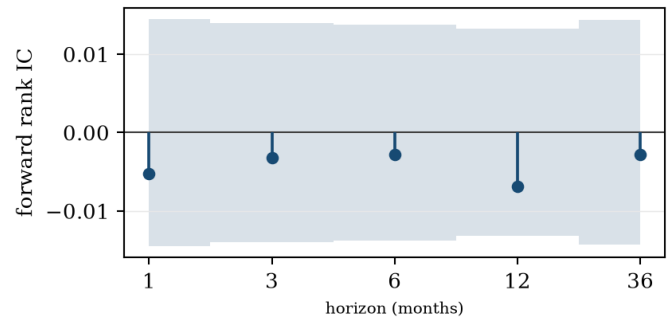


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	roe_volatility
Component signs	roe_volatility(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	212 / 167
Available window	2008-12-01 to 2026-07-01
Dropped: low breadth / common	16 / 45
Stocks, min / median / max	690 / 978 / 1518
Available-history Sharpe	0.15

66. operating_margin

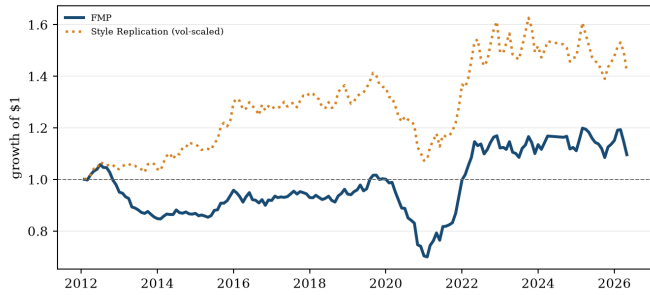


Figure: Common-window cumulative growth of \$1.

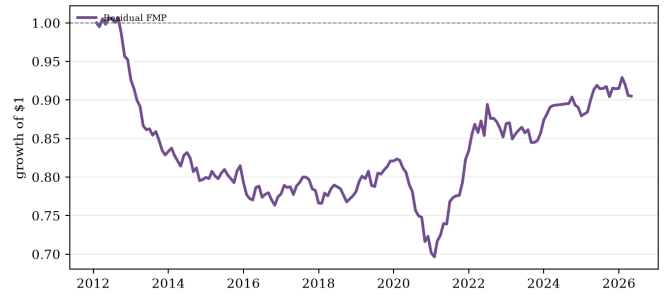


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
operating_margin	1.00%	8.27%	0.12	0.35	-33.85%	0.657	-1.31%	8.27%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
operating_margin	-0.02 (-0.56)	-0.29 (-6.36)	0.26 (5.02)	0.13 (2.65)	0.11 (1.90)	0.37 (4.89)	-0.60% (-0.40)	0.662

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
operating_margin	-0.31%	0.54%	-0.77%	0.81%	0.62%	0.71%	-0.60%	1.00%

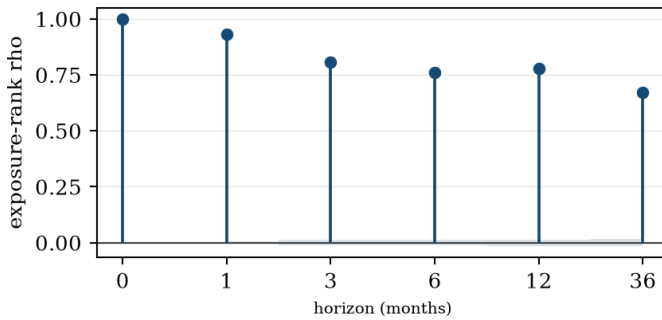


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

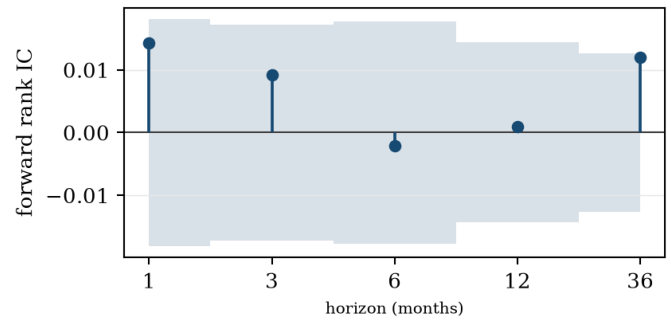


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	operating_margin
Component signs	operating_margin(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	745 / 1050 / 1649
Available-history Sharpe	0.14

67. debt_to_assets

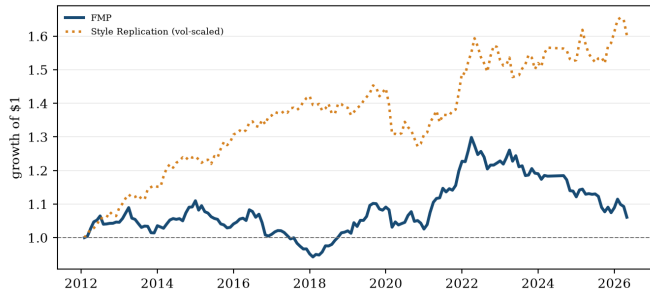


Figure: Common-window cumulative growth of \$1.

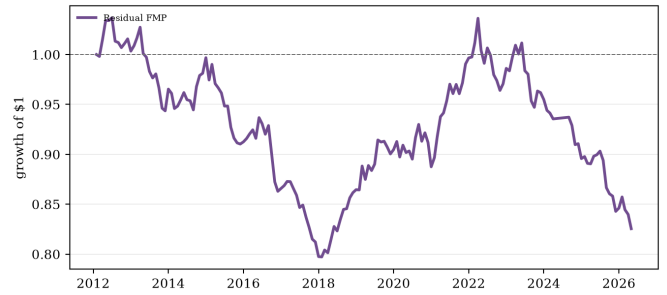


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_assets	0.55%	5.14%	0.11	0.36	-18.31%	0.216	-1.08%	5.14%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_assets	0.06 (1.88)	-0.11 (-2.00)	0.13 (3.58)	0.04 (0.80)	0.16 (2.52)	0.01 (0.10)	-1.27% (-0.92)	0.217

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_assets	0.90%	0.20%	-0.40%	0.22%	0.90%	0.01%	-1.27%	0.55%

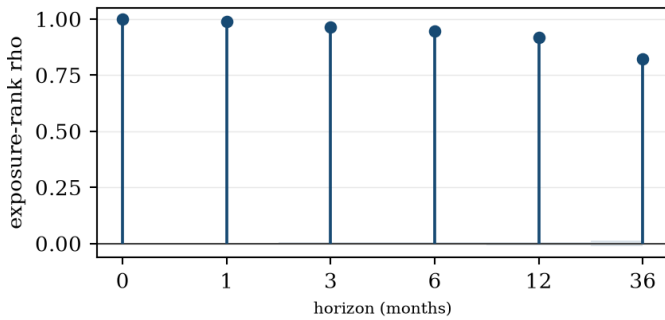


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

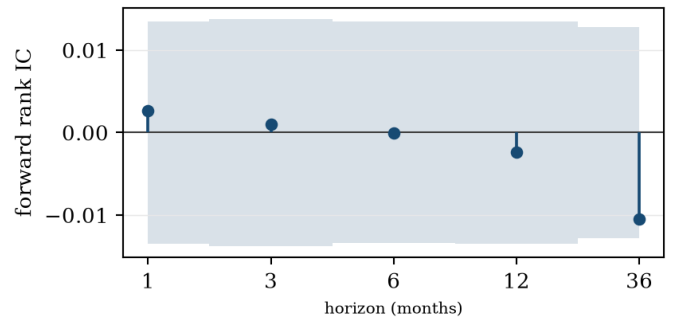


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	debt_to_assets
Component signs	debt_to_assets(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	50 / 76
Stocks, min / median / max	699 / 980 / 1650
Available-history Sharpe	0.22

68. growth_gpoa_5y

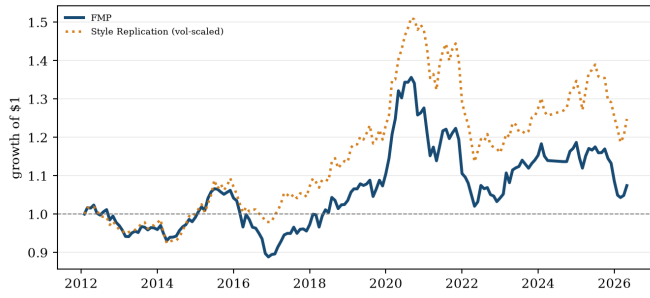


Figure: Common-window cumulative growth of \$1.

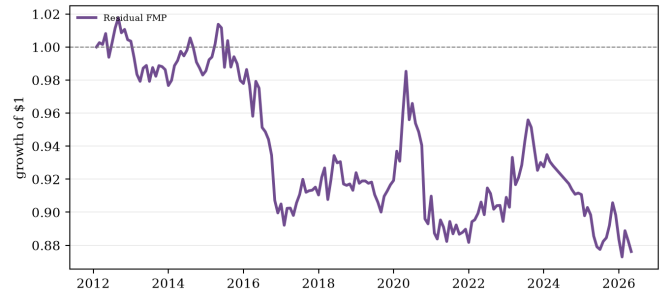


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_gpoa_5y	0.78%	7.27%	0.11	0.33	-24.76%	0.712	-0.79%	7.27%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_gpoa_5y	0.02 (0.75)	0.00 (0.02)	-0.54 (-20.31)	-0.06 (-2.03)	-0.00 (-0.04)	0.07 (1.92)	-0.87% (-0.77)	0.712

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_gpoa_5y	0.29%	-0.00%	1.64%	-0.40%	-0.01%	0.13%	-0.87%	0.78%

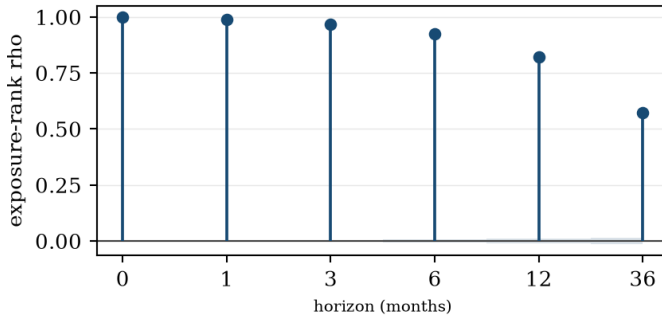


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

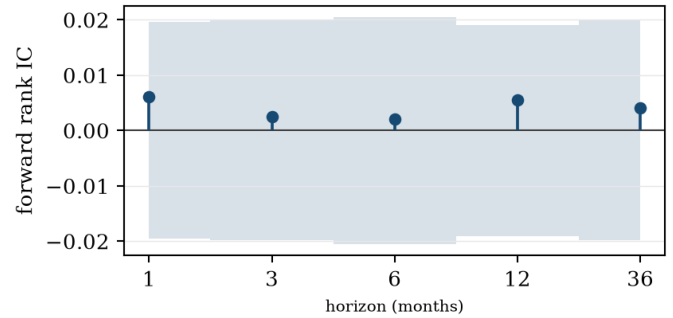


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	growth_gpoa_5y
Component signs	growth_gpoa_5y(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	173 / 167
Available window	2012-03-01 to 2026-07-01
Dropped: low breadth / common	16 / 6
Stocks, min / median / max	57 / 950 / 1477
Available-history Sharpe	0.10

69. asset_growth_yoy

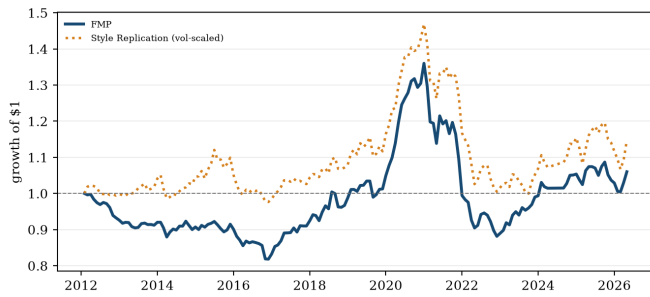


Figure: Common-window cumulative growth of \$1.

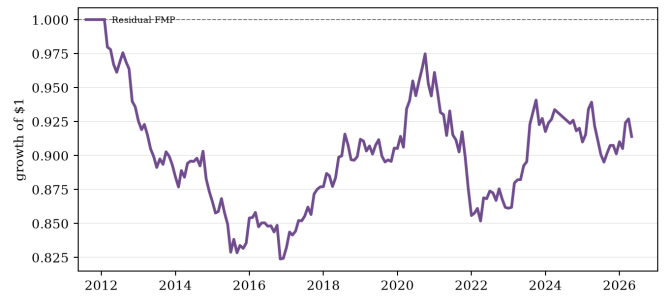


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
asset_growth_yoy	0.70%	7.50%	0.09	0.28	-35.20%	0.729	-0.38%	7.50%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
asset_growth_yoy	-0.01 (-0.48)	0.02 (0.48)	-0.49 (-12.30)	-0.05 (-1.72)	0.15 (3.06)	-0.25 (-5.06)	-0.57% (-0.55)	0.730

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
asset_growth_yoy	-0.20%	-0.04%	1.47%	-0.32%	0.83%	-0.48%	-0.57%	0.70%

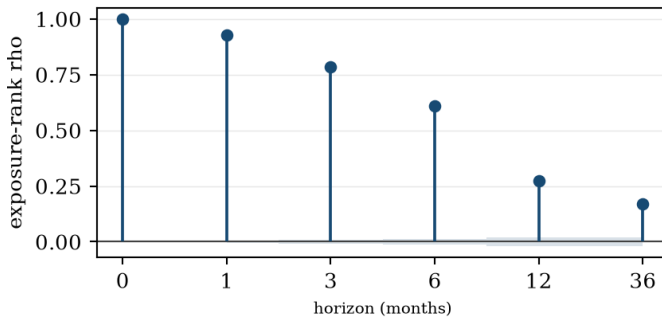


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

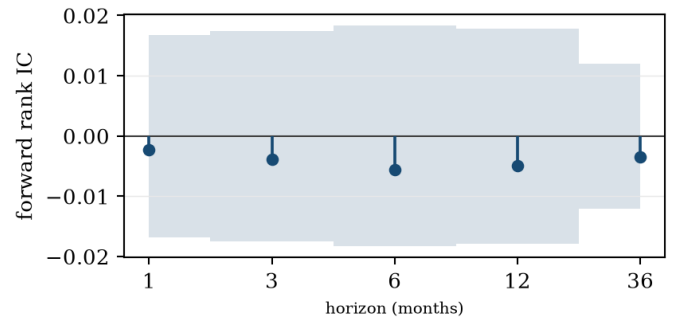


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	asset_growth_yoy
Component signs	asset_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	237 / 167
Available window	2006-11-01 to 2026-07-01
Dropped: low breadth / common	54 / 70
Stocks, min / median / max	741 / 1047 / 1664
Available-history Sharpe	-0.02

70. growth_gmar_5y

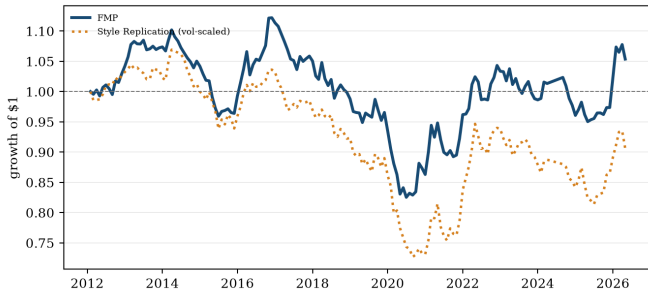


Figure: Common-window cumulative growth of \$1.

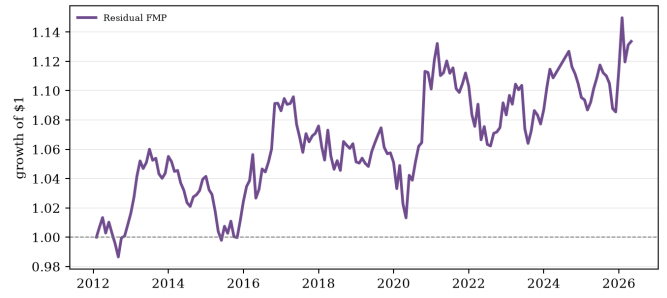


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_gmar_5y	0.56%	6.17%	0.09	0.29	-26.44%	0.607	0.97%	6.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_gmar_5y	0.04 (1.58)	0.03 (0.72)	0.45 (12.53)	0.12 (2.98)	-0.08 (-1.77)	0.04 (1.05)	0.98% (0.85)	0.607

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_gmar_5y	0.60%	-0.05%	-1.37%	0.77%	-0.44%	0.08%	0.98%	0.56%

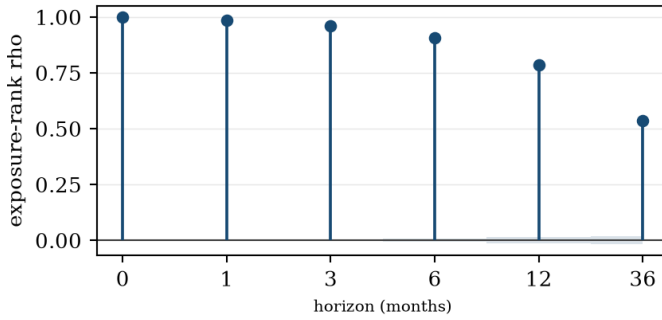


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

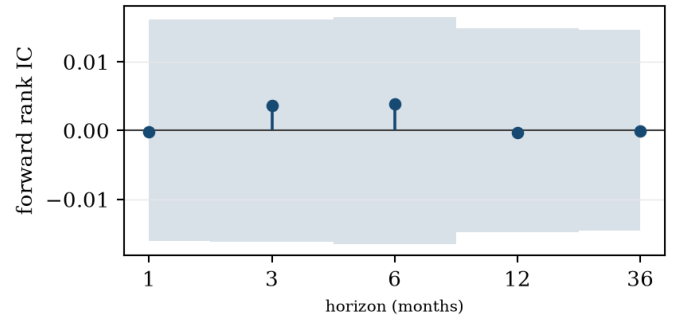


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	growth_gmar_5y
Component signs	growth_gmar_5y(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	173 / 167
Available window	2012-03-01 to 2026-07-01
Dropped: low breadth / common	16 / 6
Stocks, min / median / max	57 / 954 / 1476
Available-history Sharpe	0.04

71. ev

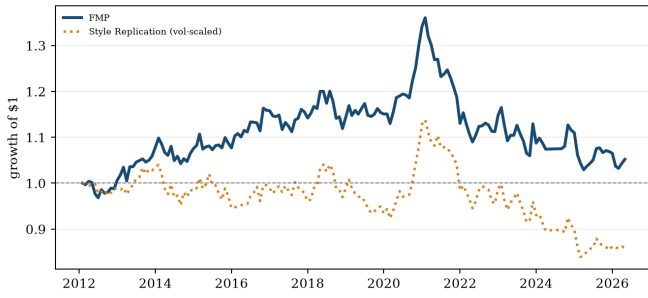


Figure: Common-window cumulative growth of \$1.

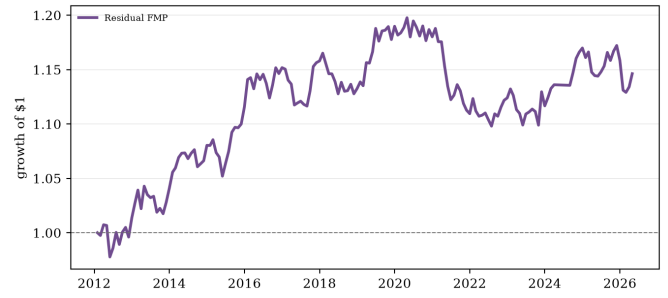


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ev	0.55%	6.15%	0.09	0.36	-24.36%	0.710	1.31%	6.15%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ev	0.03 (1.69)	0.56 (15.71)	-0.09 (-3.28)	-0.01 (-0.50)	-0.01 (-0.27)	-0.00 (-0.00)	1.04% (1.18)	0.712

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ev	0.42%	-1.04%	0.26%	-0.08%	-0.04%	-0.00%	1.04%	0.55%

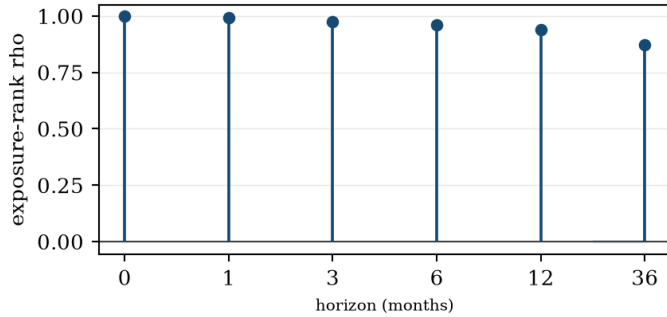


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

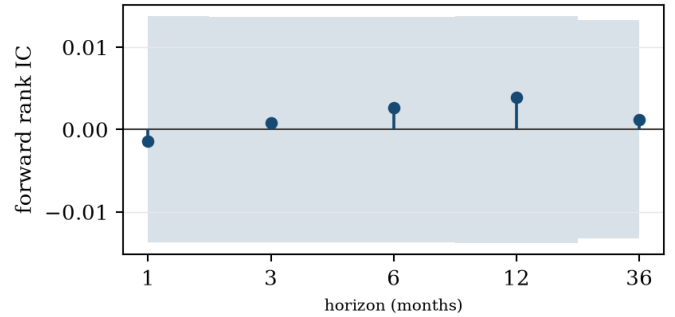


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	ev
Component signs	ev(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	50 / 76
Stocks, min / median / max	699 / 980 / 1649
Available-history Sharpe	0.28

72. vol_63d

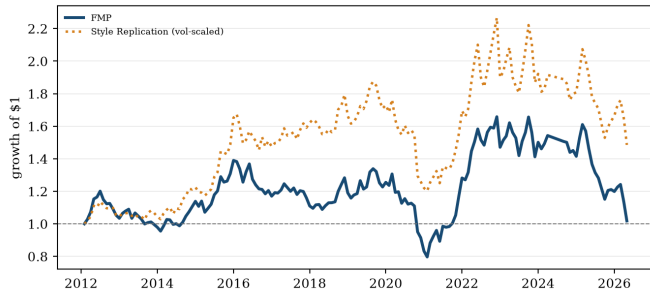


Figure: Common-window cumulative growth of \$1.

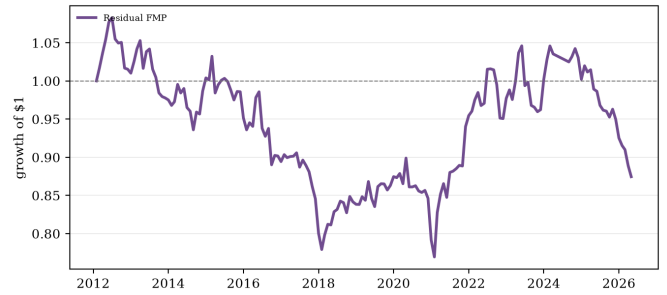


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
vol_63d	1.33%	15.47%	0.09	0.30	-42.68%	0.770	-2.23%	15.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
vol_63d	-0.21 (-3.41)	-0.43 (-4.42)	0.23 (3.47)	0.25 (3.24)	0.30 (2.50)	0.82 (6.53)	-0.70% (-0.38)	0.776

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
vol_63d	-2.99%	0.80%	-0.69%	1.59%	1.74%	1.58%	-0.70%	1.33%

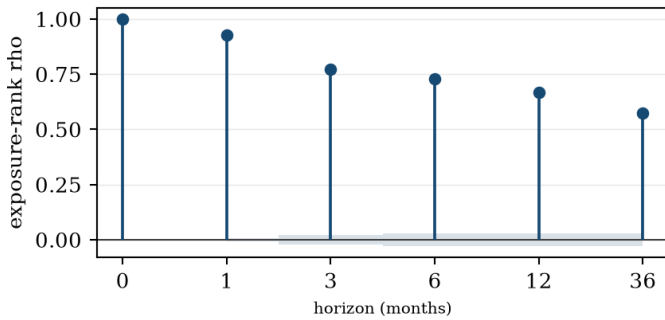


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

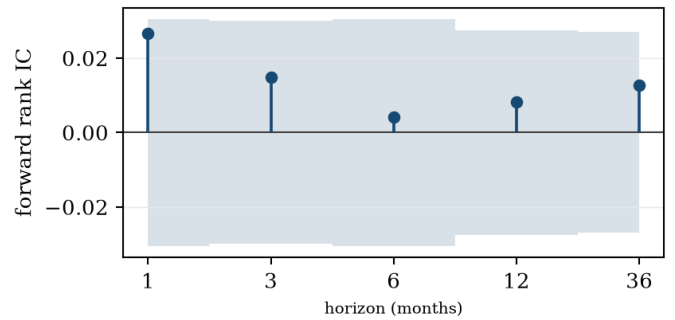


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	vol_63d
Component signs	vol_63d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	749 / 1058 / 1674
Available-history Sharpe	0.07

73. is_negative_invested_capital

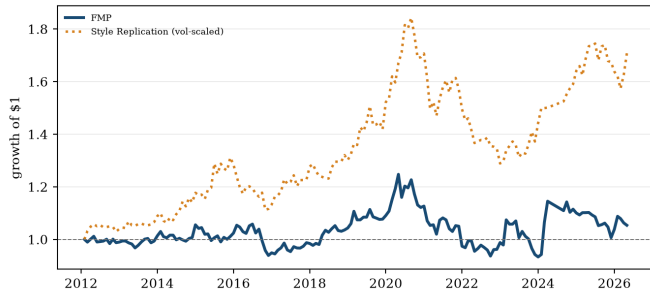


Figure: Common-window cumulative growth of \$1.

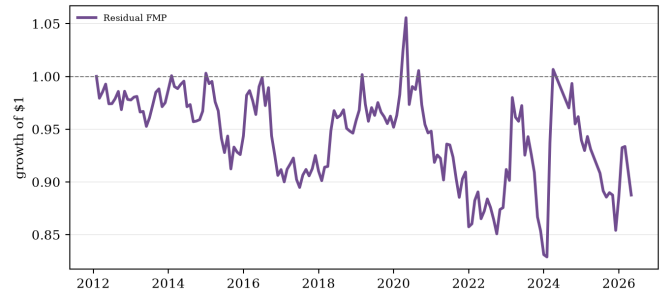


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_negative_invested_capital	0.76%	8.85%	0.09	0.34	-25.18%	0.107	-0.63%	8.85%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_negative_invested_capital	-0.02	-0.18	-0.18	0.02	0.11	-0.09	-0.52%	0.107
	(-0.35)	(-1.64)	(-2.12)	(0.26)	(1.20)	(-1.01)	(-0.23)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_negative_invested_capital	-0.22%	0.33%	0.55%	0.13%	0.64%	-0.17%	-0.52%	0.76%

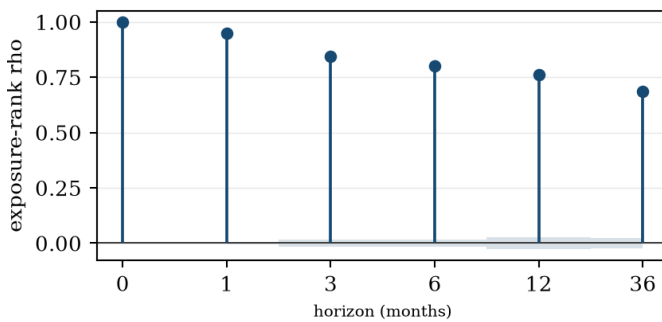


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

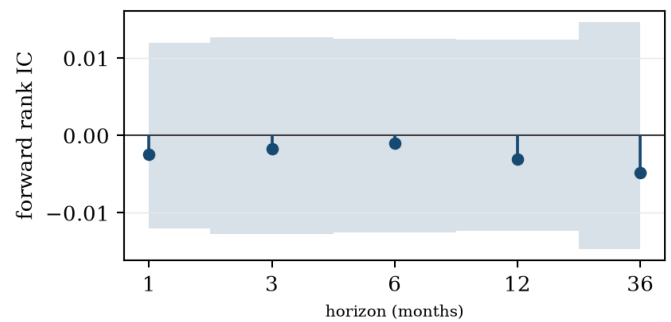


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	is_negative_invested_capital
Component signs	is_negative_invested_capital(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	241 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	47 / 74
Stocks, min / median / max	31 / 872 / 1328
Available-history Sharpe	0.23

74. dso

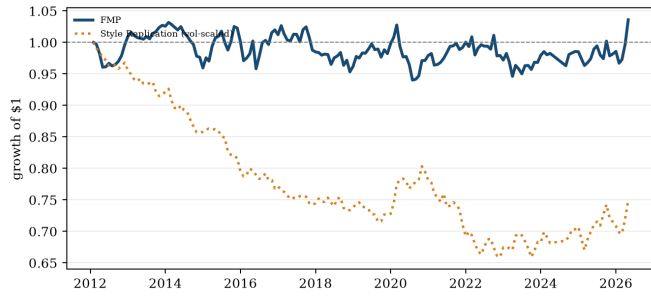


Figure: Common-window cumulative growth of \$1.

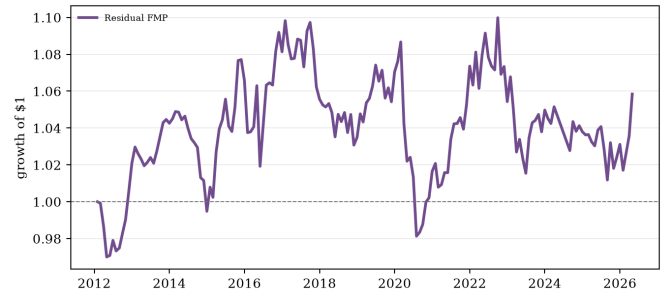


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dso	0.36%	4.57%	0.08	0.34	-8.87%	0.136	1.08%	4.57%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dso	0.03 (0.72)	0.02 (0.38)	-0.05 (-1.22)	0.02 (0.39)	-0.11 (-2.69)	-0.08 (-1.68)	0.50% (0.50)	0.150

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dso	0.38%	-0.03%	0.15%	0.12%	-0.60%	-0.15%	0.50%	0.36%

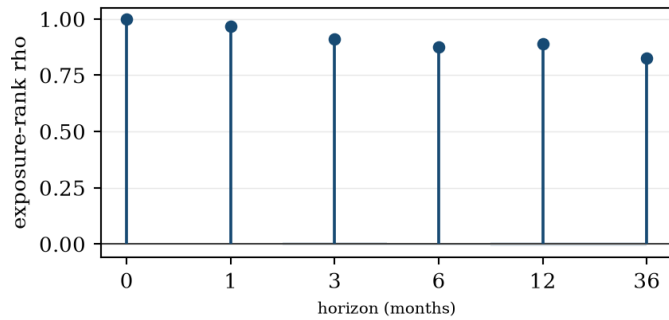


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

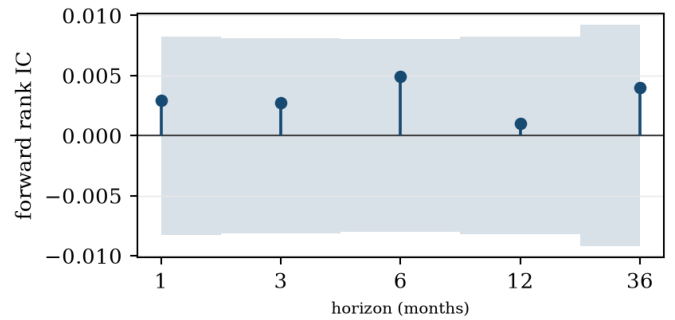


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	dso
Component signs	dso(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	15 / 66
Stocks, min / median / max	701 / 1023 / 1557
Available-history Sharpe	-0.17

75. lottery_max_5_21d

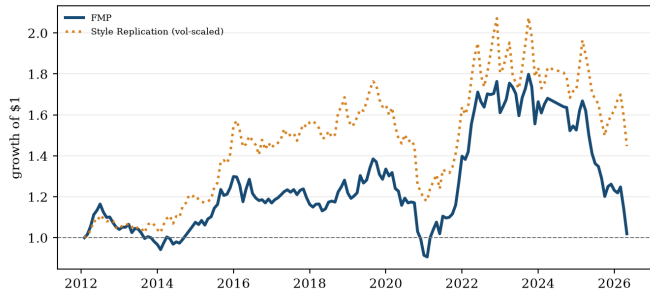


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
lottery_max_5_21d	1.04%	13.43%	0.08	0.26	-43.32%	0.725	-2.00%	13.43%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
lottery_max_5_21d	-0.15 (-2.52)	-0.41 (-4.64)	0.23 (3.05)	0.14 (1.64)	0.28 (2.28)	0.67 (5.60)	-0.61% (-0.29)	0.731

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
lottery_max_5_21d	-2.16%	0.77%	-0.69%	0.86%	1.58%	1.30%	-0.61%	1.04%

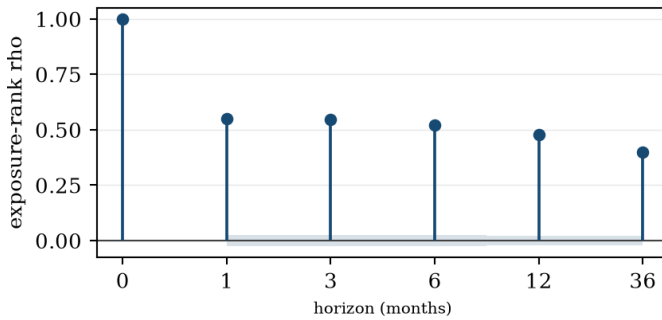


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

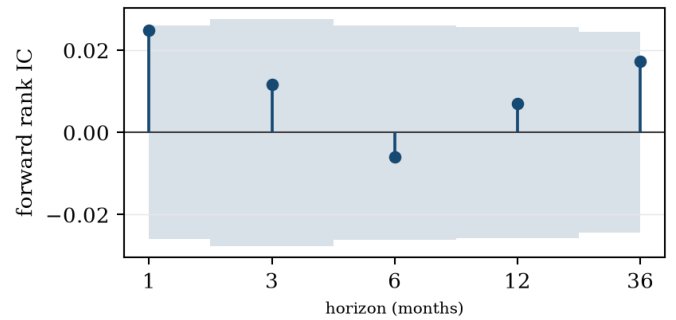


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	lottery_max_5_21d
Component signs	lottery_max_5_21d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	749 / 1060 / 1676
Available-history Sharpe	0.09

76. book_to_market

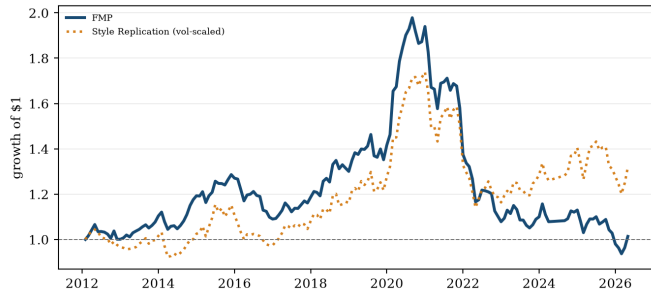


Figure: Common-window cumulative growth of \$1.

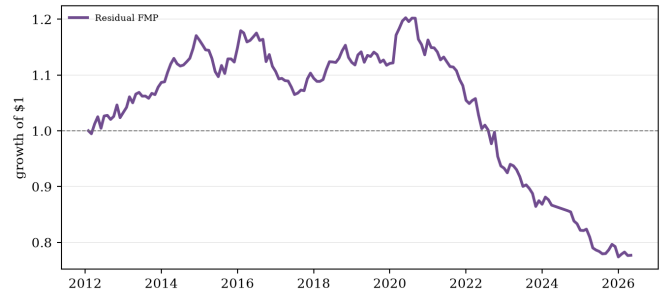


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
book_to_market	0.68%	10.81%	0.06	0.19	-52.64%	0.825	-1.66%	10.81%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
book_to_market	-0.01 (-0.45)	0.15 (2.87)	-0.76 (-15.97)	0.06 (1.68)	0.04 (0.51)	-0.00 (-0.07)	-1.71% (-1.05)	0.825

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
book_to_market	-0.20%	-0.28%	2.29%	0.38%	0.21%	-0.01%	-1.71%	0.68%

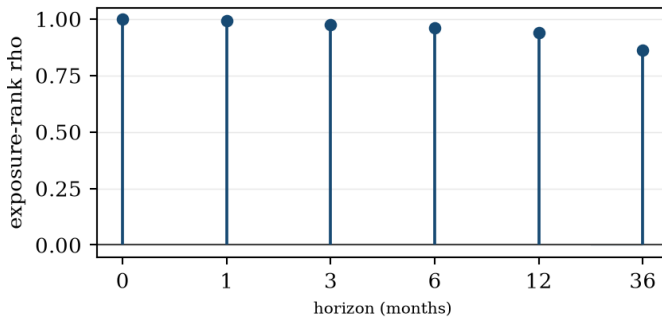


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

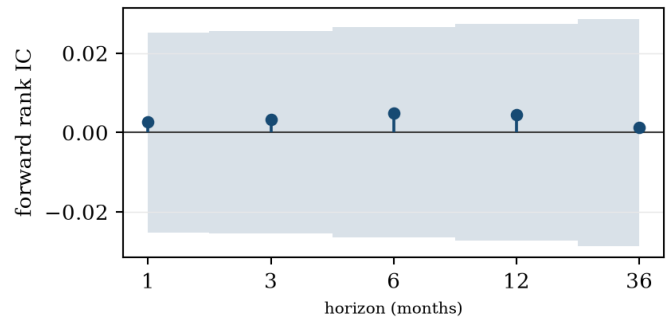


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	book_to_market
Component signs	book_to_market(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	744 / 1059 / 1677
Available-history Sharpe	0.02

77. pb

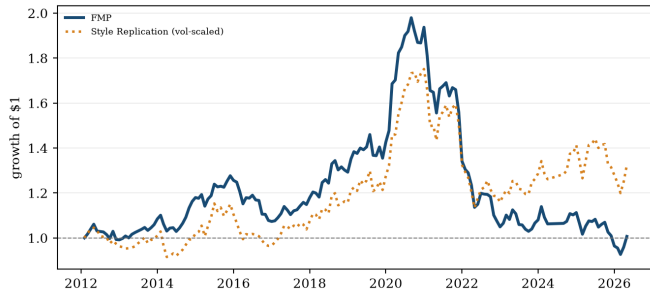


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
pb	0.68%	11.25%	0.06	0.19	-53.19%	0.822	-1.73%	11.25%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
pb	-0.01 (-0.44)	0.15 (2.81)	-0.78 (-15.34)	0.07 (2.14)	0.02 (0.26)	0.00 (0.07)	-1.78% (-1.06)	0.822

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
pb	-0.21%	-0.28%	2.36%	0.46%	0.12%	0.01%	-1.78%	0.68%

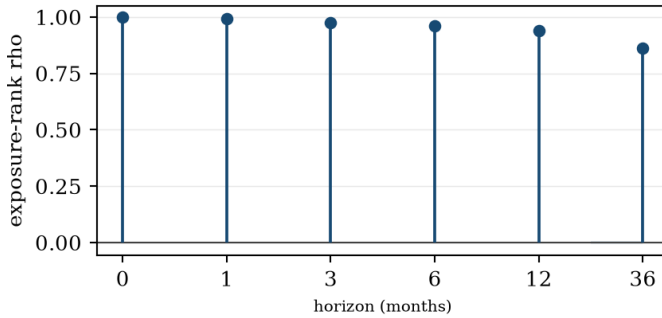


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

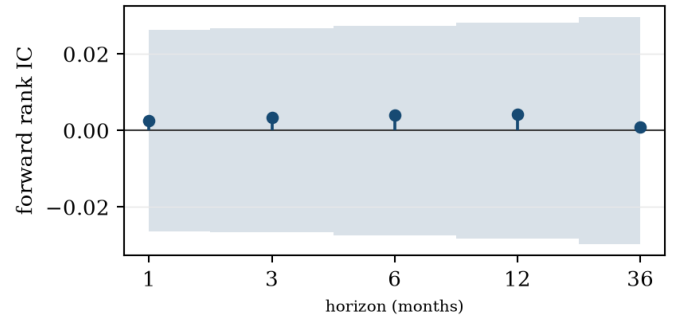


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	pb
Component signs	pb(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	729 / 1018 / 1597
Available-history Sharpe	0.01

78. gross_margin

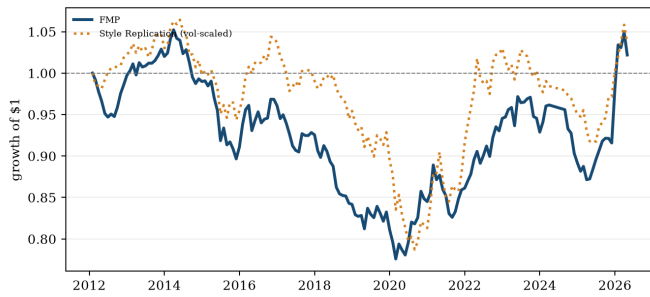


Figure: Common-window cumulative growth of \$1.

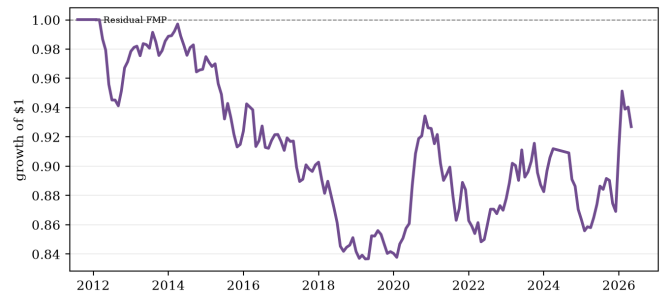


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gross_margin	0.31%	5.60%	0.06	0.18	-26.26%	0.467	0.03%	5.60%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gross_margin	0.13 (5.34)	0.13 (3.35)	0.34 (6.14)	0.12 (2.36)	-0.14 (-4.13)	0.07 (1.80)	-0.46% (-0.43)	0.475

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gross_margin	1.90%	-0.25%	-1.02%	0.79%	-0.79%	0.14%	-0.46%	0.31%

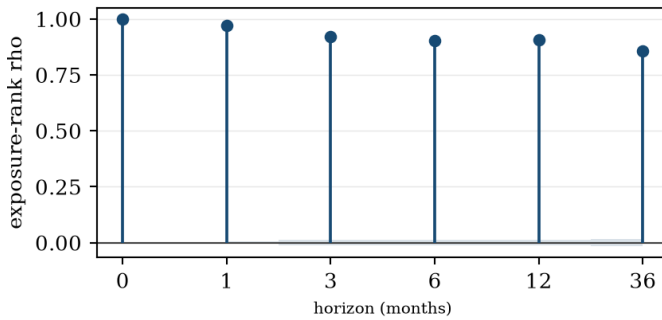


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

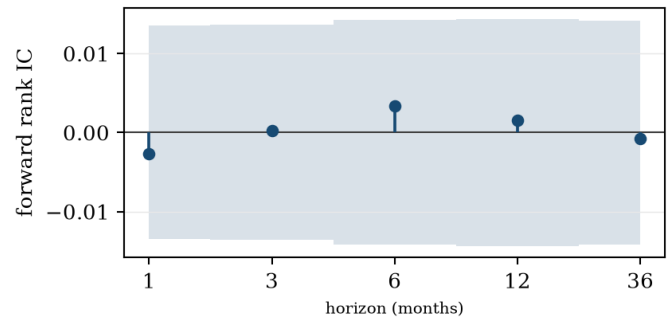


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	gross_margin
Component signs	gross_margin(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	745 / 1050 / 1647
Available-history Sharpe	0.09

79. ivol_21d

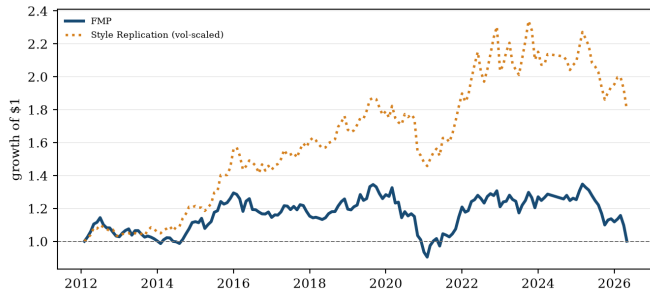


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ivol_21d	0.60%	10.83%	0.06	0.20	-32.76%	0.650	-3.28%	10.83%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ivol_21d	-0.07 (-1.48)	-0.26 (-3.06)	0.03 (0.72)	0.14 (2.45)	0.20 (1.95)	0.62 (5.82)	-2.01% (-1.36)	0.658

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ivol_21d	-1.04%	0.48%	-0.11%	0.92%	1.14%	1.20%	-2.01%	0.60%

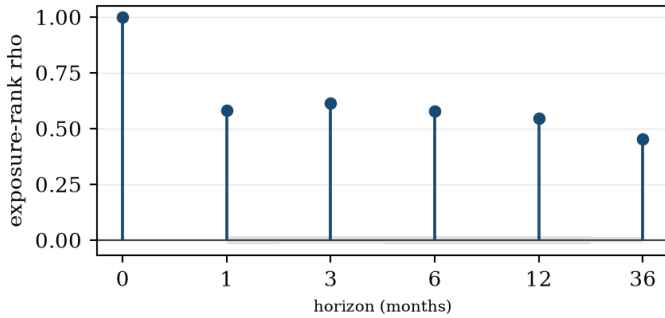


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

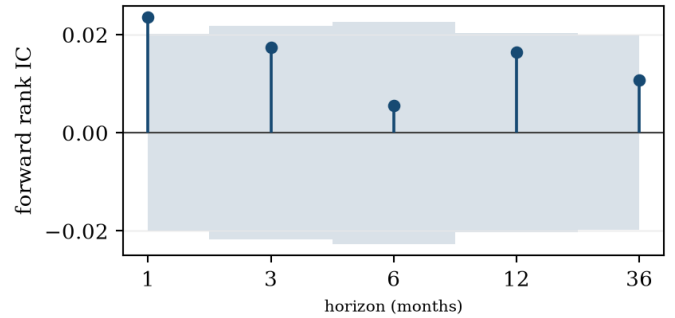


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	ivol_21d
Component signs	ivol_21d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	707 / 980 / 1564
Available-history Sharpe	0.08

80. log_market_cap

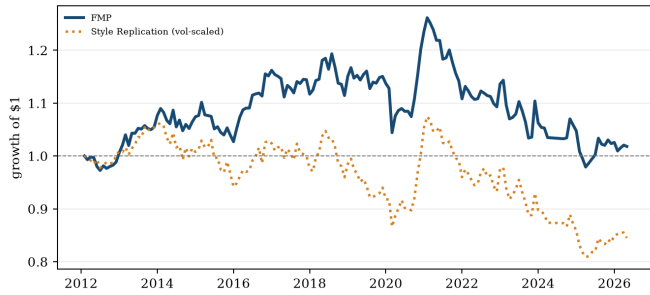


Figure: Common-window cumulative growth of \$1.

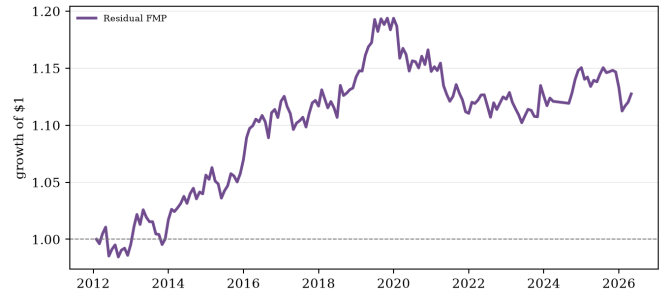


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
log_market_cap	0.32%	6.23%	0.05	0.21	-22.38%	0.775	1.22%	6.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
log_market_cap	0.06 (2.76)	0.55 (16.55)	0.07 (2.49)	-0.04 (-1.62)	0.03 (0.82)	-0.07 (-1.97)	0.91% (1.05)	0.777

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
log_market_cap	0.83%	-1.02%	-0.20%	-0.25%	0.19%	-0.14%	0.91%	0.32%

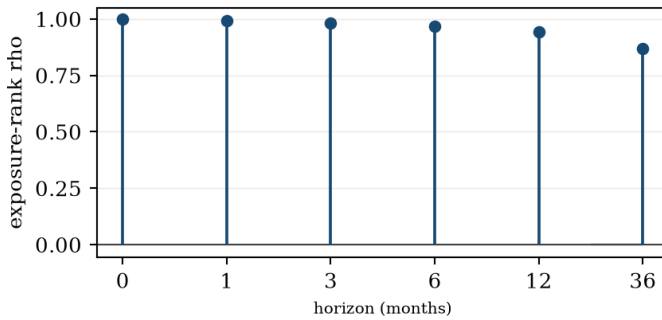


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

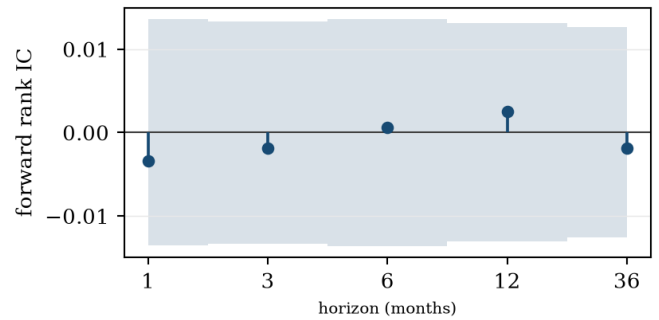


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	log_market_cap
Component signs	log_market_cap(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	749 / 1061 / 1678
Available-history Sharpe	0.28

81. market_cap

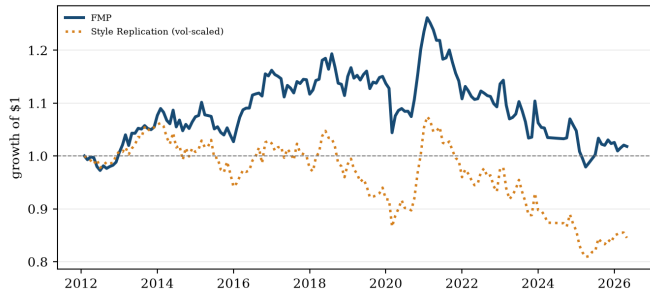


Figure: Common-window cumulative growth of \$1.

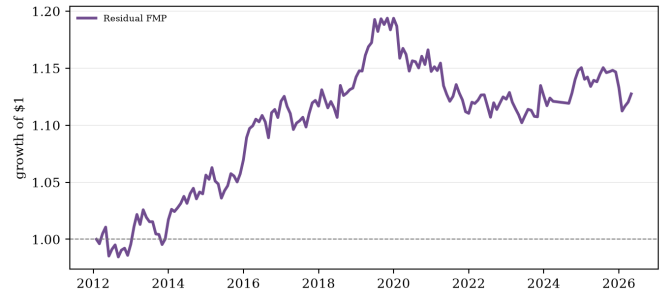


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
market_cap	0.32%	6.23%	0.05	0.21	-22.38%	0.775	1.22%	6.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
market_cap	0.06 (2.76)	0.55 (16.55)	0.07 (2.49)	-0.04 (-1.62)	0.03 (0.82)	-0.07 (-1.97)	0.91% (1.05)	0.777

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
market_cap	0.83%	-1.02%	-0.20%	-0.25%	0.19%	-0.14%	0.91%	0.32%

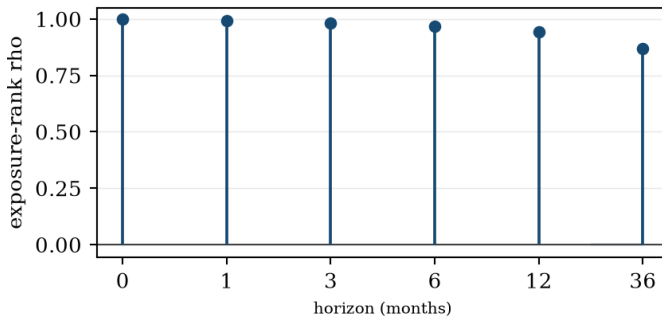


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

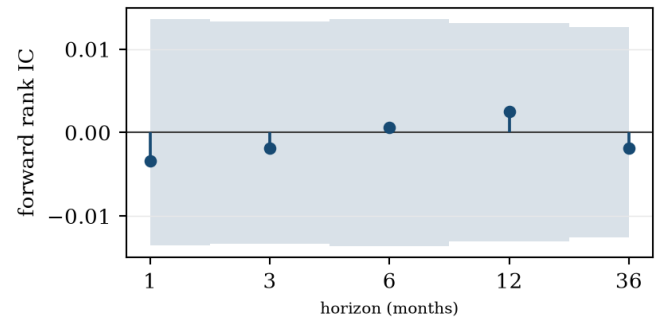


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	market_cap
Component signs	market_cap(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	749 / 1061 / 1678
Available-history Sharpe	0.28

82. altman_z

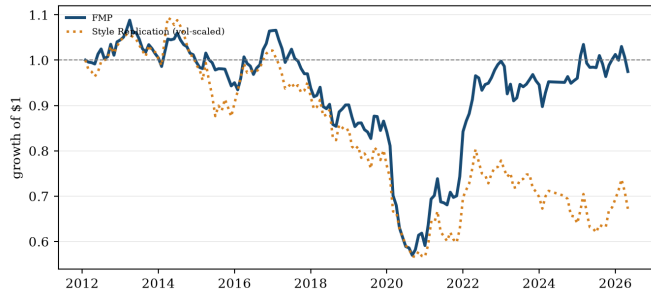


Figure: Common-window cumulative growth of \$1.

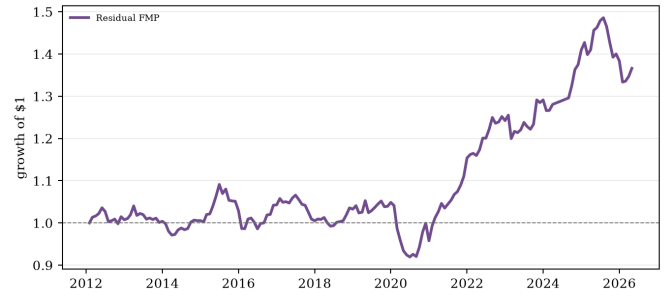


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
altman_z	0.34%	10.30%	0.03	0.10	-47.64%	0.713	2.31%	10.30%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
altman_z	-0.00 (-0.00)	-0.13 (-2.42)	0.69 (9.73)	-0.04 (-1.07)	0.02 (0.19)	-0.05 (-0.62)	2.39% (1.18)	0.713

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
altman_z	-0.00%	0.25%	-2.06%	-0.27%	0.12%	-0.09%	2.39%	0.34%

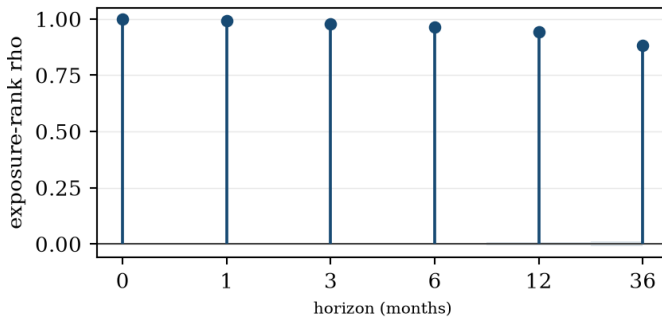


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

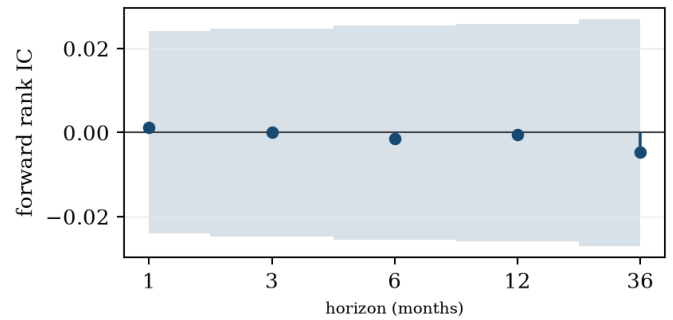


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	altman_z
Component signs	altman_z(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	15 / 66
Stocks, min / median / max	711 / 1039 / 1510
Available-history Sharpe	-0.01

83. earnings_yield

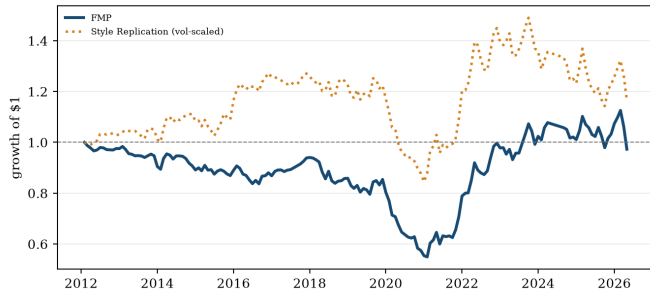


Figure: Common-window cumulative growth of \$1.

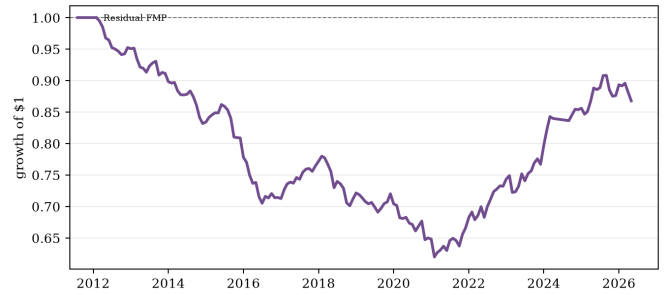


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
earnings_yield	0.35%	10.53%	0.03	0.11	-44.36%	0.758	-1.08%	10.53%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
earnings_yield	0.11 (3.45)	-0.17 (-2.51)	0.57 (10.92)	0.03 (0.68)	-0.01 (-0.16)	0.51 (7.82)	-0.88% (-0.50)	0.759

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
earnings_yield	1.52%	0.31%	-1.71%	0.18%	-0.05%	0.98%	-0.88%	0.35%

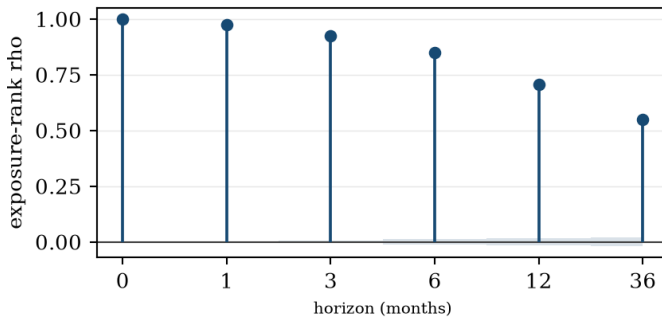


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

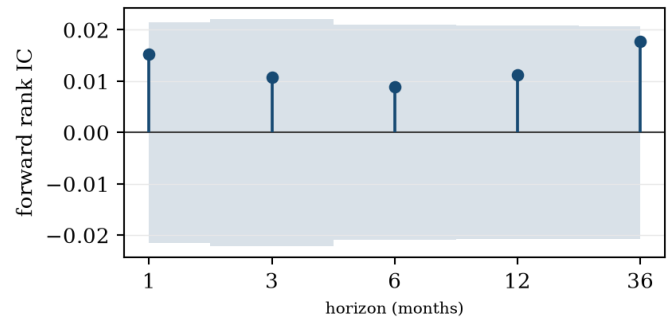


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	earnings_yield
Component signs	earnings_yield(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	15 / 66
Stocks, min / median / max	674 / 962 / 1519
Available-history Sharpe	0.20

84. earnings_to_price

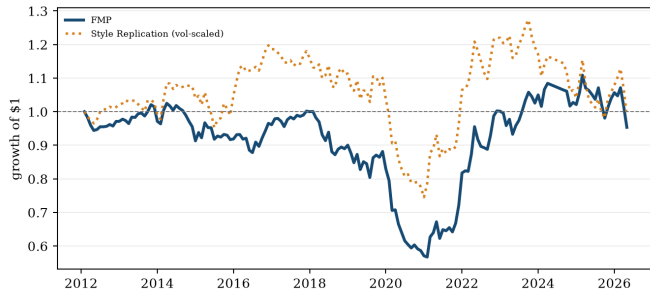


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
earnings_to_price	0.27%	11.18%	0.02	0.08	-44.65%	0.749	-0.18%	11.18%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
earnings_to_price	0.16 (5.21)	-0.11 (-1.76)	0.61 (7.99)	-0.10 (-2.05)	-0.06 (-0.82)	0.42 (6.35)	-0.19% (-0.09)	0.749

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
earnings_to_price	2.30%	0.20%	-1.85%	-0.66%	-0.34%	0.80%	-0.19%	0.27%

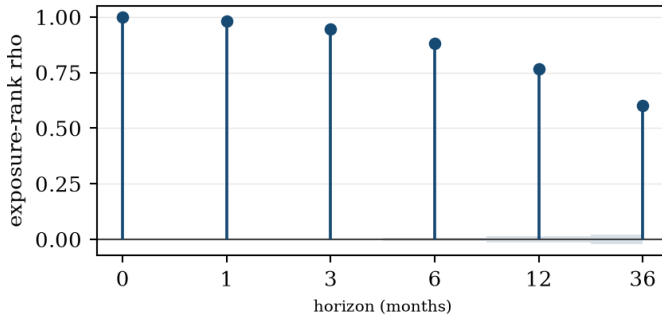


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

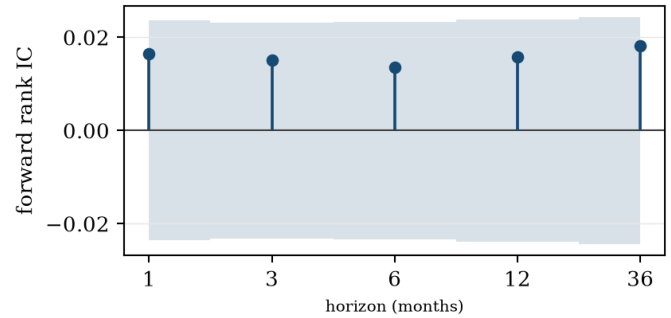


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	earnings_to_price
Component signs	earnings_to_price(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	21 / 66
Stocks, min / median / max	703 / 1034 / 1600
Available-history Sharpe	0.20

85. pe_ttm

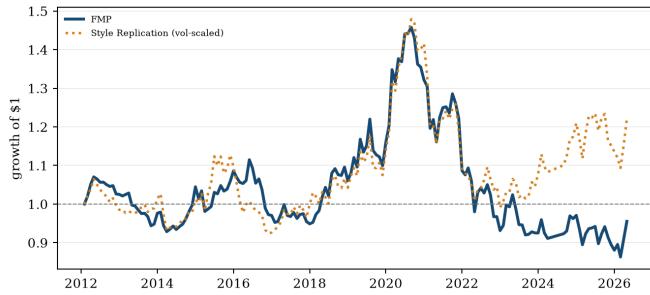


Figure: Common-window cumulative growth of \$1.

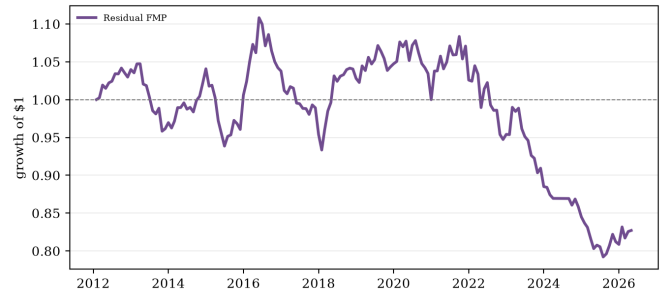


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
pe_ttm	0.22%	10.54%	0.02	0.08	-40.81%	0.709	-1.45%	10.54%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
pe_ttm	-0.13 (-3.39)	-0.07 (-1.11)	-0.55 (-7.96)	0.23 (4.02)	0.03 (0.43)	-0.10 (-1.52)	-1.20% (-0.65)	0.710

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
pe_ttm	-1.82%	0.13%	1.66%	1.45%	0.20%	-0.19%	-1.20%	0.22%

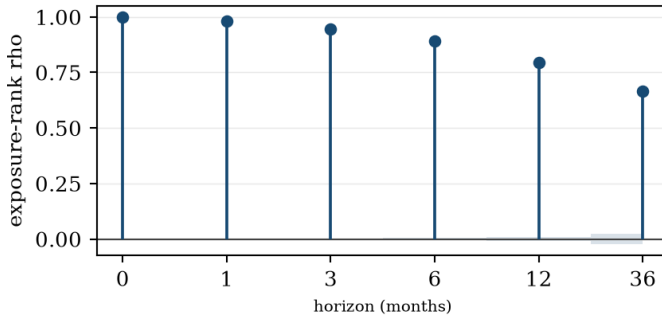


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

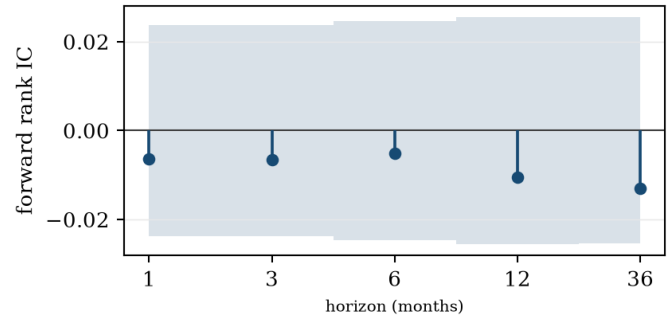


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	pe_ttm
Component signs	pe_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 167
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	21 / 66
Stocks, min / median / max	675 / 973 / 1457
Available-history Sharpe	-0.14

86. cash_ratio

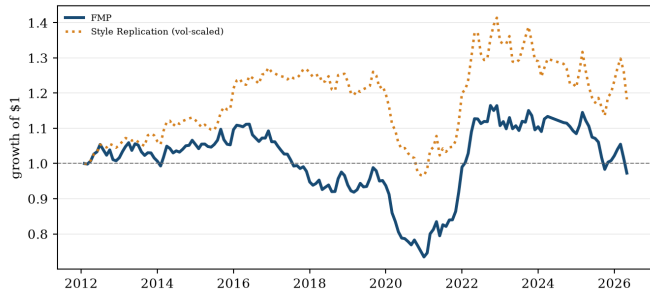


Figure: Common-window cumulative growth of \$1.

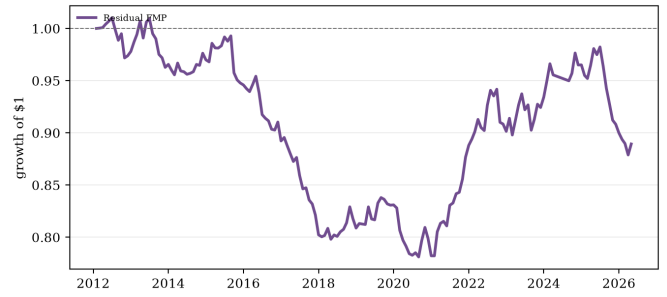


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
cash_ratio	0.09%	7.64%	0.01	0.04	-33.92%	0.702	-1.16%	7.64%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
cash_ratio	-0.01 (-0.25)	-0.10 (-2.12)	0.41 (9.50)	0.12 (3.13)	0.13 (1.89)	0.30 (4.86)	-0.76% (-0.59)	0.704

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
cash_ratio	-0.11%	0.18%	-1.25%	0.73%	0.72%	0.57%	-0.76%	0.09%

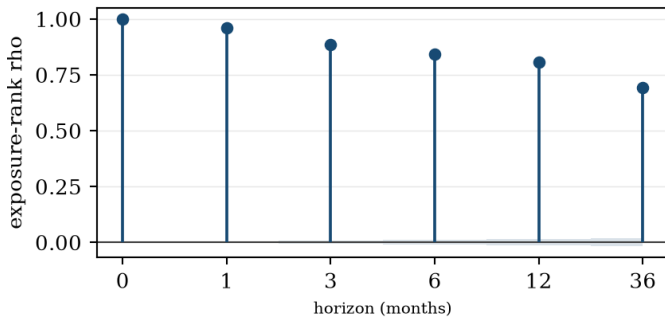


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

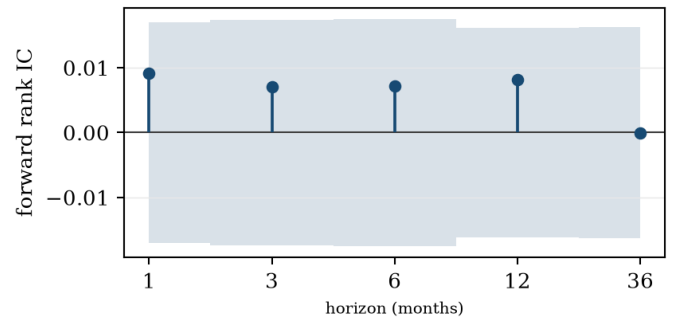


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	cash_ratio
Component signs	cash_ratio(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	737 / 1055 / 1653
Available-history Sharpe	-0.06

87. current_ratio

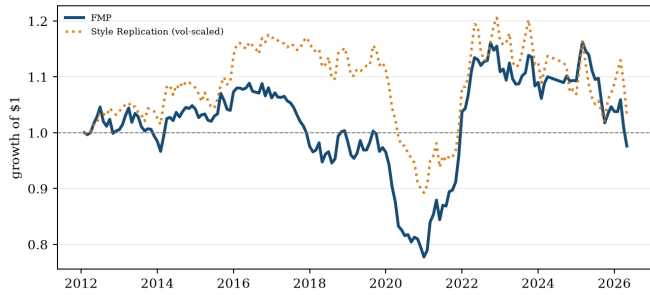


Figure: Common-window cumulative growth of \$1.

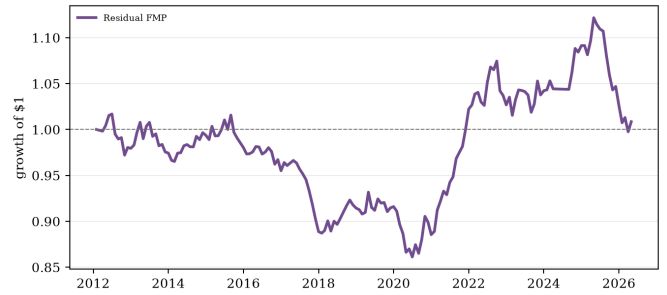


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
current_ratio	0.07%	7.03%	0.01	0.03	-28.57%	0.675	-0.32%	7.03%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
current_ratio	-0.05 (-1.67)	-0.20 (-4.84)	0.32 (6.45)	0.02 (0.35)	0.13 (2.16)	0.19 (3.59)	0.14% (0.11)	0.678

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
current_ratio	-0.68%	0.38%	-0.97%	0.10%	0.75%	0.36%	0.14%	0.07%

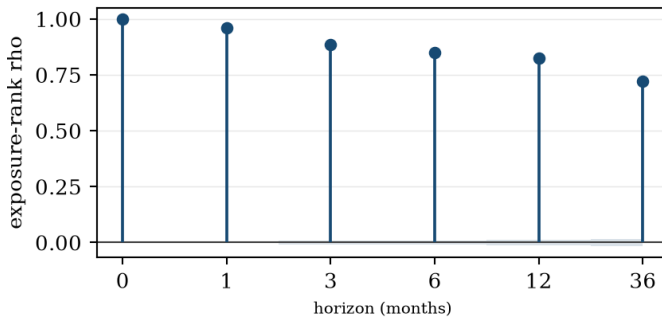


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

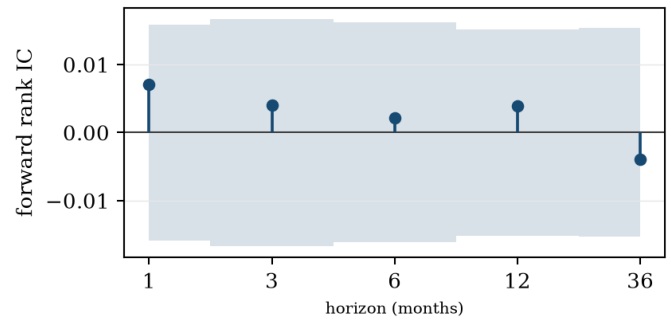


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	167
Signals	current_ratio
Component signs	current_ratio(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 76
Stocks, min / median / max	736 / 1055 / 1654
Available-history Sharpe	-0.03

88. net_operating_assets_proxy

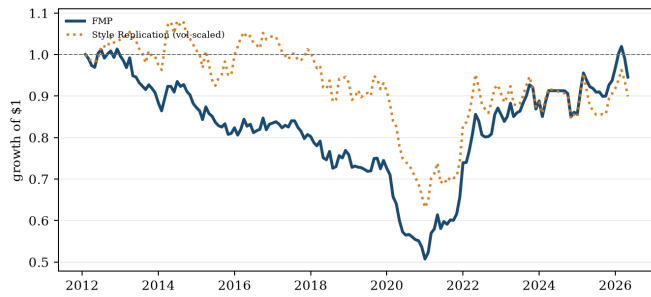


Figure: Common-window cumulative growth of \$1.

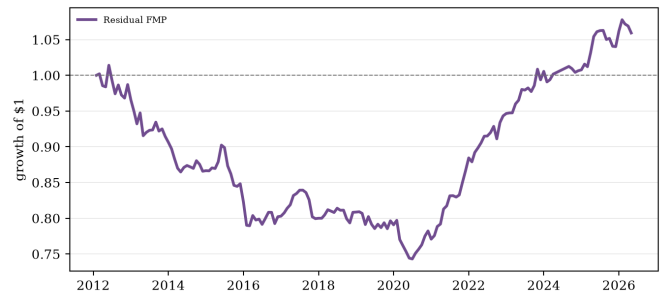


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_operating_assets_proxy	0.09%	10.00%	0.01	0.03	-49.93%	0.805	0.33%	10.00%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_operating_assets_proxy	0.01	-0.54	0.58	0.01	-0.01	0.07	0.51%	0.806
	(0.42)	(-12.00)	(12.65)	(0.35)	(-0.17)	(1.15)	(0.37)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_operating_assets_proxy	0.17%	1.01%	-1.75%	0.07%	-0.06%	0.14%	0.51%	0.09%

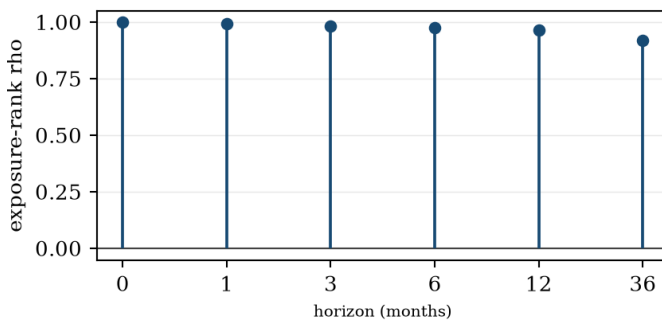


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

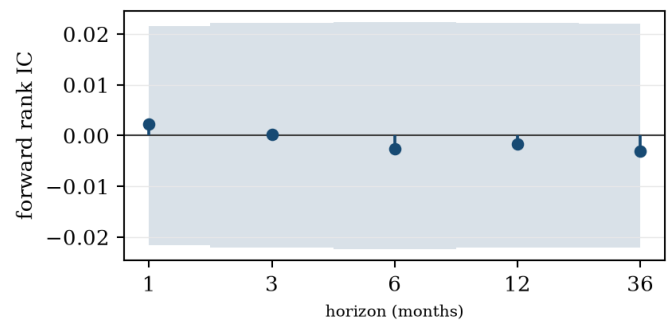


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	167
Signals	net_operating_assets_proxy
Component signs	net_operating_assets_proxy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 167
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	50 / 76
Stocks, min / median / max	699 / 980 / 1649
Available-history Sharpe	-0.08

Reproducibility

Run ID: 20260829T023500Z_all-available-registered-signals_2a27644e4e.

Gold snapshot: version=20260826T131322Z.

Configuration hash: 2a27644e4e.

Symbols requested / loaded: all / 5857.

Seed: 0. Comparison months: 167, 2012-03-01 to 2026-05-01.

Command: C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/
factor_model_runner/run.py--configsrc/experiments/factor_model_runner/all_signals_catalogue.py--n0--seed0.

Direction is selected once from raw comparison-window Sharpe and then applied before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.